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Faculty of Computers and Information

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CareerCompass: AI-Powered Career Guidance and Job Recommendation Platform

CareerCompass

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Acknowledgment

The project team would like to express sincere appreciation to Dr. Amena Mahmoud for academic supervision, technical guidance, and continuous feedback during the preparation of CareerCompass. The team also thanks the Faculty of Computers and Information at Kafr El-Sheikh University for providing the academic setting in which this graduation project was designed, implemented, tested, and documented.

The work presented in this book reflects a collaborative software engineering effort. It combines web application development, database design, AI-assisted document analysis, explainable matching, containerized deployment, testing, and technical documentation. The two supervisor-provided graduation books were used only to understand expected report structure and visual formality; no content, wording, project-specific claims, diagrams, or references were copied from them.

Abstract

CareerCompass is a graduation/demo career guidance platform that helps students and early-career users understand their CV profile, explore imported job opportunities, and compare their current skills against job requirements. The system consists of a React and Vite frontend, a Laravel API backend, a MySQL database, a FastAPI-based CV analyzer, a FastAPI/Scrapy-based job miner, MinIO-compatible private file storage, Nginx routing, and Prometheus/Grafana monitoring. The platform supports registration, login, CV upload, AI-assisted CV parsing, normalized profile and skills storage, job recommendation, gap analysis, an application tracker, and administrator dashboards for job and source diagnostics.

The AI CV Analyzer is documented as a hybrid implementation rather than a single opaque model. It combines PDF/image text extraction, OCR fallback, section segmentation, a BERT-family token-classification path for named-entity recognition, rule-based contact/date/experience extraction, skill canonicalization, domain and seniority classification, sentence embeddings, and TF-IDF-style matching. The runtime code can load an exported local NER artifact when that ignored deployment folder is present, and a Colab-oriented training notebook documents how the artifact is produced. A user-provided exported Colab PDF now records the overall NER training metrics, while the cleaned dataset content and model weights remain outside Git; therefore, the report separates recorded training-run evidence from repository-alone reproducibility and production benchmark claims.

The implementation is intentionally described as a graduation/demo system rather than a production product. The AI outputs are estimates, the job data depends on imported and demo sources, and the security posture is appropriate for demonstration but requires further production hardening. Validation was performed through Docker Compose configuration checks, backend/frontend evidence from earlier passes, Python syntax checks, containerized AI Job Miner tests, service health probes, a deterministic demo-source smoke check, and manual browser screenshots. Backend tests previously passed with 39 tests and 297 assertions, the AI Job Miner container test suite passed with 75 tests in the final cleanup pass, and frontend lint/build evidence remains recorded from the earlier validation pass. The AI CV Analyzer pytest suite was not rerun in this cleanup pass, while Python syntax compilation passed.

Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
CV	Curriculum Vitae
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
ML	Machine Learning
NLP	Natural Language Processing
OCR	Optical Character Recognition
RBAC	Role-Based Access Control
REST	Representational State Transfer
S3	Simple Storage Service compatible object storage
TF-IDF	Term Frequency-Inverse Document Frequency
UI	User Interface

Chapter 1: Introduction

1.1 Introduction to the Project

CareerCompass is an AI-assisted career guidance and job recommendation platform developed as a Computer Science graduation project for Kafr El-Sheikh University. The system helps a student upload a CV, receive a structured profile, view estimated job matches, inspect skill gaps, and save opportunities in an application tracker. The project combines web engineering, backend service design, natural language processing support, web scraping support, data persistence, and containerized operations.

The platform uses a modern web stack. The frontend is implemented with React, a component-based UI library [4], and bundled with Vite [5]. The backend uses Laravel, which provides routing, controllers, validation, middleware, Eloquent ORM, queues, testing support, and file storage abstractions [1]. The AI services are implemented with FastAPI, a Python API framework that supports type-hinted request and response handling [6]. The deployment is orchestrated through Docker Compose [11].

1.2 Background and Motivation

Many students prepare CVs and search for jobs without a clear view of how their current skills relate to job descriptions. Career guidance is often fragmented across CV feedback, job boards, learning resources, and manual advice. CareerCompass addresses this academic problem by placing those steps into one integrated demonstration system. The objective is not to replace human career advising, but to provide a practical software prototype that can parse a CV, normalize profile data, import job records, estimate matches, and present explainable gap analysis.

1.3 Problem Statement

Students frequently face three connected problems. First, they may not know whether their CV communicates their skills clearly. Second, job listings often describe requirements in different formats, making comparison difficult. Third, students may save opportunities across separate tools without a coherent tracker. CareerCompass proposes a centralized graduation/demo system that links CV data, jobs, matching, gap analysis, and tracking.

1.4 Purpose of the Project

The purpose of CareerCompass is to demonstrate how a distributed web system can support career exploration using explainable AI-assisted workflows. The project demonstrates authentication, CV upload, private storage, parsing, skill extraction, job import, recommendation, gap analysis, admin diagnostics, and monitoring in a Dockerized environment.

1.5 Project Objectives

- Provide a student-facing web interface for account creation, login, CV upload, profile review, job recommendations, gap analysis, and application tracking.
- Provide an administrator interface for dashboard statistics, job review, source diagnostics, and target role management.
- Implement a Laravel API that protects authenticated workflows using token-based access and role checks.
- Implement Python services for CV analysis and job mining support.
- Store uploaded CV files privately and expose downloads through controlled URLs.
- Validate the system with backend tests, frontend lint/build, Python tests where available, Docker checks, HTTP probes, and browser screenshots.

1.6 Proposed Solution

The proposed solution is a multi-service application. React renders the browser interface. Nginx serves as the gateway. Laravel handles REST-style APIs over HTTP [16], authentication, validation, data models, business services, and admin routes. MySQL stores normalized data [8]. MinIO provides S3-compatible object storage [9]. The CV analyzer parses PDFs and images using Python text extraction and OCR-related libraries. The job miner imports jobs from demo, API, and scraping-style sources using FastAPI, Scrapy, and HTML parsing concepts [16], [17], [18].

1.7 Project Scope

The scope covers a graduation/demo environment: local Docker deployment, student workflows, admin workflows, testing, screenshots, and documentation. It does not claim production readiness, job placement certainty, full job market coverage, legal compliance for production privacy, or complete AI accuracy.

1.8 Graduation Demo Positioning

CareerCompass should be presented as a working academic prototype with realistic engineering boundaries. The screenshots and tests show that the core workflows can run locally. However, large-scale evaluation, production identity management, cloud infrastructure, observability hardening, legal privacy review, and complete market integrations remain future work.

1.9 Report Organization

Chapter 2 analyzes requirements and users. Chapter 3 presents architecture, diagrams, database design, and deployment. Chapter 4 lists software and tools with references. Chapter 5 documents implementation modules from the repository. Chapter 6 presents the AI CV Analyzer deep technical analysis as a standalone academic contribution. Chapter 7 presents the AI Job Miner and scraping deep technical analysis. Chapter 8 presents testing and evaluation results. Chapter 9 discusses security and privacy. Chapter 10 concludes with achievements, limitations, and future work.

Chapter 2: System Analysis

2.1 Introduction

System analysis defines what CareerCompass should do, who uses it, and what constraints influence implementation. The analysis is based on the reviewed repository, including the root documentation, Docker configuration, Laravel API, React frontend, Python services, database migrations, seeders, tests, and finalization notes.

2.2 Development Methodology

The project followed an iterative engineering approach. Each feature was implemented, checked through local commands or tests, integrated with Docker services, then documented. This approach is appropriate for a graduation project because it supports incremental progress while keeping the system demonstrable. The architecture uses separate services, which follows a common microservice-style pattern where independently deployable services communicate over network APIs [29].

2.3 Existing Career Guidance Problems

The system targets common problems in student career preparation:

- CV content is not structured for direct software comparison.
- Job descriptions vary in wording and completeness.
- Students need explainable feedback rather than unexplained match numbers.
- Admin users need visibility into imported jobs and scraping sources.
- Demo systems need reliable startup, testing, and evidence for academic evaluation.

2.4 Target Users and Stakeholders

Stakeholder	Description	Main Interest
Student user	A university student or early-career user.	Upload CV, view profile, discover jobs, analyze gaps, save opportunities.
Administrator	A project operator or supervisor/demo administrator.	Review jobs, source diagnostics, users, target roles, and system health.
Supervisor	Academic supervisor evaluating the graduation project.	Correctness, completeness, originality, and honest evaluation.
Project team	Developers responsible for design and implementation.	Maintainable code, demonstrable workflows, testing, and documentation.

Table 1. Stakeholder summary.

2.5 User Roles

CareerCompass implements two practical roles. The student role can register, login, upload a CV, view recommendations, run gap analysis, and track applications. The admin role can access protected admin routes for dashboard statistics, job administration, scraping sources, target roles, and user review.

2.6 Functional Requirements

ID	Requirement	Implementation Evidence
FR-01	Register and login users.	Laravel AuthController, RegisterRequest, LoginRequest, React Login/Register pages.
FR-02	Upload a CV file.	CvUploadRequest validates PDF/JPEG/PNG and max size; Dashboard appends file field cv.
FR-03	Store CV files privately.	CvStorageService and MinIO/S3-compatible disk configuration.
FR-04	Parse CV and extract profile/skills.	CvProcessingService and AI CV Analyzer /api/parse-cv.
FR-05	Display normalized profile and skills.	UserResource, Profile page, Dashboard AI insight components.
FR-06	Import and display jobs.	JobPosting model, ScrapedJobController, JobController, AI Job Miner.
FR-07	Estimate job recommendations.	JobController and GapAnalysisService use CV/profile data and matching service.
FR-08	Analyze skill gaps.	GapAnalysisController and GapAnalysis page.
FR-09	Track applications.	ApplicationController, ApplicationTrackerService, Applications page.
FR-10	Provide admin dashboards.	Admin Dashboard, Jobs, Sources, Targets pages and admin API routes.
FR-11	Provide health and metrics endpoints.	HealthController, MetricsController, Prometheus/Grafana compose services.

Table 2. Functional requirements summary.

2.7 Non-Functional Requirements

Category	Requirement	CareerCompass Approach
Usability	The UI should guide students through CV upload and recommendations.	React pages, dashboard cards, profile score, and action buttons.
Maintainability	Code should be modular.	Controllers, requests, services, resources, models, React pages, and Python services are separated.
Reliability	The system should degrade honestly if AI services fail.	CV processing returns timeout/error/no_text statuses and preserves prior data when appropriate.
Security	Authentication, validation, private files, and admin role checks are required.	Sanctum tokens, request validation, admin middleware, signed URLs, service tokens.
Observability	Health and metrics should be available.	/api/health, /api/ready, /api/metrics, Prometheus, Grafana.
Portability	The demo should run locally.	Docker Compose services and environment examples.

Table 3. Non-functional requirements summary.

2.8 Hardware Requirements

For local demonstration, a developer machine capable of running Docker Desktop and multiple containers is required. CV parsing and OCR-like processing can be CPU-intensive; therefore, enough memory should be available for the Laravel backend, MySQL, frontend, Python services, MinIO, Prometheus, and Grafana. GPU acceleration is not required for the demonstrated flow.

2.9 Software Requirements

Layer	Software
Frontend	React, Vite, Tailwind-style CSS classes, lucide-react icons, Recharts.
Backend	Laravel 12, PHP, Composer dependencies, Sanctum-style token authentication.
AI services	Python, FastAPI, text extraction, OCR-related dependencies, Scrapy-style job mining.
Data	MySQL 8.0, MinIO/S3-compatible storage.
Infrastructure	Docker, Docker Compose, Nginx, Prometheus, Grafana.
Testing	PHPUnit/Pest-style Laravel tests, pytest for Python, ESLint and Vite build.

Table 4. Software environment summary.

2.10 Input and Output Flow

Primary inputs include user account data, uploaded CV files, imported job records, target role settings, and administrator source configurations. Primary outputs include normalized user profiles, skill lists, CV analysis metadata, estimated job matches, gap reports, application records, admin statistics, health checks, and monitoring metrics.

2.11 Use Case Summary

The main use cases are shown in Figure 5. The system separates student and administrator responsibilities while sharing the same backend API and database.

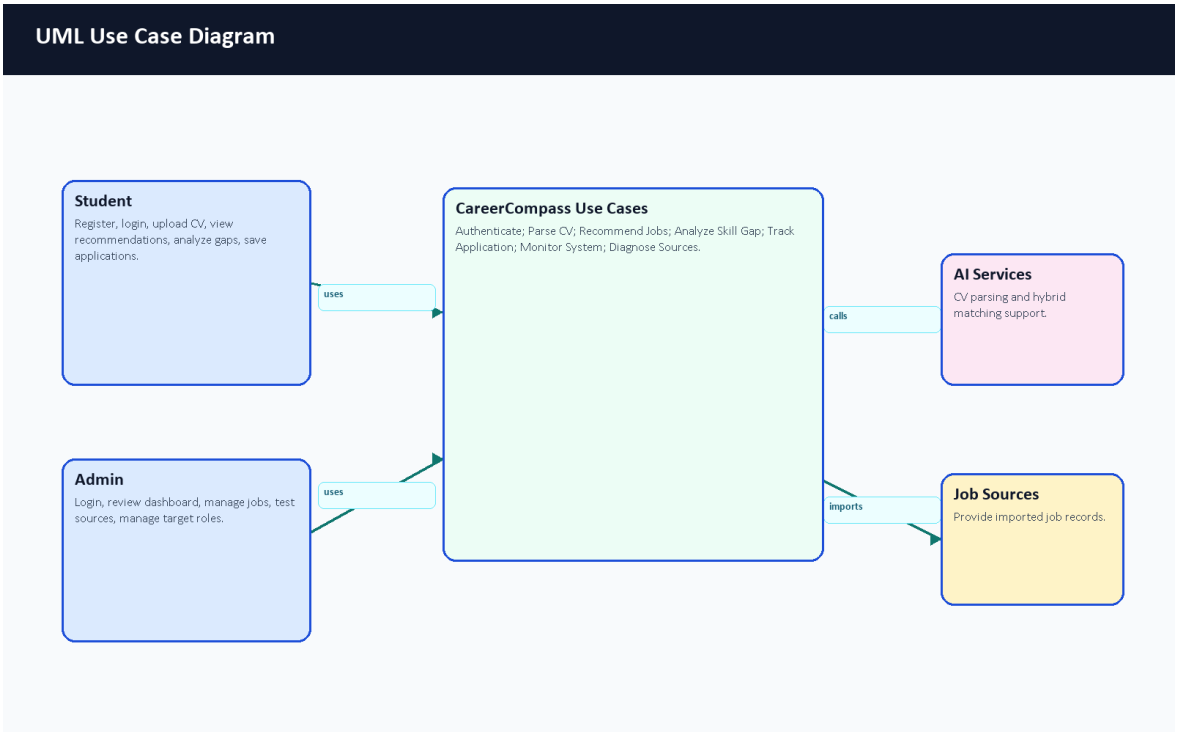


Figure 5. UML use case diagram.

Chapter 3: System Design and Architecture

3.1 Introduction

CareerCompass is designed as a Dockerized multi-service application. This design separates browser UI, API logic, AI services, data storage, object storage, reverse proxy routing, and monitoring. Docker containers help package runtime dependencies consistently [10], while Docker Compose coordinates the multi-container local deployment [11].

3.2 High-Level System Architecture

The high-level architecture is shown in Figure 1. Browser users interact with the React frontend through Nginx. The frontend calls the Laravel API. Laravel persists records in MySQL, stores CV files in MinIO-compatible storage, calls the AI CV Analyzer for parsing, calls matching logic for recommendations/gaps, and receives job imports from the job miner. This diagram was reviewed during the final evidence pass and already represents the important deployment boundaries: React, Nginx, Laravel, MySQL, MinIO, AI CV Analyzer, AI Job Miner, and monitoring.

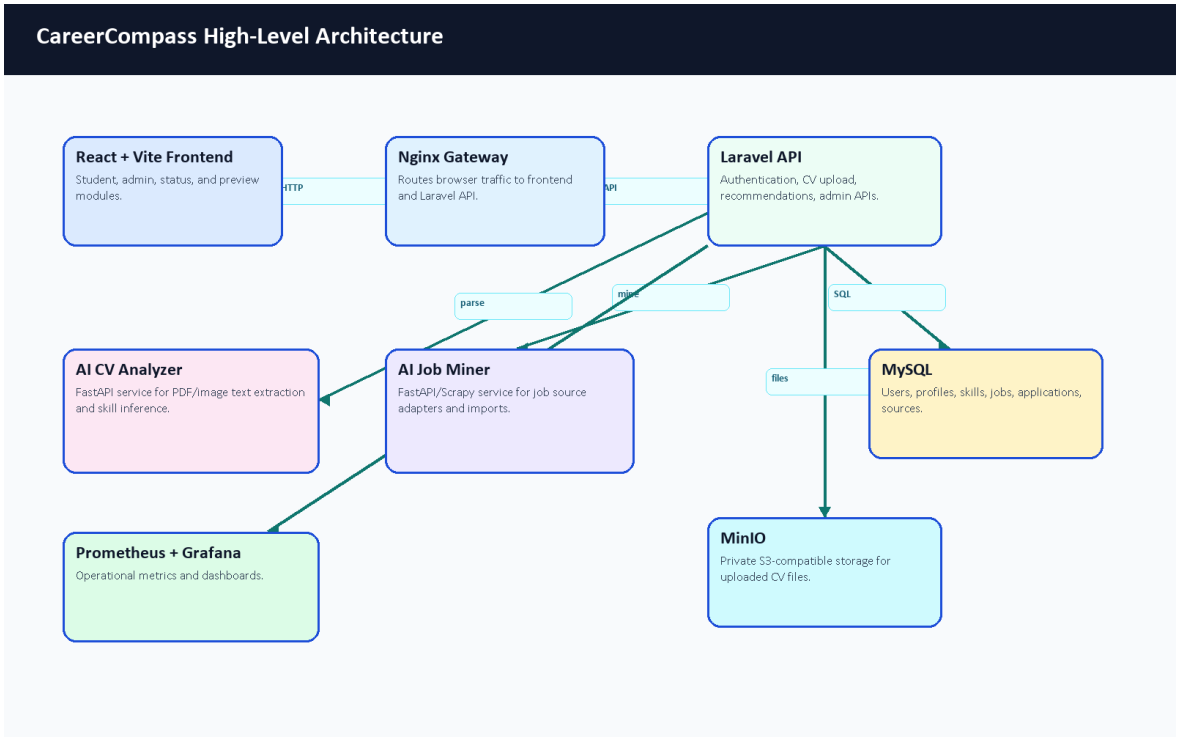


Figure 1. High-level architecture of CareerCompass.

3.3 Frontend Architecture

The frontend is implemented with React and organized around React Router routes in frontend/src/App.jsx [42]. Public pages include Home, Login, Register, About, Privacy, Terms, and System Status. Protected student routes include Dashboard, Jobs, Gap Analysis, Profile, Settings, Market Intelligence, Applications, CV Builder, Mock Interview, Learning, Career Planner, Mentorship, and Tools Hub. Protected admin routes include Admin Dashboard, Jobs, Users, Sources, and Target Roles.

Figure 63 shows how the route tree is divided. This separation matters because the student experience is focused on career guidance, while the admin experience is focused on operating data, users, sources, targets, and diagnostics. The preview modules are included as graduation and future-extension screens;

they demonstrate the intended product direction but should not be described as complete production modules.

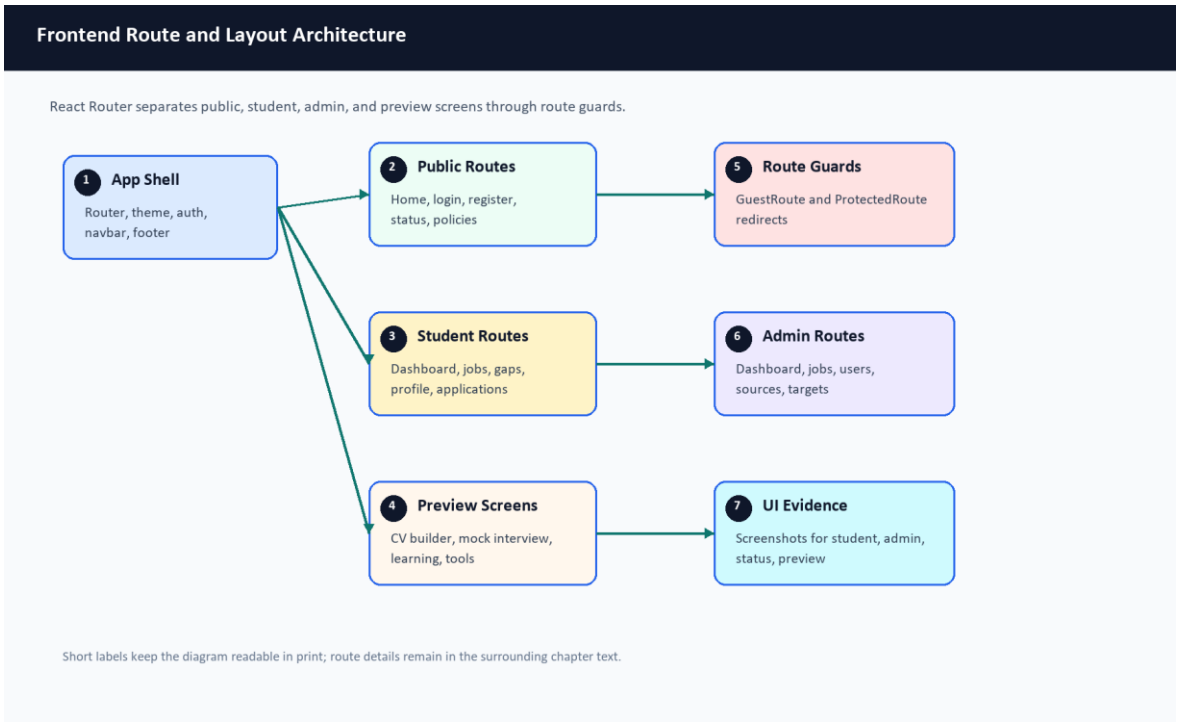


Figure 63. Frontend route and layout architecture.

The frontend API layer is located under frontend/src/api. Axios is configured in client.js, including base URL resolution, bearer token injection, request IDs, retry behavior for safe GET/HEAD requests, and 401 handling [43]. Authentication state is managed by AuthContext.jsx, which stores the user and token in local storage and refreshes the current user through /user. Route guards in ProtectedRoute.jsx and GuestRoute.jsx redirect unauthenticated users, keep admin-only routes behind role checks, and prevent logged-in users from returning to guest-only screens. Localization files exist under frontend/src/locales, and i18n.js uses browser language detection with English fallback.

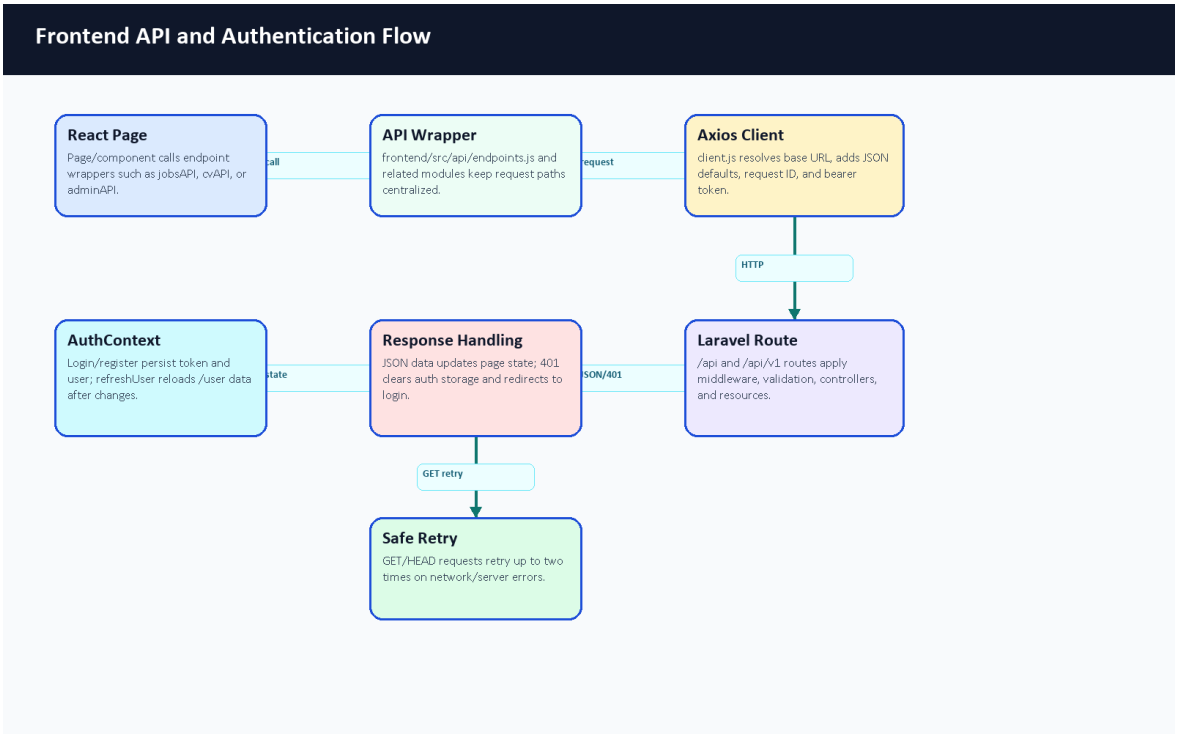


Figure 64. Frontend API and authentication flow.

3.4 Backend API Architecture

The backend is a Laravel API. Routes are defined in backend-api/routes/api.php and are registered both at /api and /api/v1. The API includes public health/readiness/metrics endpoints, guest authentication routes, public job listing routes, internal scraper import routes protected by a service token, authenticated student routes, and admin routes protected by middleware.

Laravel provides structured controllers, form requests, resources, services, models, migrations, seeders, and tests. This aligns with Laravel's documented framework responsibilities, including routing, validation, database access, queues, and testing [1], [3].

Figure 65 summarizes the normal Laravel request lifecycle and the asynchronous branch used for longer tasks such as CV processing and job mining. The main design point is that controllers do not directly own every behavior: form requests validate input, services isolate reusable work, Eloquent models persist records, resources shape JSON responses, and queue workers handle tasks that should not block the browser.

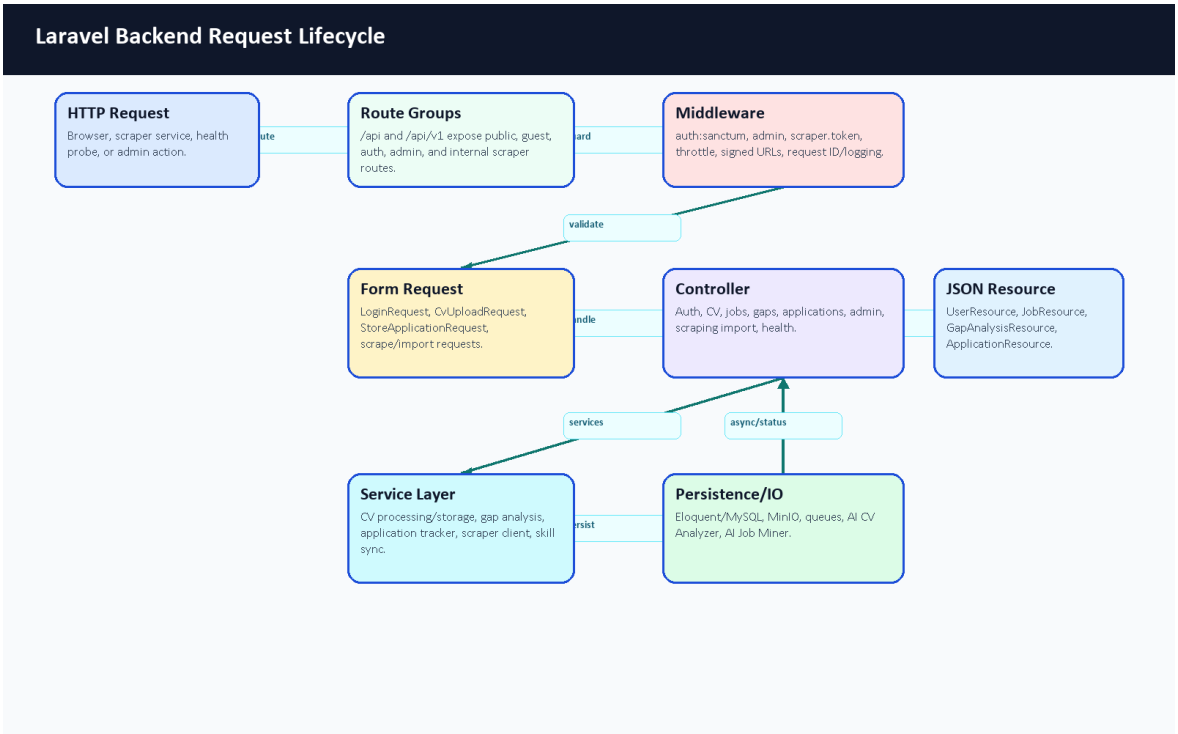


Figure 65. Laravel backend request lifecycle.

Backend Module	Main Files / Components	Responsibility	Evidence
Authentication	routes/api.php, AuthController, RegisterRequest, LoginRequest, UserResource	Registration, login, logout, current-user lookup, profile update, and token lifecycle.	Guest auth routes, Sanctum tokens, throttled login group.
CV Upload and Processing	CvController, CvUploadRequest, CvProcessingService, CvStorageService	Validate CV files, store private file metadata, call the AI CV Analyzer, and persist analysis/profile/skill updates.	/upload-cv, signed CV file route, cv_analyses storage fields.
Profile and Skills	User, UserProfile, Skill, user_skills, SkillSyncService	Keep extracted skills and profile evidence normalized for matching.	Eloquent relationships and many-to-many skill pivots.
Jobs and Recommendations	JobController, JobResource, Job, job_postings, job_skills	Public job listing, details, recommendations, and job-skill requirements.	Public /jobs routes plus authenticated recommendation routes.
Gap Analysis	GapAnalysisController, GapAnalysisService, GapAnalysisResource	Compare a user's evidence against job or role requirements.	/gap-analysis/job/{jobId} and /gap-analysis/role/{roleId}.
Application Tracking	ApplicationController, ApplicationTrackerService, Application, ApplicationResource	Save and update opportunities across saved/applied/interviewing-style statuses.	Authenticated applications resource routes.
Admin Operations	AdminDashboardController, AdminJobController, AdminUserController, ScrapingSourceController, TargetJobRoleController	Admin statistics, users, jobs, scraping sources, diagnostics, and target roles.	Admin route group protected by admin middleware.

Backend Module	Main Files / Components	Responsibility	Evidence
Scraping Import	ScrapedJobController, VerifyScraperToken, import request classes	Protected duplicate checks, imports, failure reports, and proxy access for scraper integration.	scraper.token and throttle:scraper route group.
Health and Metrics	HealthController, MetricsController, monitoring middleware	Liveness, readiness, and metrics surfaces for local operations.	/health, /ready, /metrics, and /status.
Queues and Workers	ProcessOnDemandJobScraping, market scraping jobs, Docker workers	Move long network or service calls out of request time.	Queue jobs and dedicated worker services.
File Storage	CvStorageService, signed CV file route, cv_analyses metadata	Keep uploaded CV binaries out of public MySQL fields and expose only signed access.	Private storage path, disk, checksum, MIME, and size metadata.

Table 5. Backend module responsibility summary.

Risk / Input Boundary	Control	Example Files
Invalid login or registration payload	Form requests, guest auth routes, and login throttling.	RegisterRequest, LoginRequest, routes/api.php
Invalid CV file upload	File validation, upload throttling, private storage, and status-aware error handling.	CvUploadRequest, CvController, CvStorageService
Unauthenticated user routes	Sanctum bearer-token middleware and current-user refresh.	auth:sanctum, AuthController::user
Non-admin access to admin routes	Backend admin middleware plus frontend role-based route guards.	routes/api.php, ProtectedRoute.jsx
Invalid scraper import payload	Service-token middleware, scraper throttling, and import form requests.	VerifyScraperToken, StoreScrapedJobRequest, CheckScrapedJobRequest, ReportScrapingFailureRequest
Duplicate scraped jobs	URL and title/company checks inside a transaction before create/update.	ScrapedJobController, job_postings migration
Private CV file access	Signed route and storage service rather than public direct file paths.	/cv-files/{cvAnalysis}, CvStorageService
Request tracing and debugging	Frontend request IDs and backend request-id middleware.	client.js, RequestIdMiddleware

Table 6. Laravel validation and protection mapping.

3.5 AI CV Analyzer Architecture

The AI CV Analyzer is a FastAPI service. Laravel sends CV files to this service for parsing through /api/parse-cv. The analyzer routes PDFs and images differently, enforces timeout/error fallbacks, extracts readable text, runs structured extraction, and returns fields such as predicted role, seniority, domain, skills, strengths, gaps, red flags, confidence, document statistics, and parsing status. The backend handles statuses such as success, OCR fallback, timeout, error, empty file, and no text.

The analyzer is not a pure pretrained-model wrapper and not a model built entirely from scratch. It is a hybrid pipeline. The runtime prefers a local exported token-classification model under ai-cv-

analyzer/models/ner_weights/career_compass_ner_final when that ignored deployment artifact exists; if the local artifact is unavailable, the NER engine has a fallback model path. Around that model-loading path, the team implemented practical CV-specific logic: spatial PDF parsing, OCR fallback, semantic sectioning, contact extraction, date/experience parsing, noisy-skill filtering, canonicalization, domain inference, seniority inference, and hybrid matching. PDF and OCR-related libraries are supported by external tools such as PyMuPDF, pdfplumber, and EasyOCR [22], [23], [24]. Transformer token classification and training concepts follow Hugging Face documentation [31], [32], [33].

Figure 9 summarizes the runtime path from the browser upload to Laravel persistence, FastAPI parsing, model/rule extraction, and dashboard output.

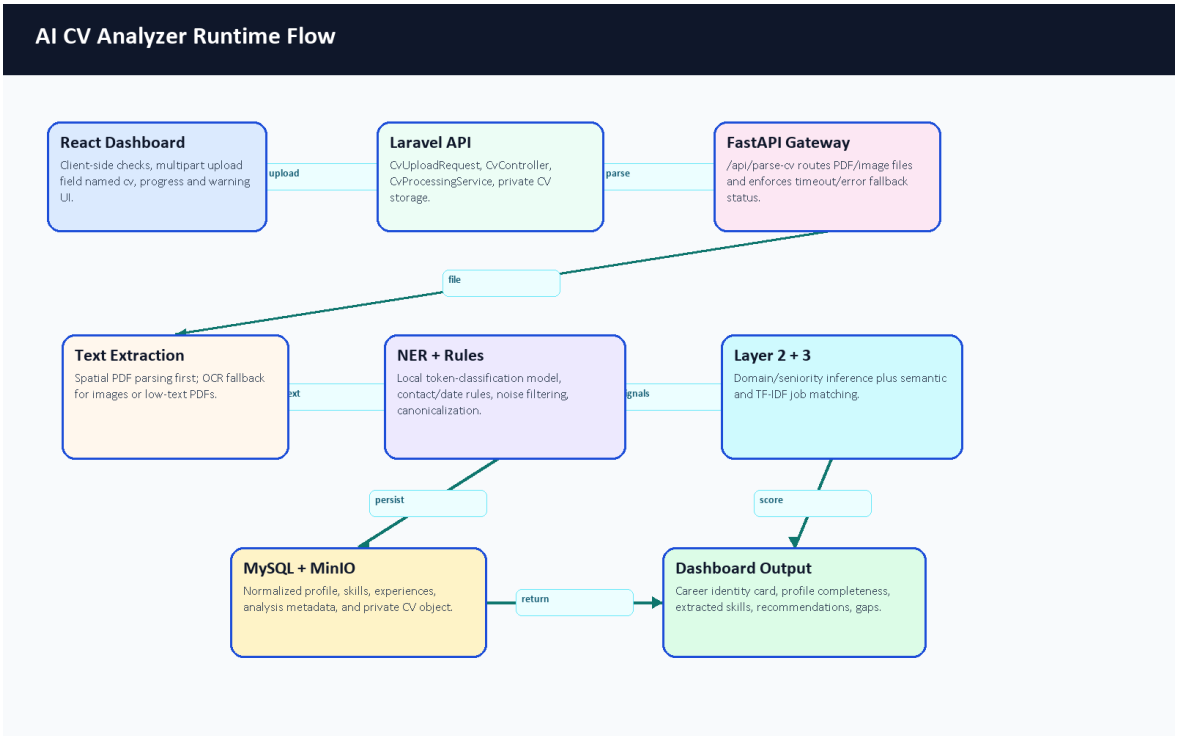


Figure 9. AI CV Analyzer runtime flow.

3.6 AI Job Miner Architecture

The AI Job Miner is a FastAPI service with scraping and import support. It includes source adapters for demo/local data, public APIs, and HTML/Scrapy-style extraction. Scrapy is a Python framework for extracting structured data from websites [17], and Beautiful Soup is commonly used to parse HTML documents [18]. CareerCompass uses a quality gate and honest source classifications so that demo/imported data is not overstated as broad labor-market reach.

3.7 Database Design

MySQL stores users, user profiles, skills, user-skill pivots, experience records, CV analyses, job postings, job-skill pivots, applications, scraping sources, target job roles, scraping jobs, failed URLs, and related metadata. MySQL is a relational database system documented by Oracle [8]. The Laravel migrations define schema constraints, indexes, foreign keys, and unique combinations such as job title/company uniqueness.

Figure 66 explains the relationship rationale behind the schema. The database is deliberately normalized around users, reusable skills, job requirements, CV analysis evidence, and job-mining operations. This

keeps matching logic explainable: the system can compare a user's normalized skills against job-required skills while preserving the CV and scraping evidence that produced those records.

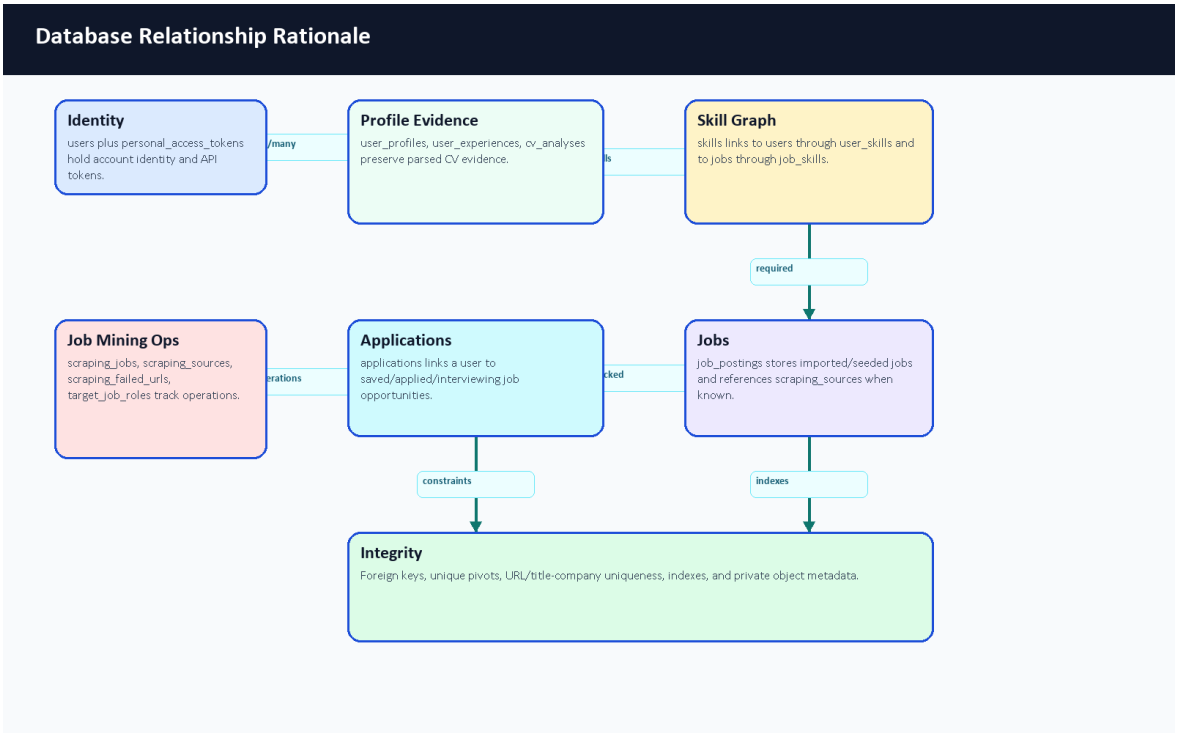


Figure 66. Database relationship rationale.

Area	Tables	Design Reason
User identity and profile	users, user_profiles, personal_access_tokens	Separates authentication/token data from CV-derived profile data.
User skills and experience	skills, user_skills, user_experiences	Supports normalized skill matching and structured career evidence.
CV analysis	cv_analyses	Stores parsing status, extracted metadata, model evidence, and private file references.
Jobs and requirements	job_postings, job_skills	Separates job records from reusable required skills.
Applications	applications	Tracks saved/applied opportunities for each user and job.
Job mining operations	scraping_jobs, scraping_sources, scraping_failed_urls, target_job_roles	Preserves operational scraping state, source configuration, target roles, and failure evidence.
Runtime infrastructure	jobs, job_batches, cache, sessions	Supports queues, batches, cache, sessions, and repeatable local demonstration behavior.

Table 7. Database design rationale.

Data Integrity Mechanism	Purpose	Code / Schema Evidence
Foreign keys	Keep user, job, application, skill, scraping, and source records consistent.	Migration foreignId(...)->constrained() and nullOnDelete() definitions.
Unique identity fields	Prevent duplicate core identities.	users.email, skills.name, user_profiles.user_id, cv_analyses.user_id.

Data Integrity Mechanism	Purpose	Code / Schema Evidence
Pivot uniqueness	Prevent repeated skill links while allowing many-to-many matching.	Unique pairs on user_skills and job_skills.
Job duplicate constraints	Reduce duplicate imported jobs at the database layer.	Unique url and unique title/company combination in job_postings.
Application uniqueness	Prevent a user from tracking the same job repeatedly.	Unique user_id/job_id pair in applications.
Operational indexes	Make status, source, and retry queries practical for dashboards.	Indexes on scraping status/type/source/created and failed-URL retry fields.
Private object references	Avoid storing CV binary files directly in MySQL.	cv_analyses storage disk/path/checksum/size/MIME metadata.
Form request validation	Reject malformed payloads before persistence.	Laravel app/Http/Requests classes.

Table 8. Data integrity mechanisms.

3.8 ERD

Figure 8 summarizes the main database tables and relationships. It is not a complete replacement for migrations, but it provides a readable graduation-book view of the data model. The final diagram was reviewed against migrations and updated to show the actual user_profiles, user_experiences, user_skills, and job_skills relationships.

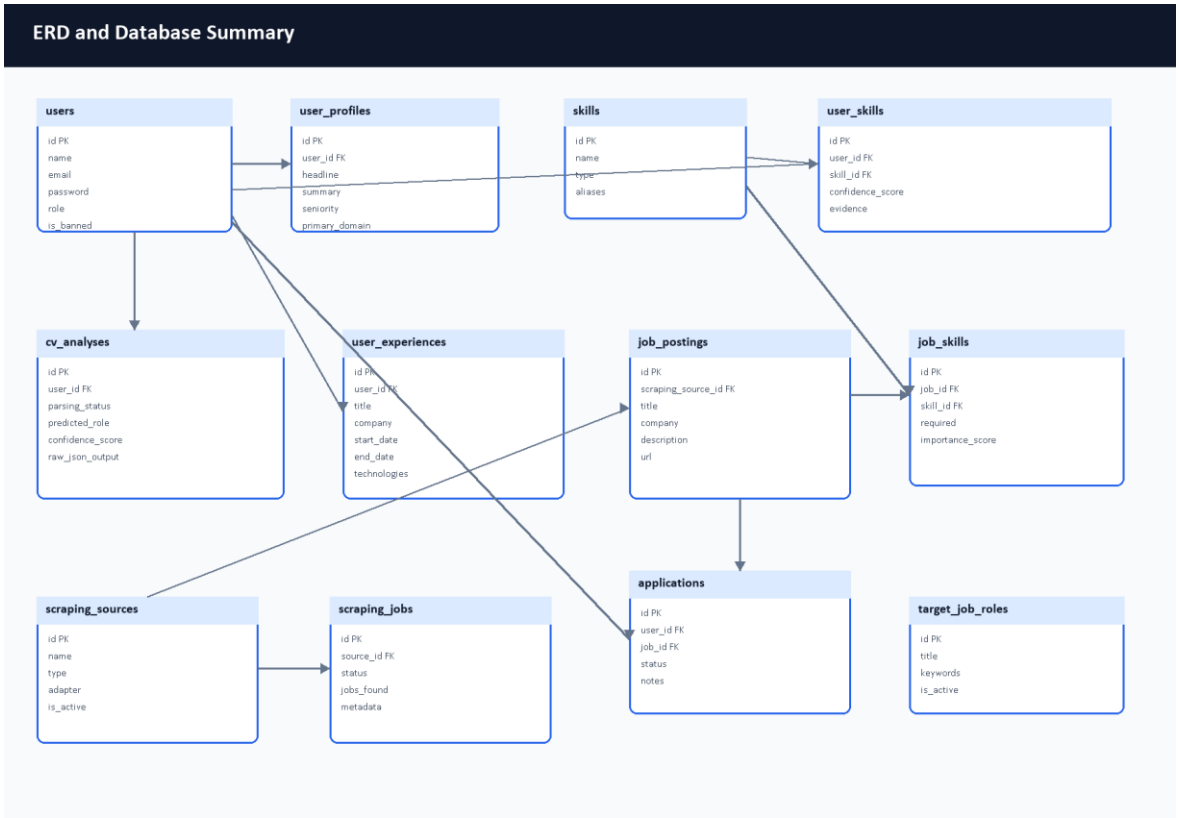


Figure 8. ERD and database summary diagram.

Relationship	Meaning

Relationship	Meaning
users -> user_profiles	One user has one profile; profile data is kept separate from authentication data.
users -> user_experiences	One user can have many structured experience records.
users <-> skills	Many-to-many relationship through user_skills, including confidence/evidence metadata.
users -> cv_analyses	CV analysis records preserve parsing status, extracted evidence, storage metadata, and model output.
job_postings <-> skills	Many-to-many relationship through job_skills, including required/importance metadata.
users <-> job_postings	Users track opportunities through applications; the application row stores user-specific status and notes.
job_postings -> scraping_sources	Imported jobs can reference their source configuration while remaining available if the source is later disabled.
scraping_jobs -> scraping_failed_urls	Failed URLs can be associated with an operational scraping run for diagnostics.
target_job_roles -> scraping workflows	Active target roles guide full scraping runs; they are operational configuration, not a direct job foreign key.

Table 9. Main ERD relationship notes.

3.9 Data Flow Diagrams

The context-level data flow is shown in Figure 3, and the expanded process-level view is shown in Figure 4. Student and administrator workflows enter the same system boundary, while external job sources and AI services interact with controlled backend processes.

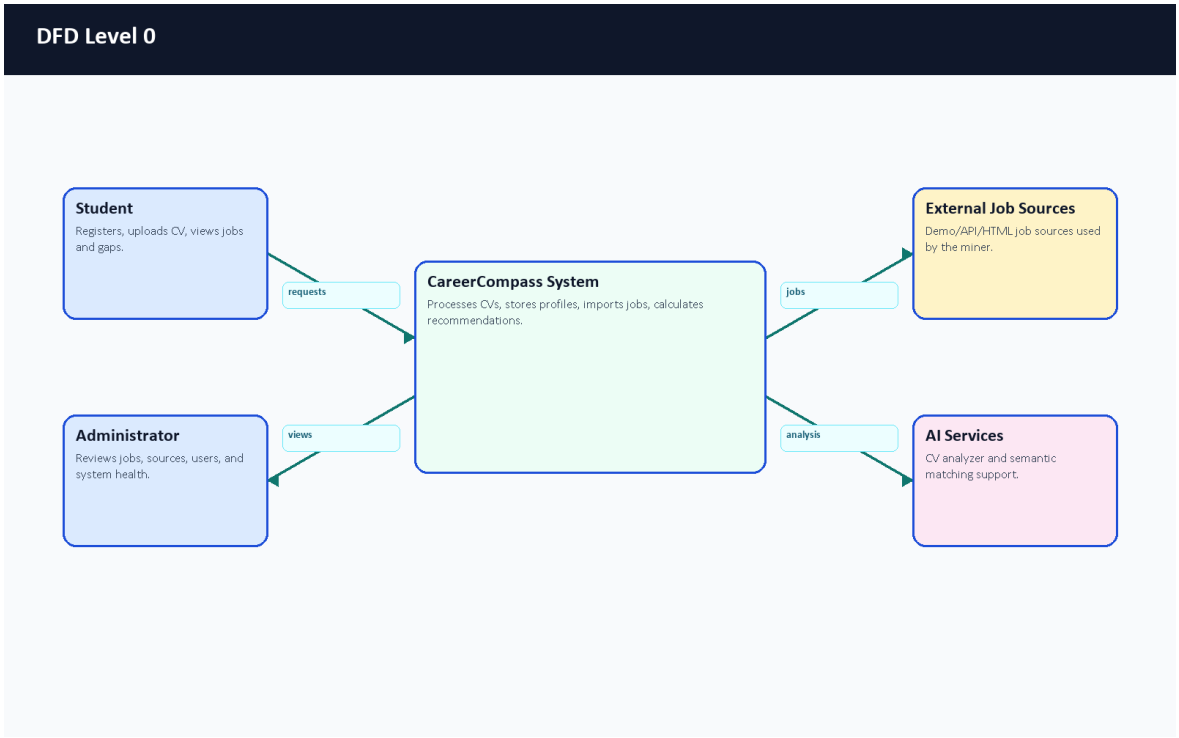


Figure 3. DFD Level 0 context diagram.

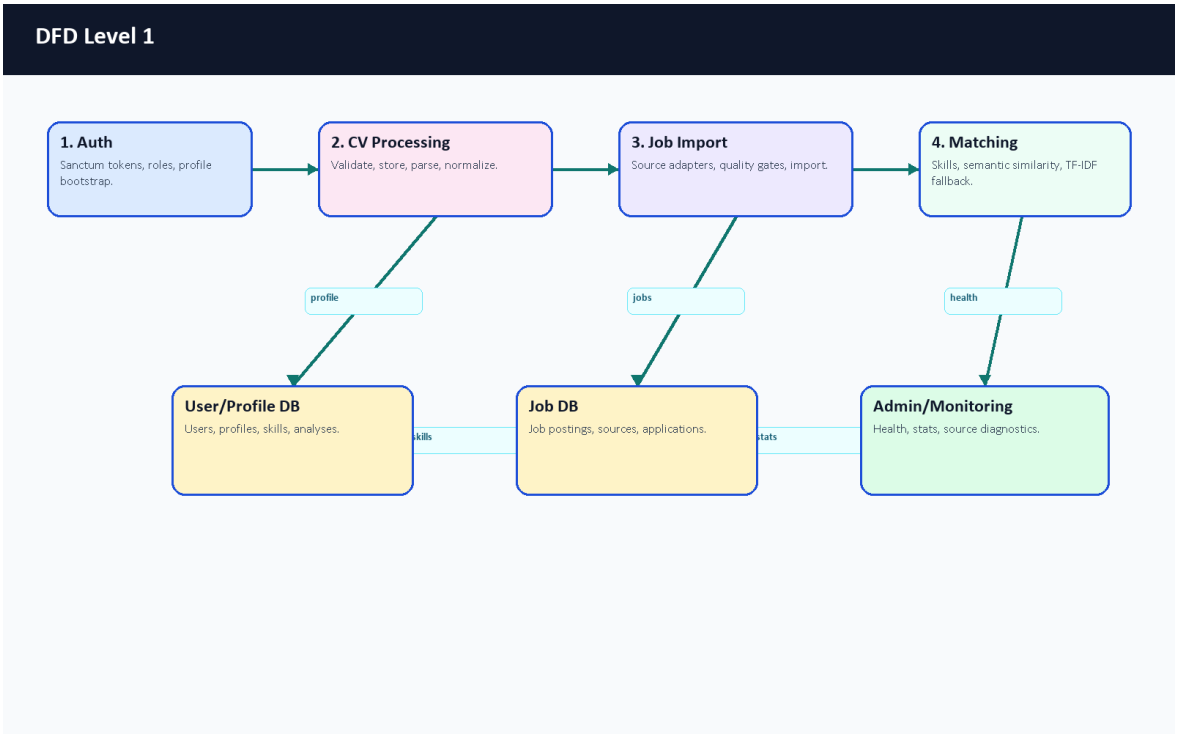


Figure 4. DFD Level 1 process diagram.

3.10 UML Use Case Diagram

The use case diagram separates student actions from administrator actions. Student workflows focus on career exploration. Admin workflows focus on operating and inspecting the imported job ecosystem.

3.11 UML Sequence Diagrams

Figure 6 shows the CV upload and analysis sequence. Figure 7 shows recommendation and gap analysis.

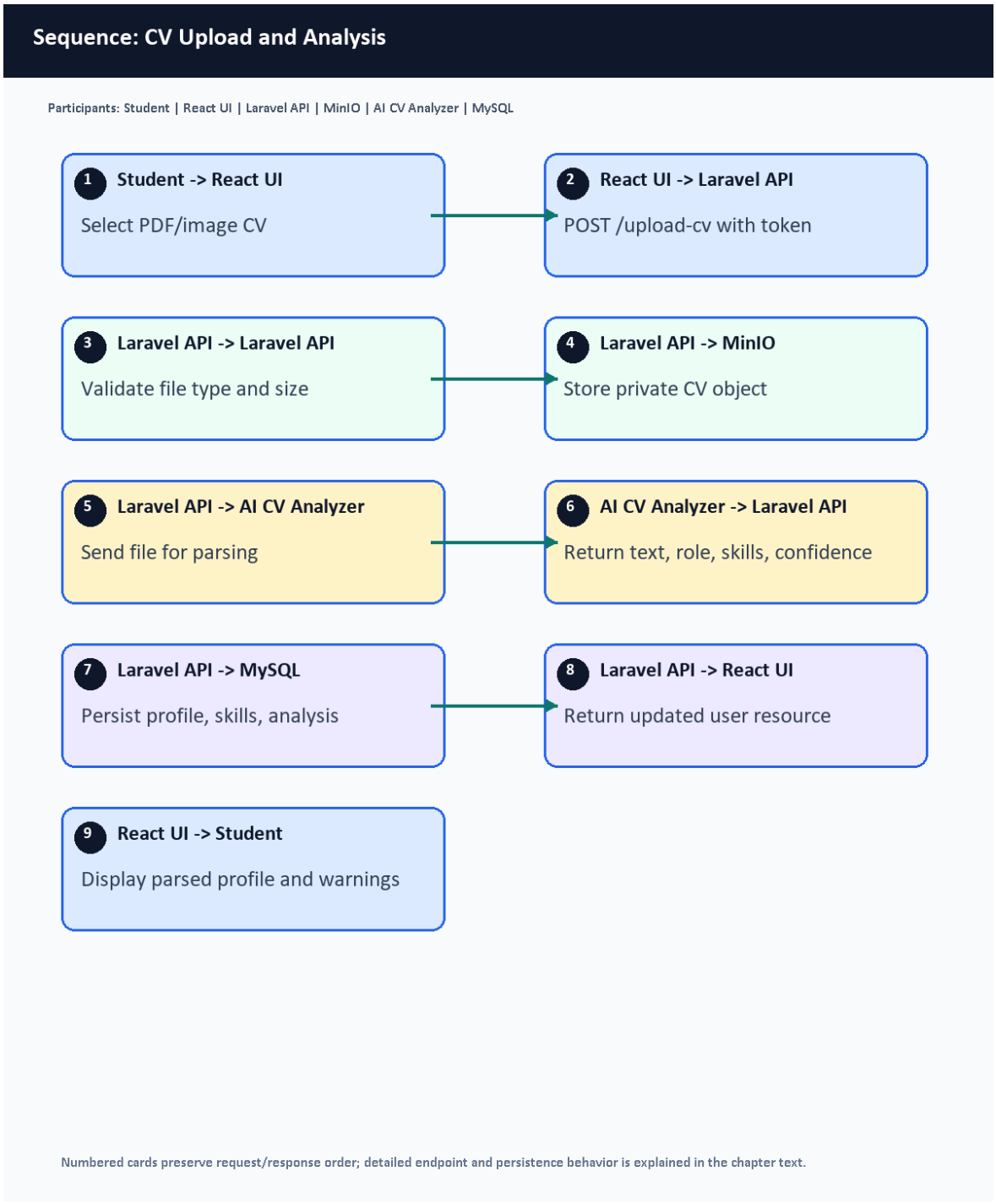


Figure 6. Sequence diagram for CV upload and analysis.

Figure 7. Sequence diagram for recommendation and gap analysis.

3.12 Deployment Architecture with Docker

The deployment is defined by Docker Compose files. Nginx exposes the application, the frontend serves built React assets, the Laravel API and workers handle backend work, MySQL stores structured data, MinIO stores private CV objects, Python services provide AI CV parsing and job mining, and Prometheus/Grafana provide monitoring. Figure 2 summarizes the container layout.

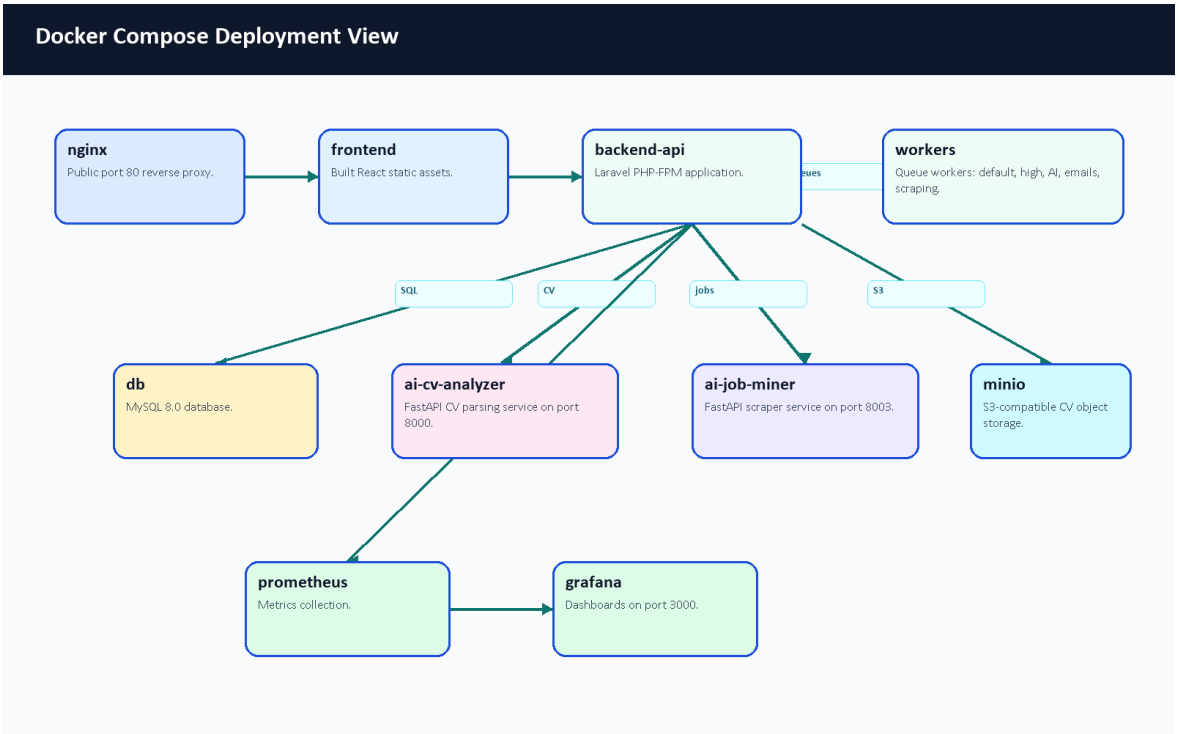


Figure 2. Docker deployment architecture.

3.13 Monitoring Architecture

CareerCompass includes live, readiness, and metrics endpoints. Prometheus is used for scraping and time-series metrics collection [13], while Grafana visualizes metrics and dashboard panels [14]. The admin dashboard also exposes application-level health information for the graduation demo.

3.14 Design Decisions and Justification

Decision	Justification
Use Laravel for the main API.	The project benefits from built-in routing, validation, Eloquent ORM, queues, resources, tests, and middleware [1].
Use React for UI.	Component-based pages support student/admin route separation and reusable cards [4].
Use FastAPI for AI services.	Python AI/NLP dependencies are easier to isolate behind typed HTTP services [6].
Use Docker Compose.	Multiple services can be started consistently for a graduation defense [11].
Use private object storage for CV files.	CV files are sensitive; private storage and signed downloads reduce accidental exposure [9], [27].
Keep AI wording honest.	Match scores and CV parsing are estimates; the demo should avoid claiming certain outcomes.

Table 10. Design decisions summary.

Chapter 4: Software and Tools Used

4.1 Laravel

Laravel is used for the backend API, controllers, middleware, validation requests, Eloquent models, migrations, resources, queues, tests, and storage integration [1]. In CareerCompass, Laravel centralizes authenticated business logic and data persistence.

4.2 React

React is used to build the browser-based interface for students and administrators [4]. The project uses React routes and components for dashboard cards, forms, pages, tables, modals, and visualization panels.

4.3 Vite

Vite is used as the frontend build tool [5]. The successful production build transformed 2904 modules during validation.

4.4 FastAPI

FastAPI is used for Python services because it supports efficient API development using Python type hints and modern web service patterns [6]. CareerCompass uses it for CV analysis and job mining service endpoints.

4.5 Python

Python is used for AI and scraping-related services [7]. It supports the ecosystem of text extraction, OCR, NLP, testing, and scraping libraries used by the AI services.

4.6 MySQL

MySQL stores relational project data including users, profiles, skills, jobs, applications, scraping sources, and CV analyses [8].

4.7 MinIO / S3-Compatible Storage

MinIO-compatible object storage is used to store CV files privately. This avoids placing uploaded CVs directly in public web directories and supports signed temporary download flows [9].

4.8 Docker and Docker Compose

Docker containers package each runtime, and Docker Compose coordinates multi-container execution [10], [11]. CareerCompass uses Compose for Nginx, frontend, backend, workers, MySQL, MinIO, AI services, Prometheus, and Grafana.

4.9 Nginx

Nginx acts as the reverse proxy and public gateway for the local deployment [12]. It routes browser and API traffic to the correct containers.

4.10 Prometheus and Grafana

Prometheus collects metrics [13], and Grafana visualizes metrics and dashboards [14]. CareerCompass uses these tools to support monitoring-oriented evaluation.

4.11 GitHub Actions

GitHub Actions is configured for repository automation and CI/CD-style validation [15]. The report treats workflow definitions as part of the development process, while final PR status should be reviewed after the draft PR is opened.

4.12 NLP / AI Libraries

The project uses concepts and libraries related to text extraction, OCR, transformer token classification, TF-IDF, cosine similarity, and sentence embeddings. TF-IDF and cosine similarity are documented by scikit-learn [19], [20]. Sentence Transformers provides sentence embedding models and utilities [21]. PyMuPDF, pdfplumber, and EasyOCR support PDF/image text extraction and OCR-style workflows [22], [23], [24]. The token-classification path builds on BERT-style contextual representations [37].

Hugging Face Transformers is used for model loading and token-classification style inference/training [31], [32]. The training notebook uses Hugging Face Trainer concepts, token-label alignment, the seqeval metric family, and a BERT token-classification configuration [33]. Optional dynamic quantization is supported in the local model-loading code as an inference optimization concept [38]. The deployed matching layer also uses sentence embeddings and a custom TF-IDF fallback so that a service outage or weak semantic signal does not become a silent failure.

4.13 Synthetic Data and Training Tools

The repository includes a model-training workflow designed around synthetic annotated CV snippets. The data-generation script uses the Gemini API through Google AI developer tooling to generate labeled examples, while the training notebook is designed for Google Colab GPU execution [34], [35], [36]. This report treats the external AI tooling as training-support tooling, not as a runtime dependency for private CV uploads.

4.14 Testing Tools

Backend tests use Laravel/PHP testing tools and PHPUnit concepts [25]. Python service tests use pytest where available [26]. Frontend validation uses ESLint and Vite build checks.

4.15 Development and Version Control Tools

Git and GitHub are used for version control and pull-request-based collaboration. Docker Desktop provides the local container runtime. Browser screenshots were captured through a Chrome DevTools Protocol helper to provide evidence images for this report.

Chapter 5: System Implementation

5.1 Introduction

This chapter documents the implemented CareerCompass modules based on repository files. The implementation is not a generic career platform; it is a Laravel/React/FastAPI/Docker system with specific routes, services, pages, models, seeders, and tests.

5.2 Authentication and User Management

Authentication is implemented in backend-api/app/Http/Controllers/Api/AuthController.php. Registration creates a user with role user, loads profile and related resources, and returns a token. Login validates credentials, checks the banned state, revokes old tokens, creates a new token, and returns a user resource. The frontend stores the token as auth_token and the user object in local storage through frontend/src/context/AuthContext.jsx.

The register request restricts emails to selected public email domains and validates password format. The admin role is protected by the IsAdmin middleware. Admin seed data creates a demo-only administrator account through AdminUserSeeder.

5.3 Student Dashboard

The student dashboard is implemented in frontend/src/pages/user/Dashboard.jsx. It presents the current profile state, CV upload/update controls, profile completeness, career identity, AI insights, and next actions. Before CV upload, it prompts the user to add a CV. After upload, it displays parsed CV availability, role inference, profile score, experience, and action buttons.

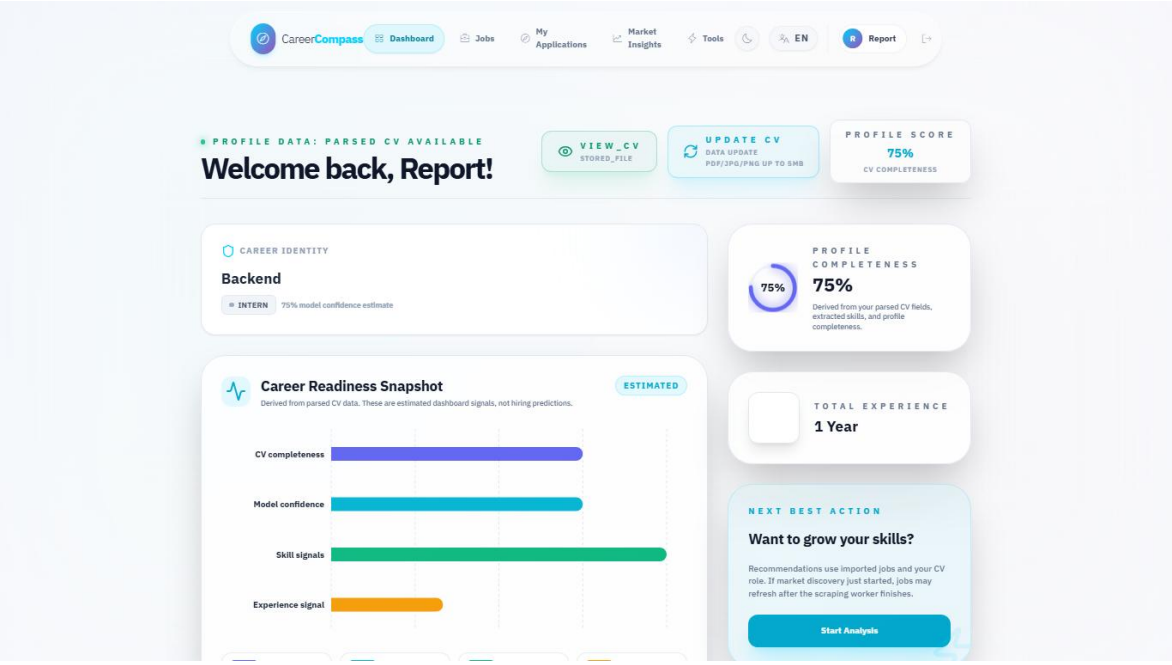


Figure 34. Student dashboard before CV upload.

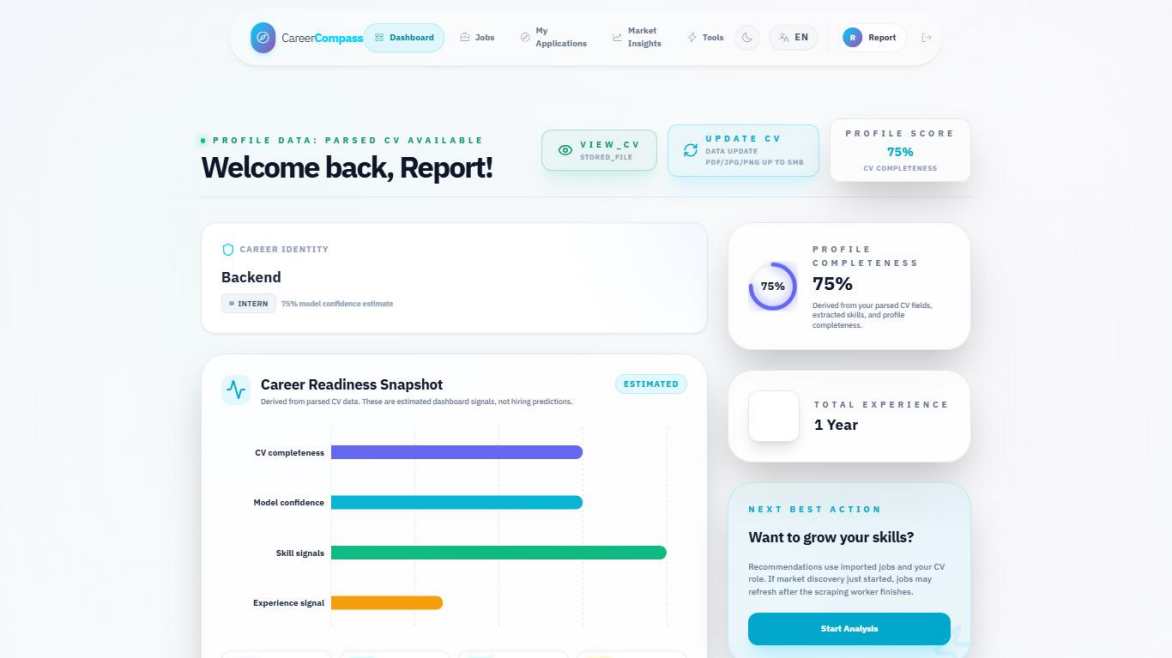


Figure 36. Dashboard after successful CV parsing.

5.4 CV Upload and Storage

CvUploadRequest requires a cv file and accepts PDF, JPEG, JPG, and PNG files up to 5 MB. The frontend appends the selected file as cv in a FormData object. CvController calls the CV processing service, persists the file path and metadata, and returns a unified user resource.

CV storage is handled as a private file workflow. The system supports signed download URLs, which is a better demo posture than public file exposure. OWASP recommends validating uploaded file type, extension, size, and storage handling carefully [27].

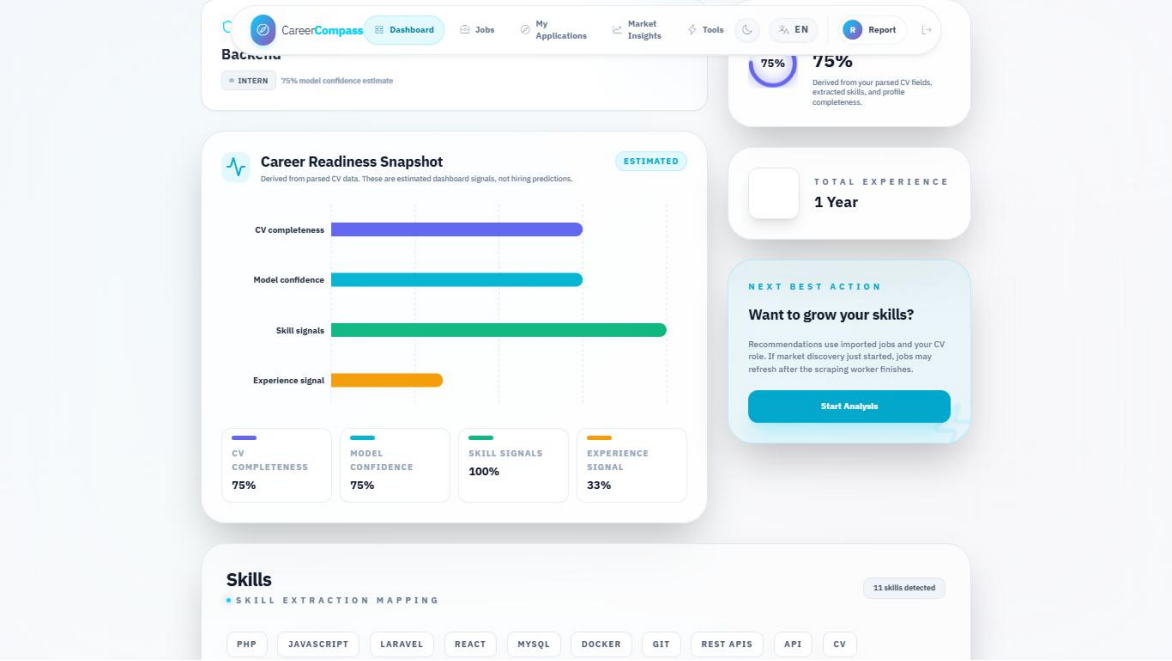


Figure 35. CV upload user interface.

5.5 CV Parsing and Skill Extraction

The CV processing flow sends the file to the AI CV Analyzer, receives parsed data, synchronizes skills, updates profile fields, and stores CV analysis metadata. The implementation handles multiple parsing statuses honestly. If analysis times out, fails, or finds no readable text, the backend returns warnings and preserves existing profile details rather than silently replacing data with low-quality output.

5.5.1 Analyzer Runtime Components

The analyzer is implemented as layered Python code rather than one monolithic function. main.py exposes FastAPI endpoints, CVOrchestrator coordinates extraction, AdvancedNEREngine runs transformer-based named-entity recognition, and supporting engines handle contacts, sections, experience blocks, canonicalization, domain classification, seniority classification, semantic embeddings, and hybrid job matching. Figure 11 summarizes these extraction components.

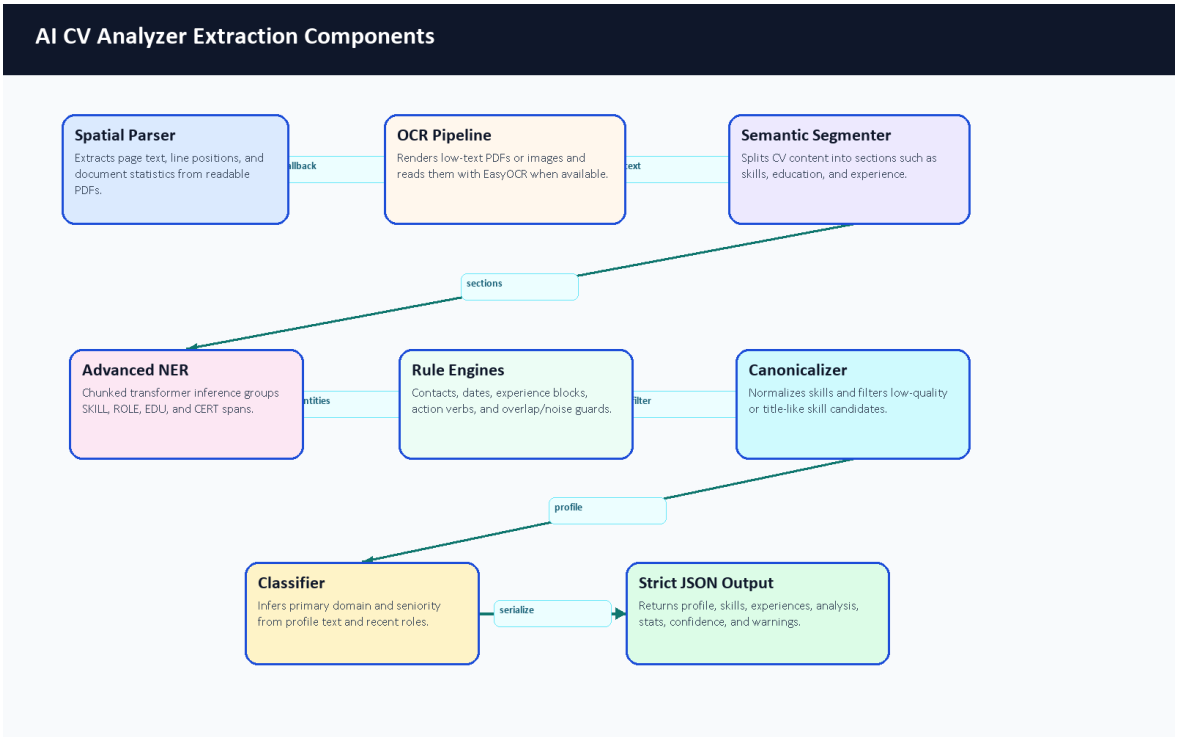


Figure 11. AI CV Analyzer extraction components.

Component	Repository Evidence	Responsibility	Output Used By
FastAPI gateway	ai-cv-analyzer/main.py	Receives /api/parse-cv and /api/hybrid-match requests; handles timeout/error fallbacks.	Laravel CvProcessingService and GapAnalysisService
Laravel CV service	backend-api/app/Services/CvProcessingService.php	Sends uploads to the analyzer, stores private CV objects, persists normalized analysis.	User profile, skills, experiences, recommendations
Spatial/OCR extraction	spatial_parser.py, ocr_pipeline.py	Reads PDF text first and falls back to image/OCR when needed.	Section segmenter and NER pipeline

Component	Repository Evidence	Responsibility	Output Used By
Advanced NER	advanced_ner.py and optional ignored local model folder	Loads the exported local token-classification model when deployed; chunks long CVs and groups entity spans.	Skills, roles, education, certifications
Rule engines	contact, experience, date, noise-filtering helpers	Extract contact details, experience blocks, dates, and remove title-like or noisy skill candidates.	Profile and experience persistence
Canonicalization/classification	Layer 1 and Layer 2 modules	Normalize skills and infer primary domain plus seniority.	Dashboard identity card and matching
Hybrid matching	Layer 3 matching modules	Combines semantic scores, skill text similarity, domain alignment, constraints, and TF-IDF fallback.	Recommendations and gap analysis
Frontend display	Dashboard.jsx, AiInsights.jsx	Shows upload status, confidence-style signals, role/seniority, and extracted skills.	Student-facing CV feedback

Table 11. AI CV Analyzer components.

5.5.2 Model Type and Customization

The NER part is a fine-tuned transformer token-classification design, not a model trained from randomly initialized architecture. The notebook uses bert-base-cased as the base checkpoint and defines a CV-specific BIO label set. The runtime checks for the exported local model at ai-cv-analyzer/models/ner_weights/career_compass_ner_final and uses it when available. That folder is ignored by Git, so committed repository evidence should be described as code and training workflow evidence. Local ignored metadata inspected during this documentation update identified a BERT token-classification configuration, 512-token maximum position setting, 28,996-token vocabulary, 12 transformer layers, 768 hidden size, and a cased tokenizer; the binary model weights were not copied into the report.

The project customization is mainly in the data, labels, orchestration, and post-processing. The team defined CV-specific labels, generated synthetic annotated examples, cleaned entity spans, aligned character spans to tokens, exported a local model, and connected it to a CV-specific extraction pipeline. The pipeline then adds deterministic rules for contacts, dates, experience blocks, noisy skill rejection, canonical skill names, domain/seniority inference, and safer fallback statuses.

Label	BIO Forms	Meaning in Training Data	Runtime Note
O	O	Token outside a labeled entity.	Used by the model to ignore ordinary text.
SKILL	B-SKILL, I-SKILL	Technical skill such as Laravel, React, Docker, SQL, or PyTorch.	Returned as skills after filtering and canonicalization.
ROLE	B-ROLE, I-ROLE	Job title or role such as Backend Developer or Data Analyst.	Used for predicted role and role evidence.

Label	BIO Forms	Meaning in Training Data	Runtime Note
EDU	B-EDU, I-EDU	Degree, major, faculty, university, or education phrase.	Used as education/profile evidence.
CERT	B-CERT, I-CERT	Certification such as AWS Cloud Practitioner.	Used as certification evidence.
SOFT	B-SOFT, I-SOFT	Soft skill phrase such as leadership or communication.	Present in training configuration; the runtime NER grouping mainly returns SKILL, ROLE, EDU, and CERT.

Table 12. NER entity label schema.

5.5.3 Synthetic Dataset Generation

The helper material under D:/Graduation/model-analys-helper was reviewed as documentation support. The top-level helper folders found were docs, layer1, layer2, and layer3. No raw datasets, screenshots, or secrets were copied into the graduation-book folder. The important training workflow is already represented in the repository by ai-cv-analyzer/training/generate_tech_dataset.py, clean_dataset.py, and train_ner.ipynb.

The dataset generator is designed to call the Gemini API through API keys stored outside source control. It asks for synthetic technical CV snippets across backend, frontend, DevOps, mobile, AI/data, cybersecurity, cloud, QA, and networking contexts. It intentionally includes positive labeled examples and negative decoy examples so that the model learns both what to tag and what to ignore. Because this uses external generation and keys, it was inspected rather than executed for this documentation update.

Step	Repository Evidence	Description	Documentation Decision
API-key loading	generate_tech_dataset.py	Reads GEMINI_API_KEYS from .env and rotates keys/models during generation.	Do not commit keys; no secrets were found in helper docs.
Synthetic sample generation	Generator system prompt	Requests batches with technical domains, noise, varied CV formats, and labeled entities.	Documented as synthetic data, not real student CV data.
Negative decoys	Generator distribution comments	Includes examples with no entities to reduce false positives.	Preserved in the description because it affects model behavior.
Cleaning	clean_dataset.py	Normalizes whitespace, deduplicates exact text, validates entity text, filters long skill spans.	Documented as a data-quality gate before training.
Output	Training docs and scripts	Expected cleaned JSON/JSONL input for notebook training.	Dataset files were not present in the repository, so metrics cannot be reproduced from repo files alone.

Table 13. Synthetic dataset generation workflow.

5.5.4 Training Notebook Workflow

The training notebook is structured for Google Colab rather than local execution. It installs model-training dependencies, loads cleaned JSON data, defines labels, tokenizes examples, aligns entity spans to token

labels, initializes AutoModelForTokenClassification from bert-base-cased, trains with Hugging Face Trainer, evaluates with sequence-labeling metrics, then exports career_compass_ner_final for deployment [31], [32], [33], [36].

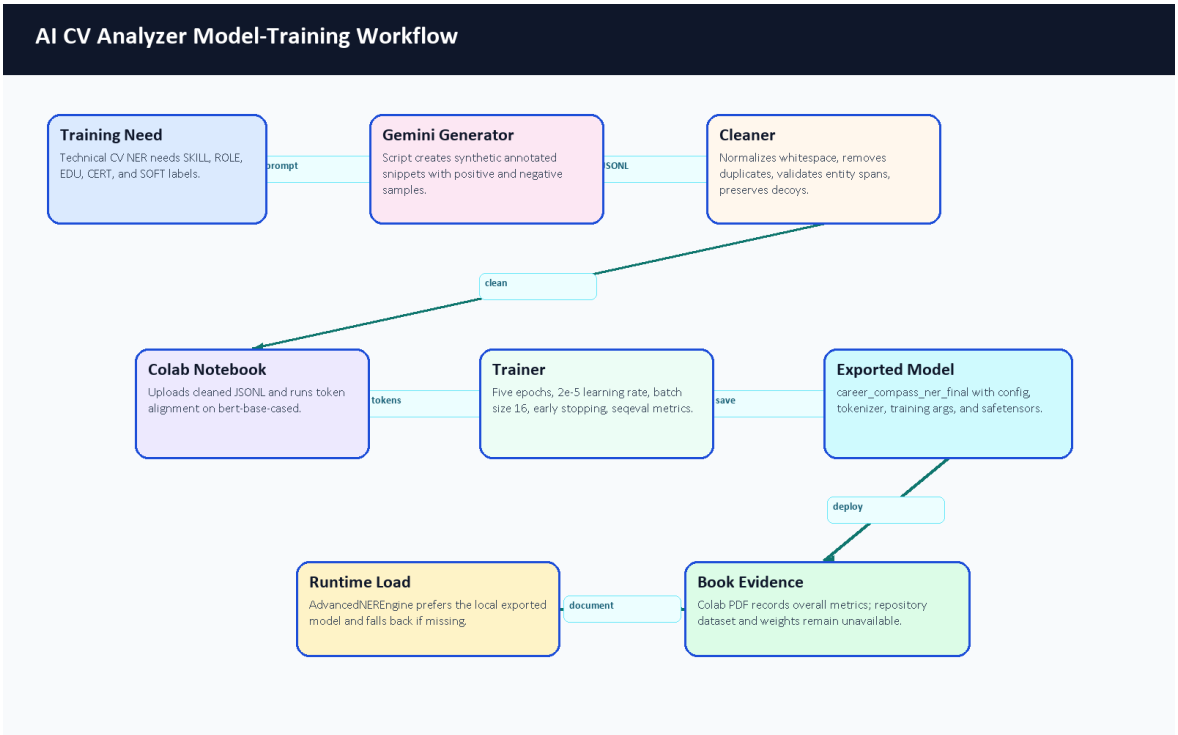


Figure 10. AI CV Analyzer model-training workflow.

Setting	Value Found in Notebook or Docs	Purpose	Evidence Limitation
Base checkpoint	bert-base-cased	Provides pretrained language representations for token classification.	Confirmed by training notebook and Colab PDF.
Labels	O plus B/I for SKILL, ROLE, EDU, CERT, SOFT	Encodes CV entity spans using BIO tagging.	Label map is reproducible from notebook/config.
Split	90 percent train, 10 percent test, seed 42	Creates a repeatable train/evaluation split.	Colab PDF records 41,319 train rows and 4,592 test rows; dataset content is not committed.
Max length	512 tokens	Fits BERT token-classification input limits.	Long runtime CVs are handled through chunking separately.
Epochs and learning rate	5 epochs, 2e-5	Standard fine-tuning style schedule for a small NER task.	No final epoch table was available.
Batch size	16 train/eval	Balances GPU memory and throughput on Colab/T4-style runtime.	Visible in the exported Colab PDF.
Metrics	precision, recall, F1, accuracy via sequence labeling	Evaluates entity extraction quality when labels are available.	Colab PDF records overall metrics; per-label report is not visible.
Export	career_compass_ner_final zip/model folder	Produces the deployable local model artifact.	Export success is visible in the PDF, but model weights are ignored by Git.

Table 14. Model training configuration.

5.5.5 Layer 1: CV Understanding Pipeline

Layer 1 is responsible for turning a noisy CV file into a structured candidate profile. The runtime begins at main.py, which accepts the uploaded file, chooses PDF or image handling, applies timeout/error wrappers, and delegates the actual extraction to CVOrchestrator. The orchestrator first tries ordered PDF text extraction. The spatial parser reads words from PDF pages, groups words into rows using an adaptive tolerance, splits row segments when large x-axis gaps imply columns, removes (cid:...) artifacts, and falls back to plain PDF extraction when the spatial output loses too much text.

If the file has little or no readable text, the OCR path renders PDF pages to images and uses EasyOCR after grayscale/blur preprocessing. After text recovery, the semantic segmenter finds CV sections, contact extraction parses email/phone/location fields, the NER engine extracts entity candidates, experience logic estimates date ranges and career signals, and the canonicalizer normalizes skills before the result is validated through strict Pydantic schema classes.

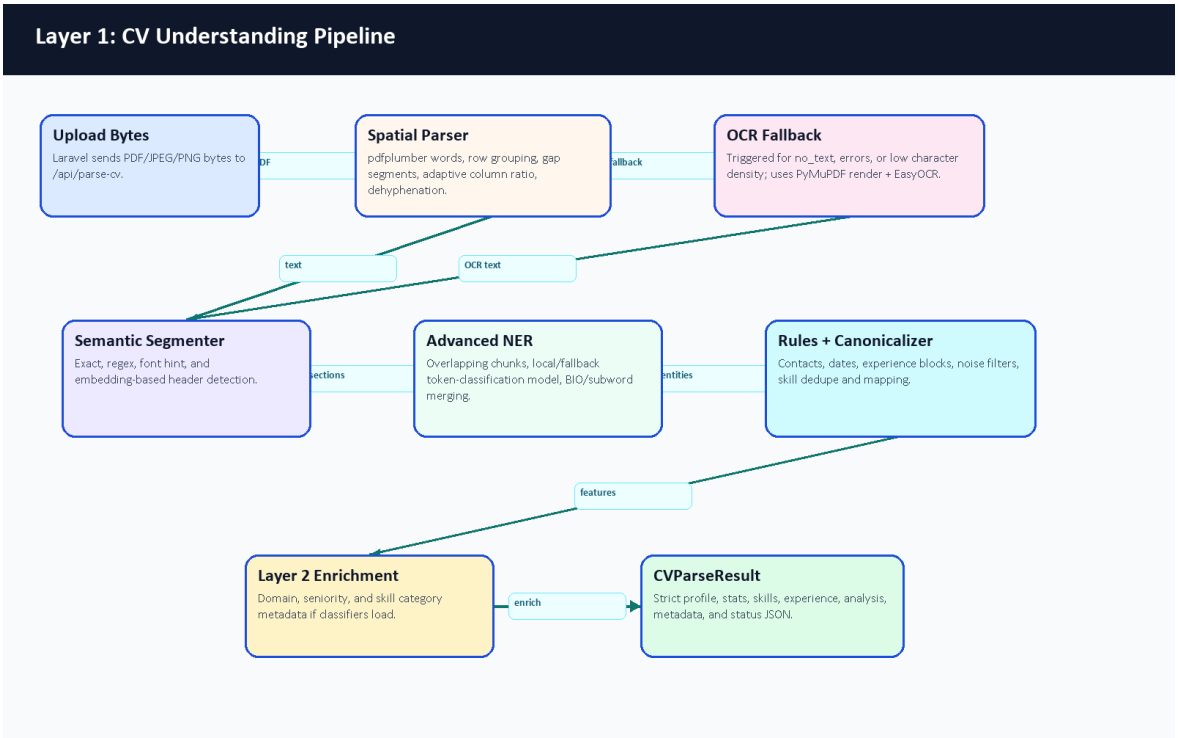


Figure 12. Layer 1 CV understanding pipeline.

Layer 1 Component	Main Files	Important Behavior	Risk or Fallback
API gateway	main.py	/api/parse-cv, /api/hybrid-match, timeout handling, health and metrics endpoints.	Timeout results are returned as explicit status dictionaries.
Spatial parser	core/layer1_understanding/spatial_parser.py	Word extraction, row grouping, column ordering, dehyphenation, plain-text fallback.	Falls back when spatial output is too weak.
OCR fallback	core/layer1_understanding/ocr_pipeline.py, orchestrator OCR helpers	Renders image-like PDFs, preprocesses pages, and extracts text when normal PDF parsing fails.	Triggered for short/no-text inputs.

Layer 1 Component	Main Files	Important Behavior	Risk or Fallback
Section segmenter	core/layer1_understanding/section_segmenter.py	Header detection from patterns and optional semantic header matching.	Missing headers fall back to profile-style grouping.
Contact and experience engines	contact_extractor.py, experience_engine.py	Extract emails/phones/location, date ranges, total years, skill durations, gaps, overlaps, and action-verb strength.	Ambiguous dates are treated conservatively.
NER and canonicalization	advanced_ner.py, canonicalizer.py	Extracts skills/roles/education/certifications, filters noise, deduplicates, and maps skills to canonical names.	Fallback model path exists when local deployment artifact is missing.
Output schema	schema.py	Strict typed response for profile, skills, experience, confidence, stats, and parsing status.	Invalid shapes are prevented before backend persistence.

Table 15. Layer 1 component details.

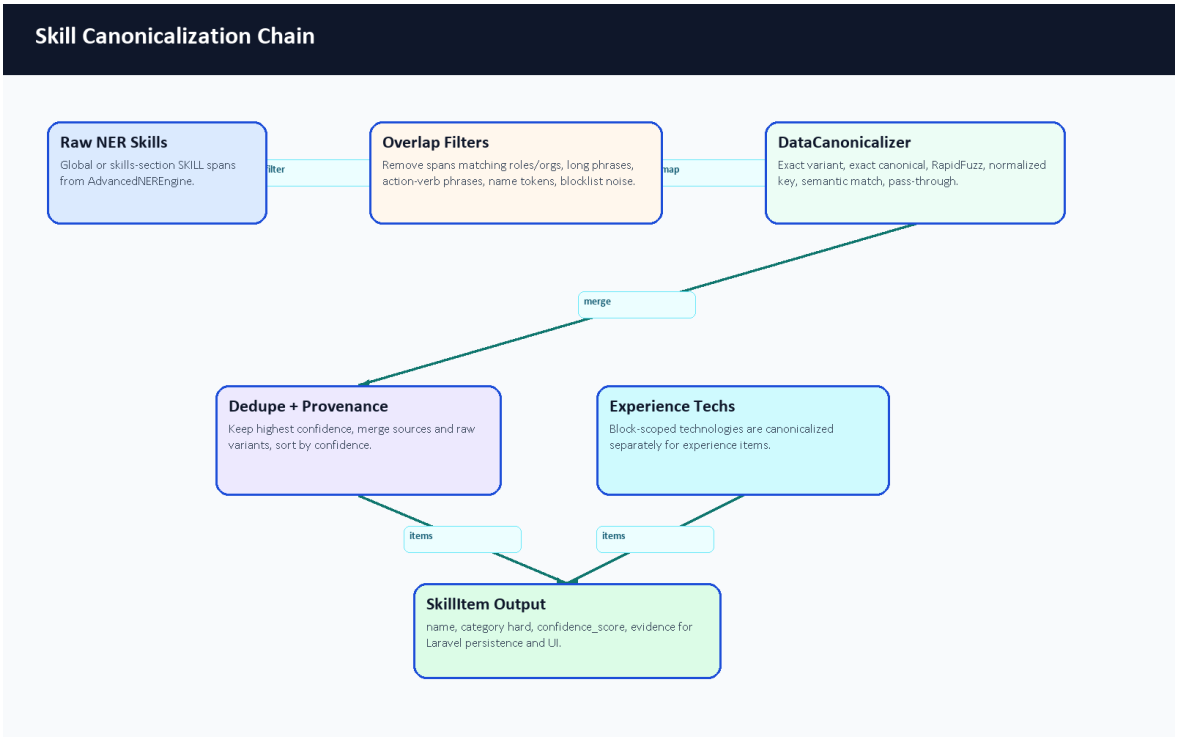


Figure 17. Skill canonicalization chain.

5.5.6 Layer 2: Classification Engine

Layer 2 enriches the extracted CV with a domain and seniority interpretation. The classification orchestrator reads the Layer 1 result, then combines title, experience, summary, skill categories, and taxonomy descriptions. DomainEngine compares CV context against taxonomy descriptions using semantic embeddings when available. SkillEngine separates hard, soft, and management-oriented skills using taxonomy rules. SeniorityEngine combines years of experience, title keywords, semantic title/summary hints, and action-verb strength.

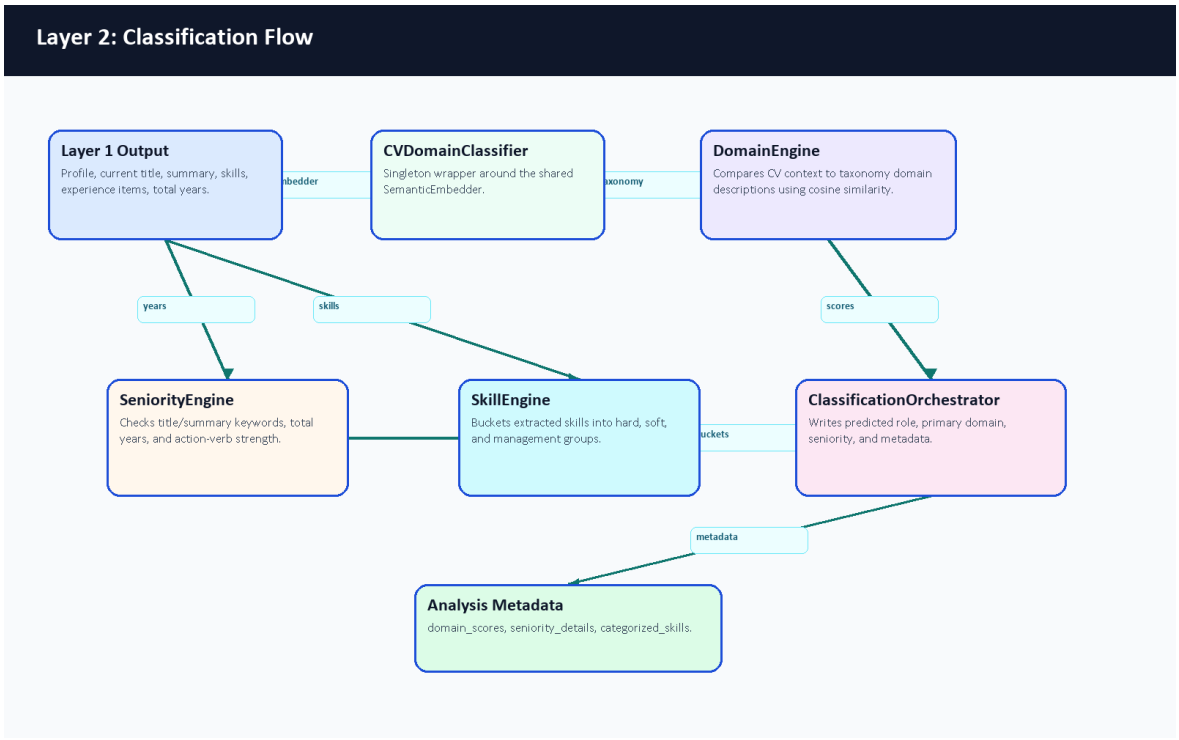


Figure 13. Layer 2 classification flow.

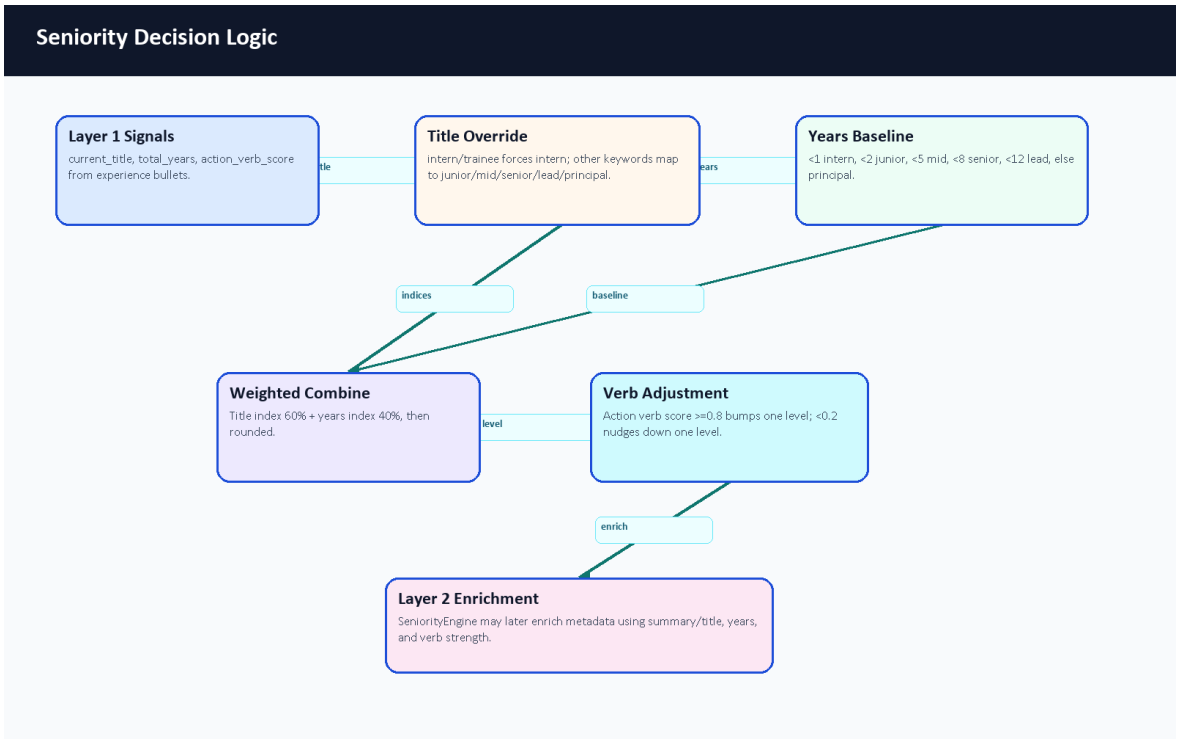


Figure 16. Seniority decision logic.

Layer 2 Component	Main Files	Input	Output
Classification orchestrator	core/layer2_classification/orchestrator.py	Parsed CV profile, skills, experience, and summary.	Adds primary domain, seniority level, skill categories, and confidence-style signals.
Domain engine	core/layer2_classification/domain_engine.py, data/taxonomy.json	Title, summary, and first experience titles.	Primary technical domain selected from taxonomy descriptions.

Layer 2 Component	Main Files	Input	Output
Seniority engine	core/layer2_classification/seniority_engine.py	Experience years, title, summary, and action verbs.	Intern, Junior, Mid-Level, Senior, or Lead / Manager estimate.
Skill engine	core/layer2_classification/skill_engine.py	Canonical skill names and taxonomy terms.	Hard, soft, and management skill buckets.
Taxonomy loader	core/layer2_classification/utils.py	JSON taxonomy file.	Shared configuration for domain and skill classification.

Table 16. Layer 2 classification engine details.

5.5.7 Layer 3: Matching Engine

Layer 3 compares a candidate profile with a job description. JobDescriptionEngine parses job text into seniority, required years, mandatory skills, bonus skills, domain, and summary. IntelligentMatcher calculates semantic similarity, skill-text similarity, and domain alignment using adaptive weights that change by seniority level. ConstraintValidator subtracts penalties for missing mandatory skills, experience shortfalls, and seniority mismatch. FitAnalysisGenerator turns the numeric result into strengths, gaps, red flags, and a verdict.

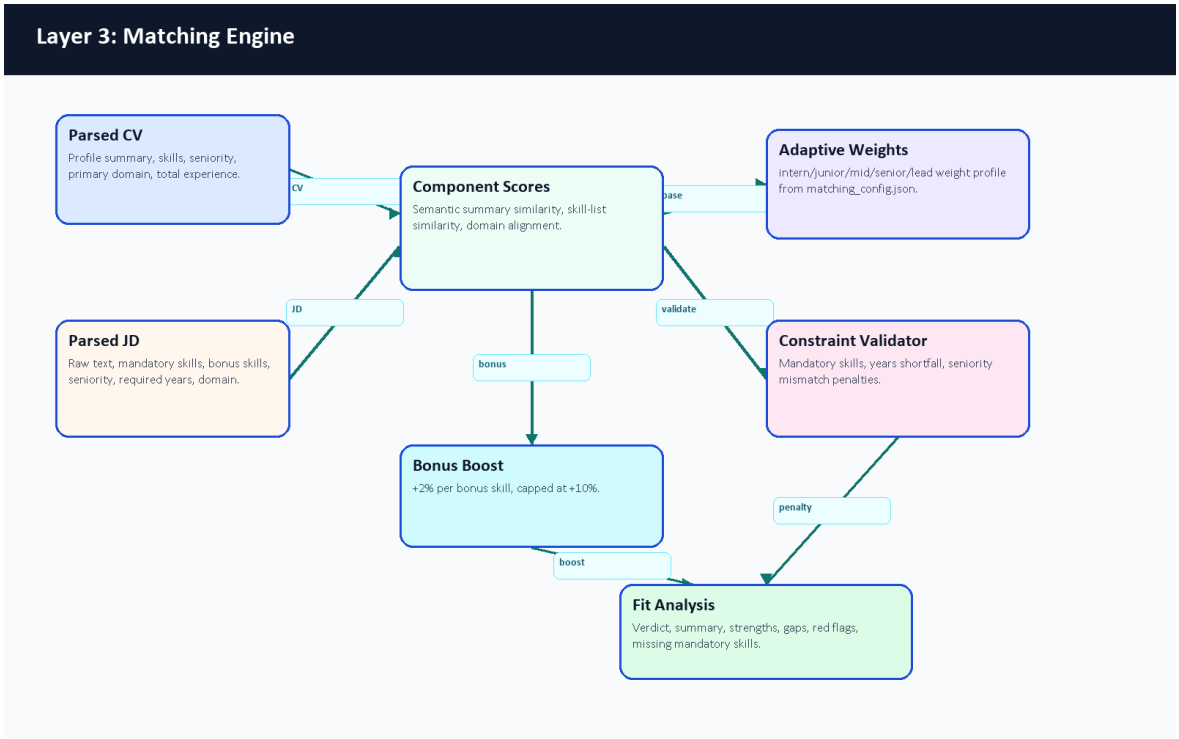


Figure 14. Layer 3 matching engine.

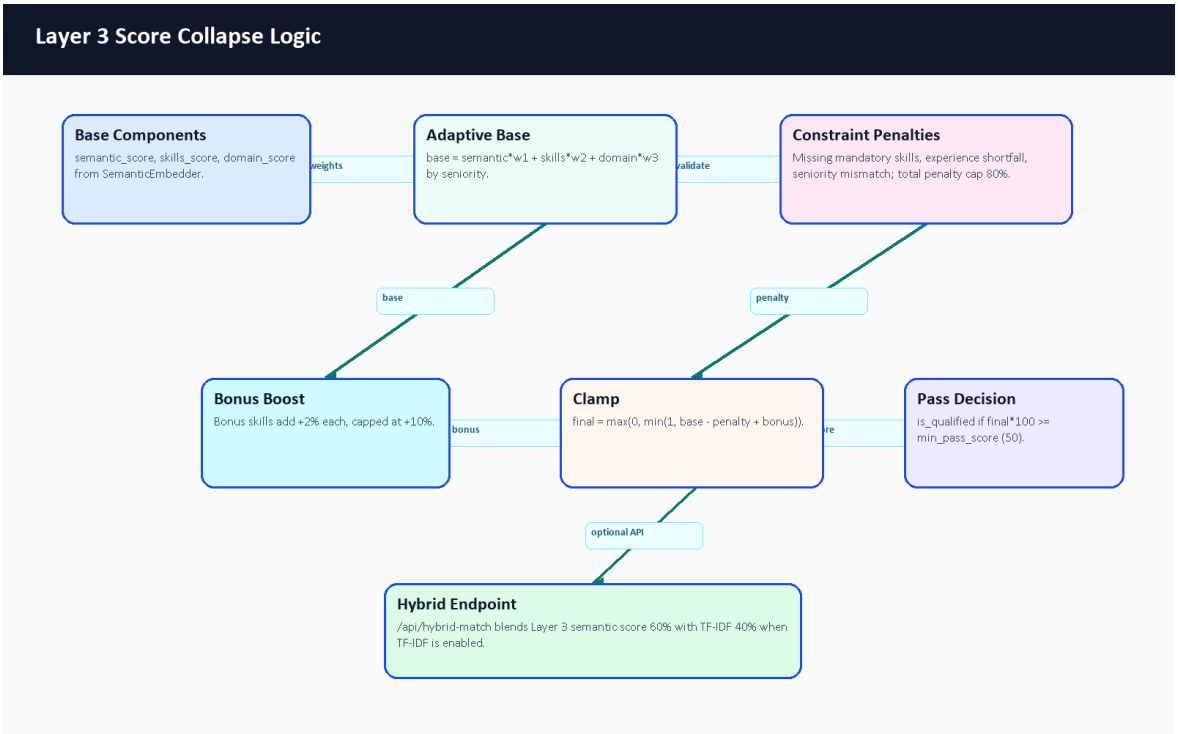


Figure 18. Layer 3 score collapse logic.

Layer 3 Component	Main Files	Scoring Contribution	Explanation Contribution
JD parser	job_description_engine.py	Extracts requirements that become matching inputs.	Explains what the system understood from the job post.
Semantic embedder	embedder.py	Summary similarity and domain-similarity fallback.	Captures meaning beyond exact keyword overlap when dependencies are available.
Intelligent matcher	similarity.py, matching_config.json	Combines semantic, skill, and domain scores using seniority-aware weights.	Produces score breakdown and qualification flag.
Constraint validator	constraint_validator.py	Applies capped penalties for mandatory gaps, experience gaps, and seniority mismatch.	Lists missing mandatory skills and mismatch reasons.
Fit analysis generator	fit_analysis_generator.py	Converts score ranges into verdict categories.	Generates strengths, gaps, and red flags for the UI.
Ranking orchestrator	ranking_orchestrator.py	Applies matcher repeatedly across candidates/jobs.	Sorts candidates or opportunities by explainable fit.

Table 17. Layer 3 matching engine details.

5.5.8 Semantic Embedding, Caching, and TF-IDF Fallback

The semantic embedder uses the configured sentence-transformer model name, with all-MiniLM-L6-v2 as the default. The class is implemented as a singleton so that model loading is not repeated for every request. It keeps a bounded embedding cache of 2,000 entries and can optionally apply dynamic quantization through EMBEDDER_QUANTIZE. If the embedding stack is unavailable, it returns zero vectors and the calling code falls back to transparent lower-confidence behavior instead of pretending semantic comparison succeeded.

The FastAPI /api/hybrid-match endpoint also combines semantic-style matching with the pure Python TF-IDF matcher when the TF-IDF module is importable. In that endpoint, semantic/adaptive scoring contributes 60 percent and TF-IDF contributes 40 percent. This gives the demo a useful exact-keyword safety check for skills such as Laravel, Docker, MySQL, React, AWS, or Kubernetes.

Method	Repository Evidence	Strength	Limitation
Sentence embeddings	core/layer3_matching/embedder.py	Captures meaning even when words differ.	Requires sentence-transformer dependencies and model loading.
Domain similarity	core/layer3_matching/similarity.py	Allows related domains to receive partial credit.	Low domain similarity is cut off below the configured threshold.
Skill text similarity	core/layer3_matching/similarity.py	Compares extracted CV skills against mandatory and bonus job skills.	Quality depends on upstream extraction and canonicalization.
TF-IDF fallback	core/layer3_matching/tfidf.py, /api/hybrid-match	Lightweight deterministic keyword overlap.	Does not understand synonyms or phrase meaning.
Constraint penalties	constraint_validator.py	Prevents high scores when hard requirements are missing.	Penalty weights need larger evaluation data for calibration.

Table 18. Semantic embedding and TF-IDF fallback comparison.

5.5.9 NER Token Processing and BIO Tagging

The training notebook uses character-span annotations and converts them into token labels. Each text sample is tokenized with offsets; special tokens are assigned -100 so they are ignored by the loss; tokens whose offsets fall inside an entity span are assigned B- or I- labels. The cased BERT tokenizer is appropriate for CV text because names, certificates, role titles, and technology names often rely on capitalization. At runtime, long CV text is chunked with overlap, model predictions are merged, subword prefixes are cleaned, and duplicate/noisy entities are filtered before canonicalization.

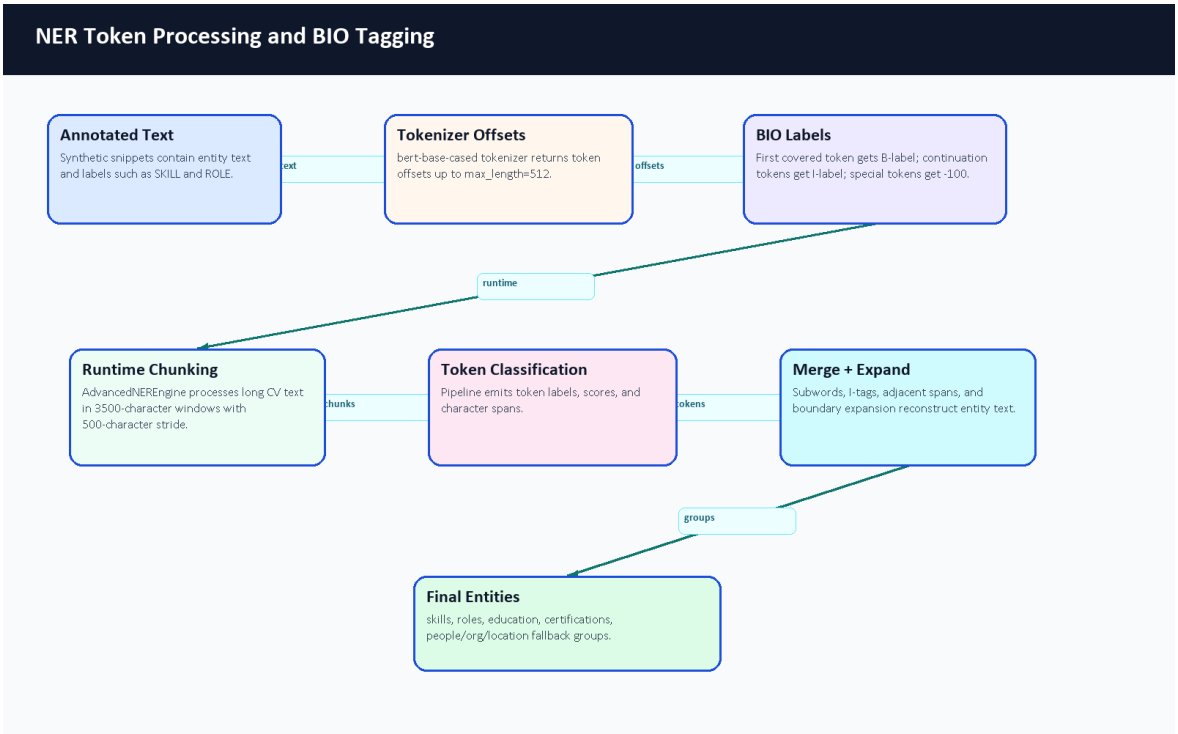


Figure 15. NER token processing and BIO tagging.

Simplified Text Token	BIO Label	Why It Matters
Experienced	O	Ordinary descriptive word, not extracted as an entity.
Backend	B-ROLE	Start of a role phrase.
Developer	I-ROLE	Continuation of the role phrase.
with	O	Connector word.
Laravel	B-SKILL	Skill entity.
Docker	B-SKILL	Skill entity.
MySQL	B-SKILL	Skill entity.

Table 19. Simplified BIO tagging example.

5.5.10 AI CV Analyzer Source Code Inventory

The AI CV Analyzer was audited as source code, not only as a running service. The inventory below summarizes the relevant repository areas. Full support notes are stored under docs/graduation-book/model-analysis/.

Area	Representative Files	Main Responsibility
FastAPI service	main.py, Dockerfile, requirements.txt, .env.example	Service startup, endpoints, container runtime, and documented configuration variables.
Layer 1 understanding	core/layer1_understanding/*.py, core/layer1_understanding/data/config.json	PDF/image text extraction, OCR fallback, sectioning, NER, contact extraction, experience analysis, and canonicalization.
Layer 2 classification	core/layer2_classification/*.py, core/layer2_classification/data/taxonomy.json	Domain, seniority, and skill-category enrichment.

Area	Representative Files	Main Responsibility
Layer 3 matching	core/layer3_matching/*.py, core/layer3_matching/matching_config.json	Job parsing, semantic/TF-IDF matching, constraints, fit explanation, and ranking.
Training workflow	training/generate_tech_dataset.py, clean_dataset.py, train_ner.ipynb	Synthetic labeled dataset generation, cleaning, token alignment, Trainer setup, metrics code, and export.
Diagnostics and tests	scripts/verify_phase*.py, tests/test_service_api.py, tests/trace_cv.py, manual tests	Phase checks, service API tests with fakes, tracing, and manual validation helpers.
Documentation	Layer README and EXPLAIN files	Developer explanations for analyzer layers and matching logic.
Ignored deployment assets	.env, models/ner_weights/...	Local secrets and model weights are intentionally ignored; only safe metadata was inspected locally.

Table 20. AI CV Analyzer source inventory summary.

Algorithm or Workflow	Primary File(s)	Notes
Parse-CV request routing	main.py	File validation, image/PDF branching, timeout and error wrappers.
Ordered PDF extraction	core/layer1_understanding/spatial_parser.py	Adaptive row/column ordering and dehyphenation.
OCR fallback	core/layer1_understanding/ocr_pipeline.py, CVOrchestrator	Used when PDF text is absent or too short.
NER extraction	core/layer1_understanding/advanced_ner.py	Singleton model loading, chunked inference, entity merging, and cleanup.
Skill normalization	core/layer1_understanding/canonicalizer.py	Exact, fuzzy, and semantic mapping toward canonical names.
Experience calculation	core/layer1_understanding/experience_engine.py	Date ranges, total years, skill durations, gaps, overlaps, and action verbs.
Seniority inference	core/layer1_understanding/orchestrator.py, core/layer2_classification/seniority_engine.py	Combines title keywords, years, semantic hints, and action verbs.
Domain classification	core/layer2_classification/domain_engine.py	Embedding comparison against taxonomy descriptions.
Job parsing	core/layer3_matching/job_description_engine.py	Seniority, years, mandatory/bonus skills, domain, and summary extraction.
Hybrid match scoring	core/layer3_matching/similarity.py, core/layer3_matching/constraint_validator.py	Weighted semantic/skill/domain score minus constraint penalties plus bonus boost.
TF-IDF fallback	core/layer3_matching/tfidf.py, main.py	Pure Python sparse cosine score used in hybrid endpoint.
Fit explanation	core/layer3_matching/fit_analysis_generator.py	Verdict, strengths, gaps, and red flags.
NER training	training/train_ner.ipynb	BERT base checkpoint, BIO labels, token alignment, Trainer, seqeval metrics.
Synthetic data generation	training/generate_tech_dataset.py	Gemini-based generation with key rotation and negative decoys.

Table 21. Algorithm-to-file mapping.

5.6 Profile and Skills Management

The profile page reads normalized user data, profile fields, experiences, skills, and CV analysis. The system distinguishes user fields, profile fields, extracted skills, predicted role, seniority, and completeness score. Skill synchronization is handled through backend services rather than only frontend state.

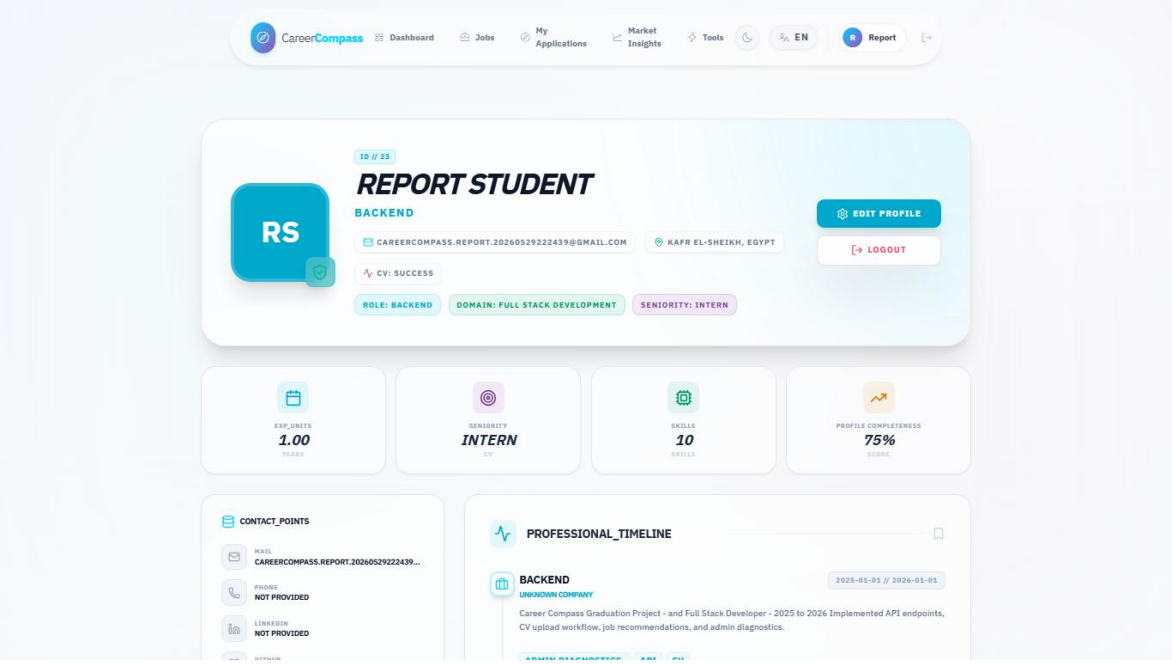


Figure 37. Extracted profile and skills page.

5.7 Job Data Model

Jobs are represented in the backend through job posting models and migrations. Fields include title, company, description/requirements, URL, source, and metadata. The seeders and import controllers enforce quality gates and uniqueness rules, including a title/company uniqueness constraint that prevented duplicate seed insertion during validation.

5.8 AI Job Miner and Scraping Sources

The job miner exposes a FastAPI service and imports candidate jobs through configured sources. This implementation chapter keeps the feature overview short: Laravel remains the system of record, the Python service handles adapter work, and admin pages expose source diagnostics, source status, testing, and target role management. Chapter 7 expands this subsystem with runtime diagrams, queue flow, API contracts, import/deduplication logic, failed-URL handling, security boundaries, evaluation evidence, and ethical limitations.

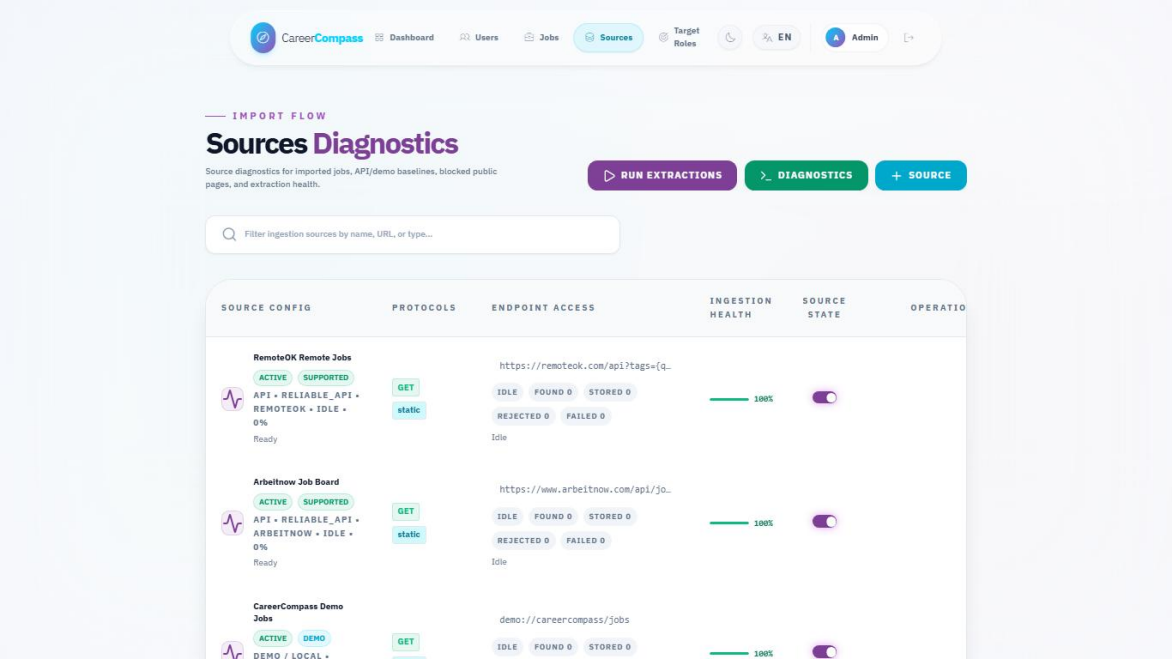


Figure 46. Admin sources diagnostics page.

5.9 Job Recommendations

The jobs page requests recommended jobs when no manual search query is active. Recommendations are based on CV/profile context when available. Matching combines normalized database data with semantic and TF-IDF-style comparison where available. TF-IDF represents text using term frequency and inverse document frequency weighting [19], while cosine similarity compares vector orientation [20].

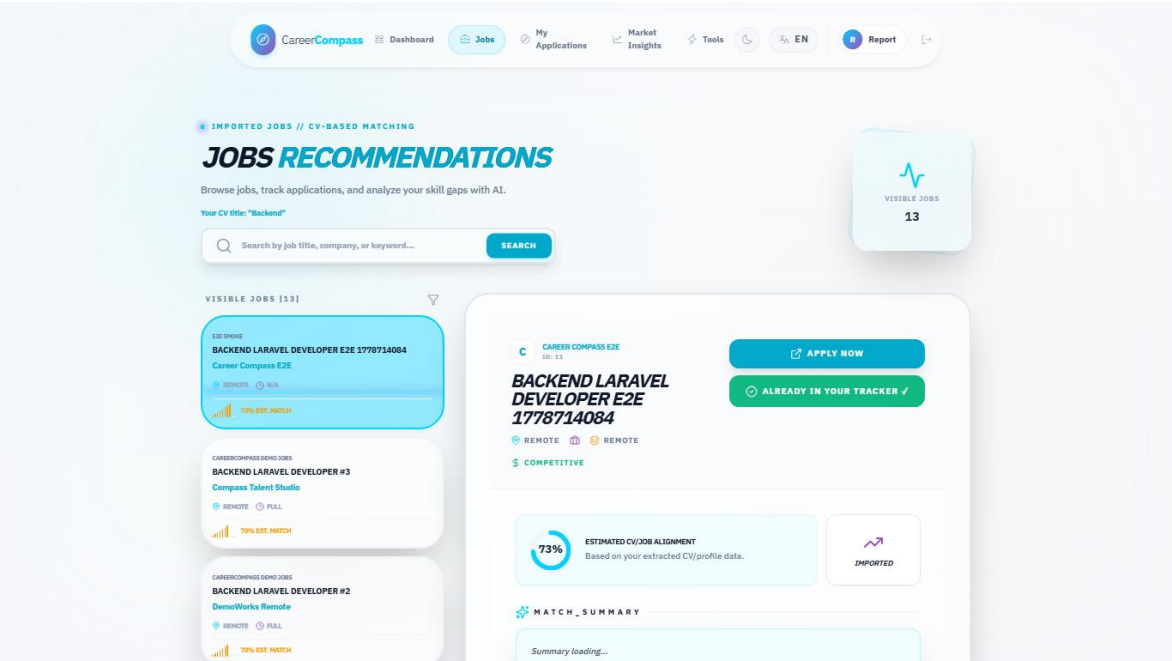


Figure 38. Jobs recommendations page.

5.10 Gap Analysis

Gap analysis compares a selected job or target role against the user's profile and extracted skills. It returns matched skills, critical/missing skills, recommendations, match percentage, and roadmap-like guidance. The frontend displays these outputs in an explainable layout rather than a single opaque score.

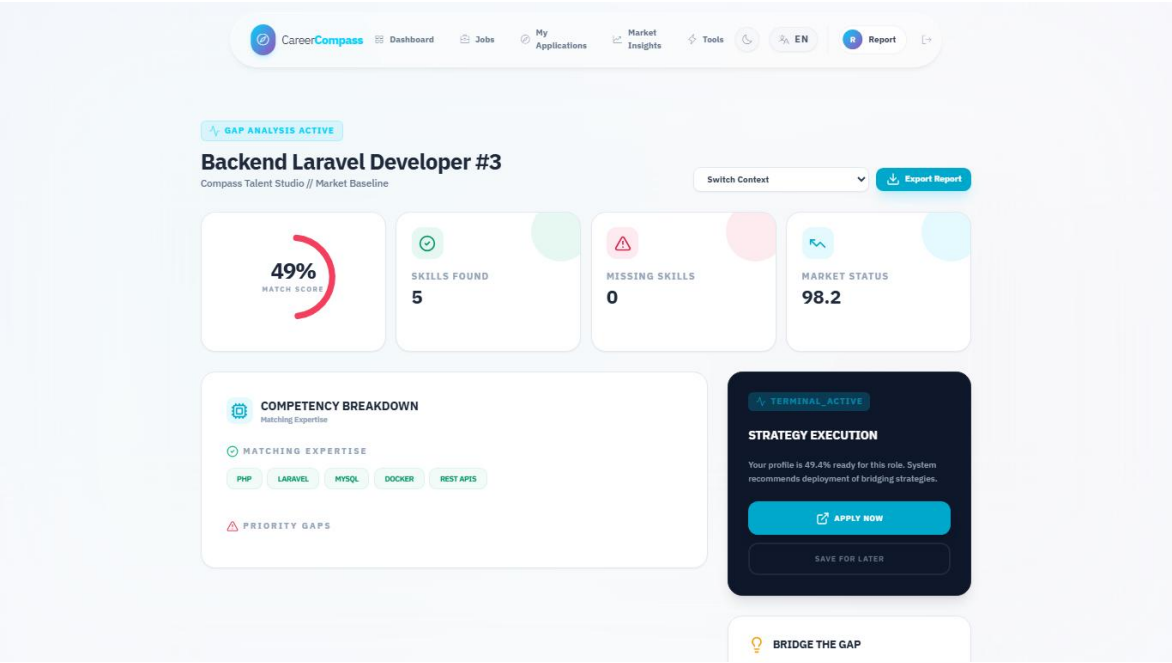


Figure 40. Gap analysis page.

5.11 Application Tracker

The application tracker is implemented through ApplicationController, ApplicationTrackerService, and frontend/src/pages/user/Applications.jsx. Students can save a job, update status, view counts, and delete tracked items. The backend validates job existence and allowed statuses.

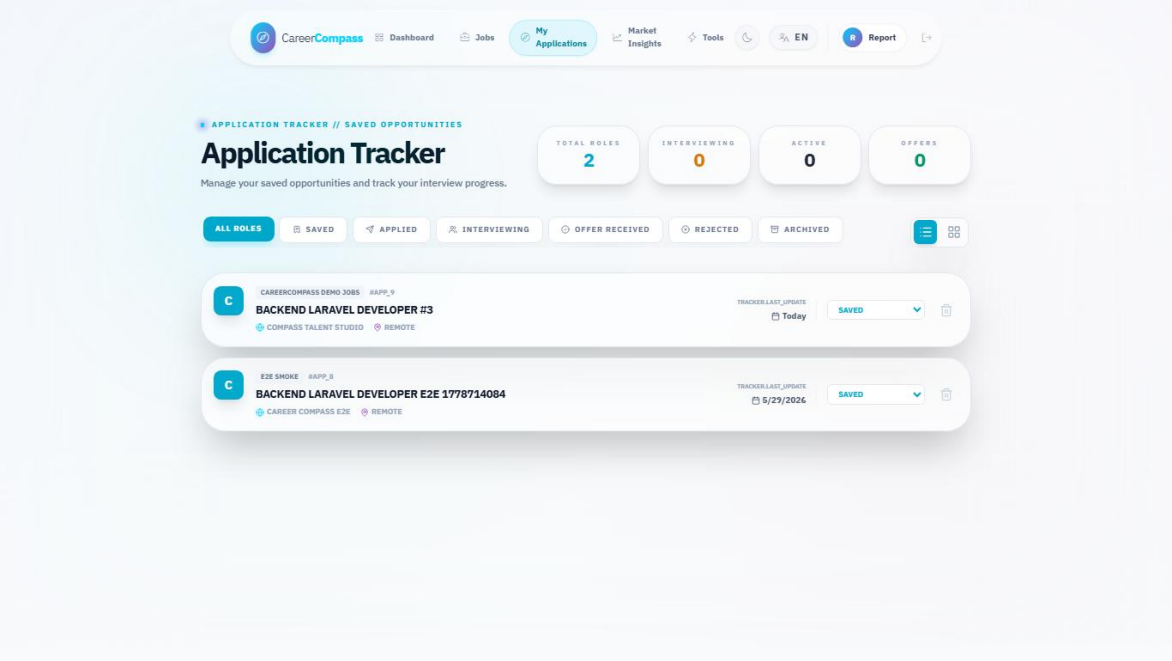


Figure 41. Applications tracker page.

5.12 Admin Dashboard

The admin dashboard summarizes users, imported jobs, active sources, target roles, health status, and scraping batch progress. It is protected by admin middleware and uses admin API routes.

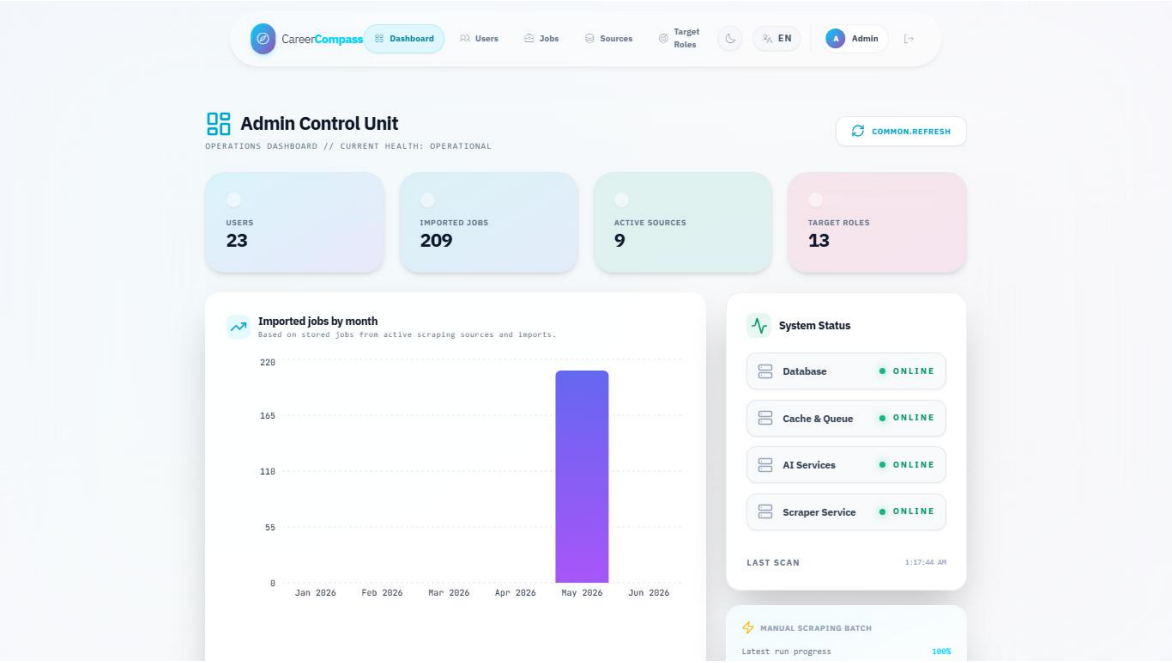


Figure 44. Admin dashboard.

5.13 Admin Source Diagnostics

The source diagnostics page lists configured scraping sources, supports source testing, and displays quality and scraping status information. The target roles page manages role names used by scraping and market discovery.

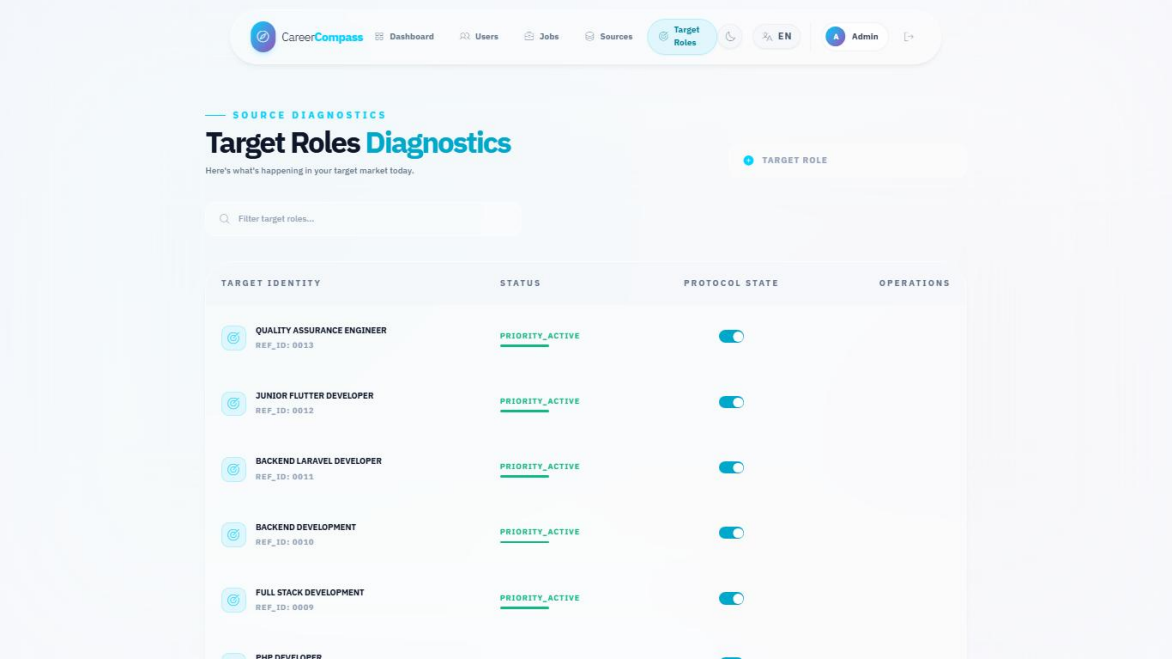


Figure 47. Admin target roles page.

5.14 System Health and Monitoring

Health endpoints include live and readiness checks. The system status page presents service state to users, while admin health data supports operational monitoring. Metrics are available for Prometheus and dashboards are available through Grafana.

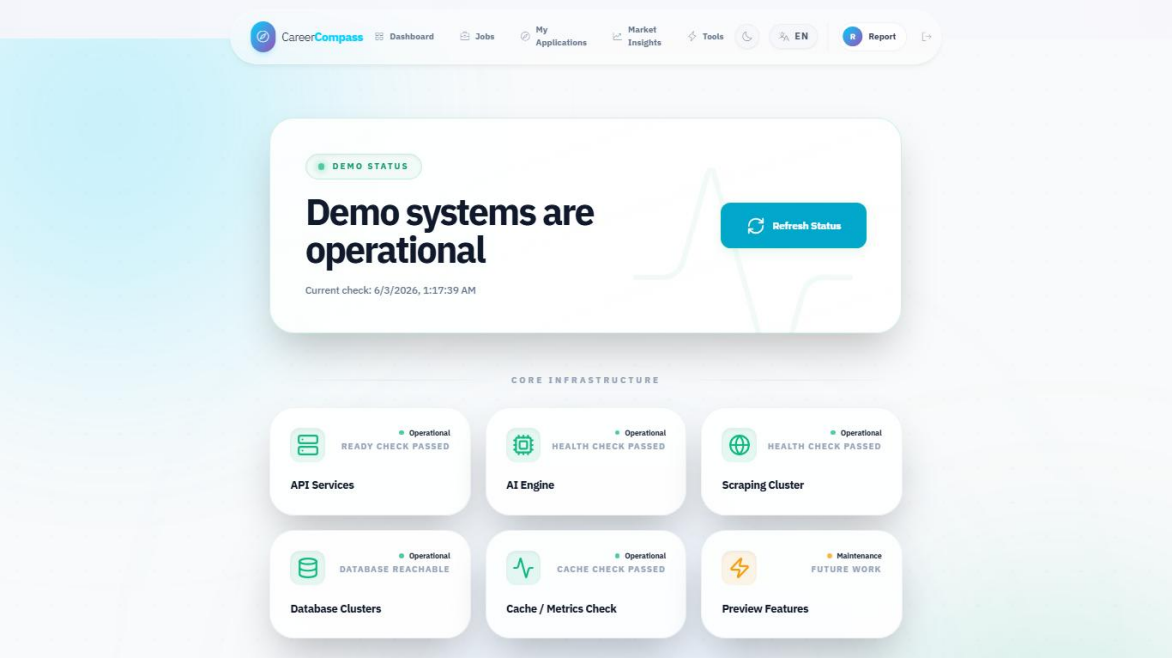


Figure 43. System status page.

5.15 Error Handling and Fallbacks

The code includes explicit handling for CV processing failures, AI gateway connection failures, validation errors, missing user data, empty job data, and unavailable services. The job recommendation and gap analysis code includes fallback behavior when AI services are not available.

5.16 Internationalization and UI Preview Modules

The frontend contains English and Arabic locale files. Several menu items intentionally appear as preview modules so the interface can show the intended product direction without presenting those areas as completed core deliverables. The completed graduation/demo core remains CV upload, parsed profile and skills, recommendations, gap analysis, application tracking, admin diagnostics, and system status.

Module	Current Status	Reason Included	Future Work
CV Builder	Preview placeholder	Shows where guided resume authoring would fit beside CV upload.	Add editable sections, export templates, and validation tests.
Mock Interview	Preview placeholder	Demonstrates a possible practice workflow after gap analysis.	Add question banks, scoring rubrics, and consent-aware recording rules.
Learning Paths	Preview placeholder	Connects missing skills to future study plans.	Integrate curated resources and progress tracking.
Career Planner	Preview placeholder	Shows longer-term roadmap direction.	Add milestone planning and advisor review.
Mentorship	Preview placeholder	Shows possible human-support extension.	Add mentor profiles, matching rules, and moderation.

Module	Current Status	Reason Included	Future Work
Tools Hub	Preview screen	Groups preview tools in one visible place.	Replace placeholders with independently tested modules.
Market Intelligence	Supporting/preview view	Helps explain job-market context from imported jobs.	Add stronger statistics, freshness indicators, and source quality labels.

Preview module clarification table.

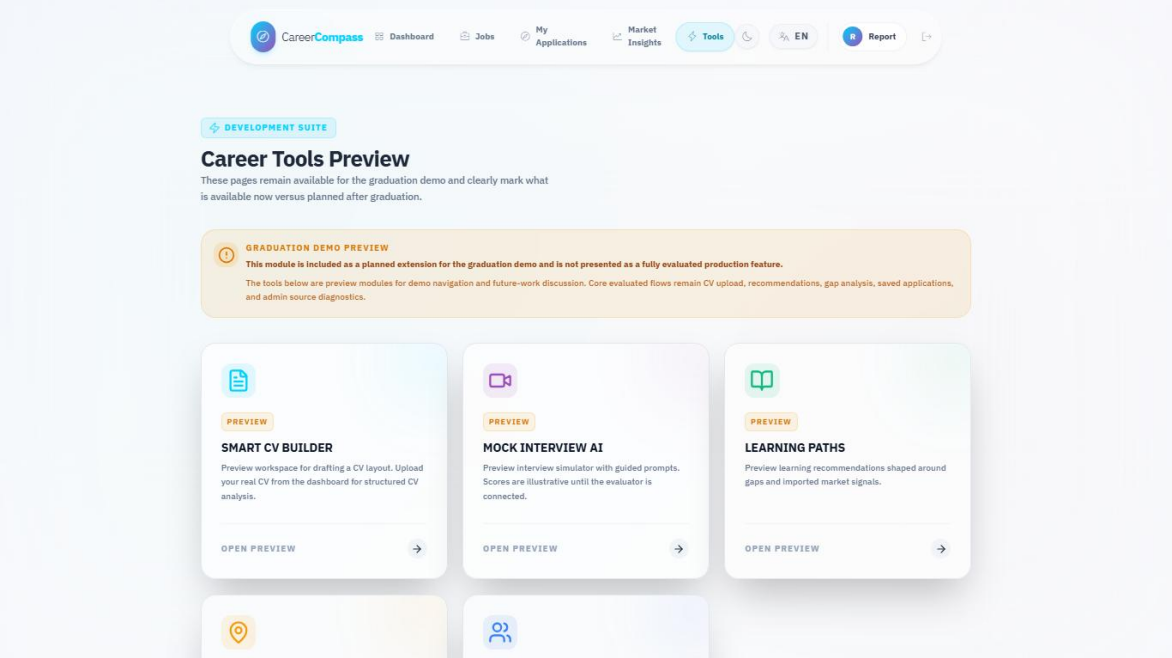
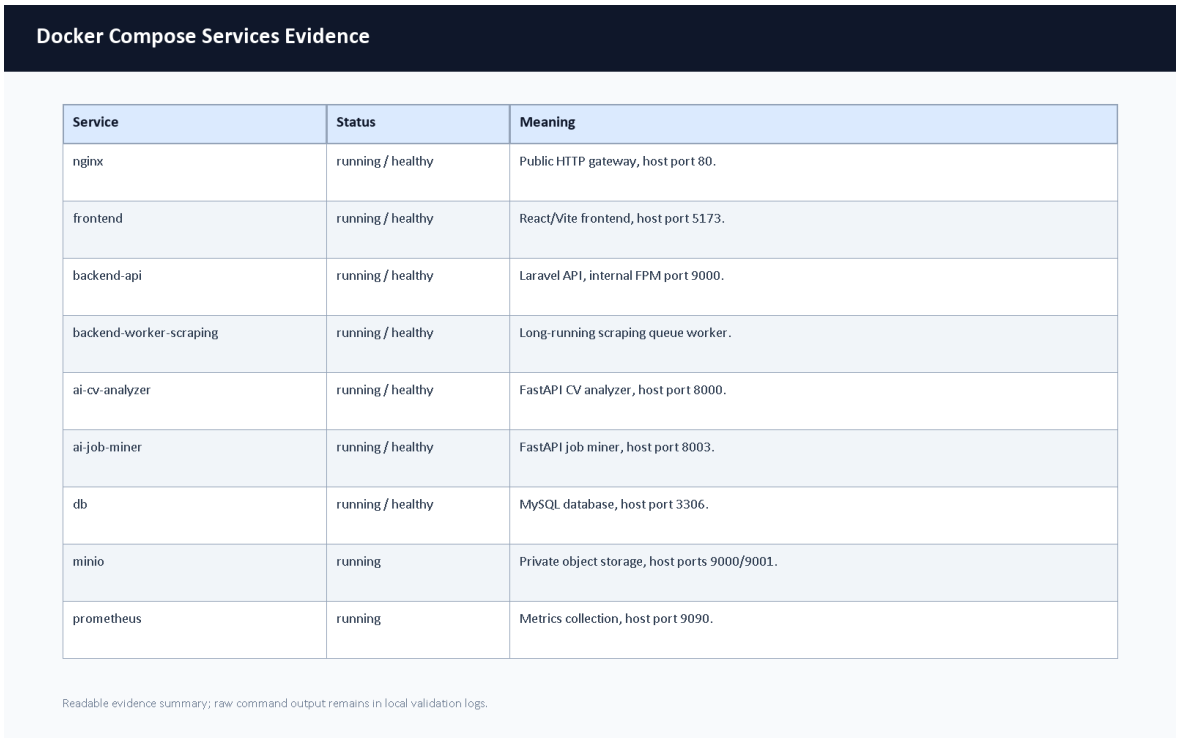


Figure 42. Tools Hub preview page.

5.17 Dockerized Runtime Flow

The runtime starts through Docker Compose. Nginx exposes the app, frontend and backend containers serve UI/API flows, backend workers process queues, Python services support AI workflows, MySQL and MinIO persist state, and monitoring services observe the stack.



Docker Compose Services Evidence

Service	Status	Meaning
nginx	running / healthy	Public HTTP gateway, host port 80.
frontend	running / healthy	React/Vite frontend, host port 5173.
backend-api	running / healthy	Laravel API, internal FPM port 9000.
backend-worker-scraping	running / healthy	Long-running scraping queue worker.
ai-cv-analyzer	running / healthy	FastAPI CV analyzer, host port 8000.
ai-job-miner	running / healthy	FastAPI job miner, host port 8003.
db	running / healthy	MySQL database, host port 3306.
minio	running	Private object storage, host ports 9000/9001.
prometheus	running	Metrics collection, host port 9090.

Readable evidence summary; raw command output remains in local validation logs.

Figure 48. Docker services evidence.

Chapter 6: AI CV Analyzer Deep Technical Analysis

6.1 Introduction

The AI CV Analyzer is one of the main technical contributions of CareerCompass. It should not be understood as a thin wrapper around one pretrained model. The implemented analyzer is a layered hybrid pipeline that combines document-processing logic, NER, deterministic extraction rules, semantic enrichment, score composition, and explanation generation. This chapter separates that AI contribution from the general implementation chapter so that supervisors and examiners can evaluate the design as an academic system component.

6.2 AI Design Philosophy

CareerCompass does not use a pure NER model because CVs are noisy, multi-format documents. They can contain multiple columns, icons, section headers, table-like blocks, scanned pages, mixed date formats, and skill aliases. NER can extract entity candidates, but NER alone does not naturally explain seniority, primary technical domain, job-fit constraints, or recommendation reasons.

The system also does not use a pure rule-based parser. Rules are deterministic and useful for validation, but fixed rules are brittle when skill names, job titles, section headings, and CV layouts vary. A rule set can recognize known patterns, but it struggles with semantic similarity, synonyms, and role/domain interpretation.

The implemented design is therefore hybrid. NER extracts structured candidates, deterministic rules improve consistency and safety, canonicalization reduces noisy variants, Layer 2 adds domain and seniority interpretation, Layer 3 compares candidate and job evidence, and the explanation layer turns scores into strengths, gaps, red flags, and verdicts. TF-IDF fallback keeps the matching endpoint useful when heavier semantic components are unavailable.

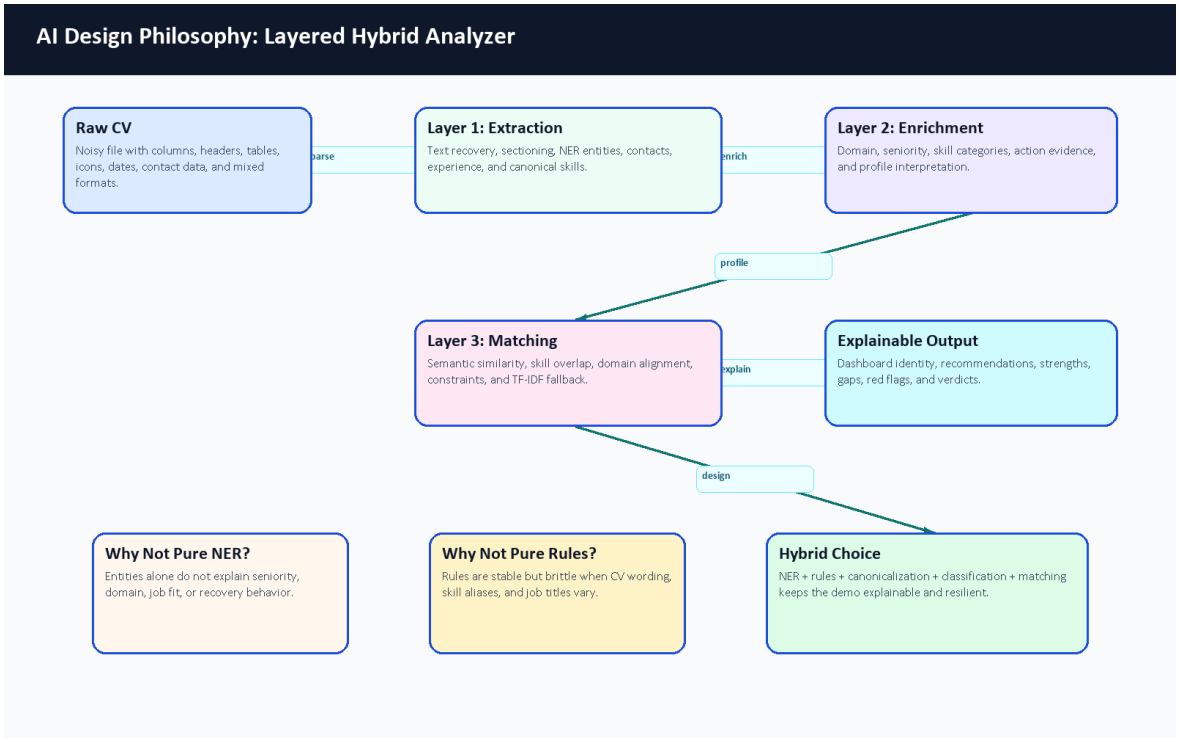


Figure 19. AI design philosophy for the layered hybrid analyzer.

Design Option	Advantage	Limitation	CareerCompass Decision
Pure NER	Learns entity patterns from data.	Does not solve file recovery, seniority, domain, matching, or explanation by itself.	Used only as one extraction component.
Pure rules	Predictable and easy to inspect.	Brittle when CVs use new wording, layouts, and aliases.	Used for safety, contacts, dates, validation, and fallback behavior.
Hybrid layered AI	Combines learned extraction, deterministic checks, semantic signals, and explanation.	More components must be tested and documented.	Chosen because it fits noisy CVs and graduation-demo transparency.

Table 22. AI design alternatives comparison.

6.3 Complete CV Processing Flow

The end-to-end flow begins when the student uploads a PDF or image CV. The frontend validates the file before sending it to Laravel. Laravel sends the file to the FastAPI analyzer, stores the private CV object, persists successful structured outputs, and records parsing status. The analyzer first recovers text, then segments sections, extracts entities, estimates experience, canonicalizes skills, classifies the profile, and supports matching for recommendations and gap analysis.

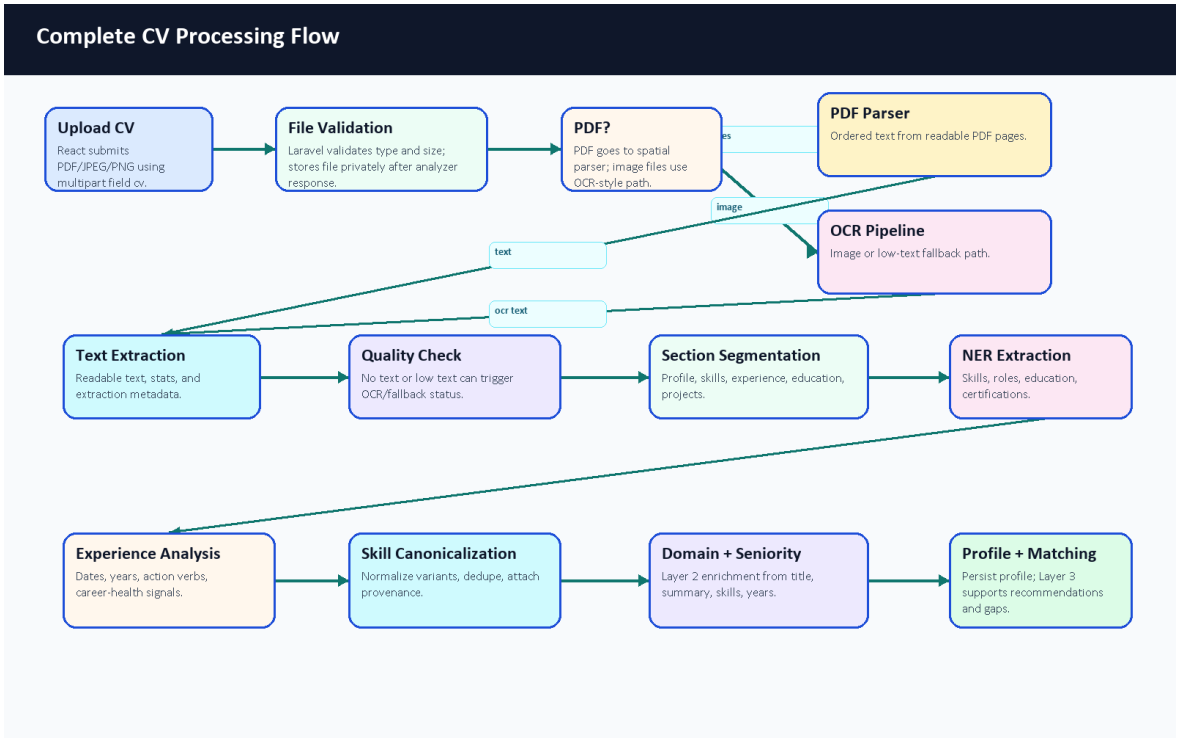


Figure 20. Complete CV processing flow.

The flowchart is intentionally more detailed than the high-level architecture diagram. It shows that the AI service performs several recoverable steps before returning data. The output is not only a list of words; it includes profile fields, skills, experience signals, domain, seniority, confidence-style values, metadata, and status.

6.4 Fault Tolerance and Recovery

CV parsing can fail for normal reasons: scanned PDFs, image-only files, weak text extraction, unsupported content, or service timeouts. The analyzer and backend are designed to report these states explicitly instead of silently overwriting good profile data with empty extraction results.

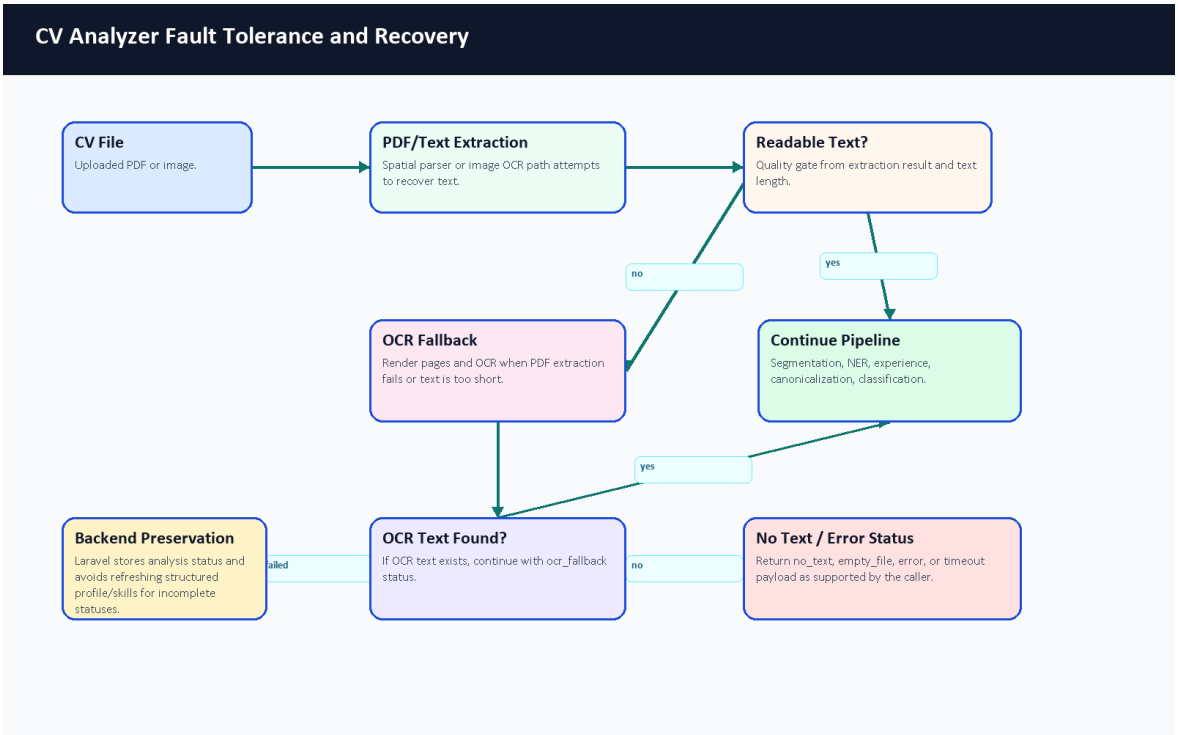


Figure 21. CV analyzer fault tolerance and recovery flow.

The FastAPI schema supports success, ocr_fallback, empty_file, no_text, and error statuses. The API-level timeout path returns a timeout payload. Laravel treats timeout, error, empty_file, and no_text as incomplete statuses and avoids refreshing profile, experience, and skills for those results. The backend still records analysis status and file metadata, then warns the frontend so the user can retry with a clearer document.

6.5 Confidence and Readiness Signals

CareerCompass uses confidence-style and readiness signals rather than a certified probability of hiring success. In Layer 1, _aggregate_confidence averages positive confidence-style values and caps the result at 1.0. Skills, profile, experience, and analysis sections can each carry confidence values. Laravel stores parser confidence_score and converts it into a completeness_score when available. The dashboard then visualizes completeness, model confidence, skill count, and experience as an estimated Career Readiness Snapshot.

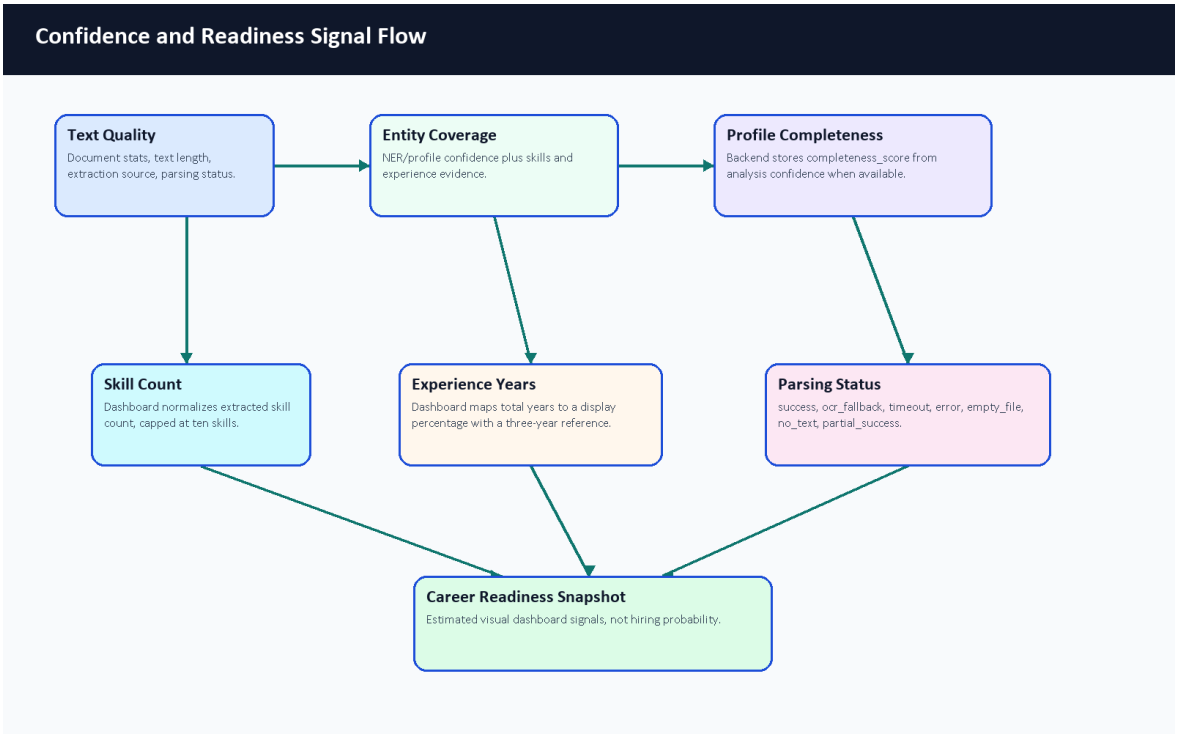


Figure 22. Confidence and readiness signal flow.

The current code-derived signal boundary is:

```
AggregateConfidence =
    min(1.0, average(positive confidence signals))

AnalysisConfidence =
    AggregateConfidence([
        skills_section.confidence_score,
        experience_section.confidence_score
    ])

BackendCompleteness =
    round(analysis.confidence_score * 100)

DashboardSkillSignal =
    min(extracted_skill_count * 10, 100)

DashboardExperienceSignal =
    min((total_experience_years / 3) * 100, 100)
```

These formulas are intentionally described as signals. They help the dashboard explain whether enough structured CV evidence was extracted, but they are not probabilities of employability, hiring success, or model correctness.

Signal	Source File or Component	Meaning	Used In	Limitation
parsing_status	main.py, schema.py, CvProcessingService.php	Whether parsing succeeded, used OCR, failed, timed out, or found no text.	Upload feedback, stored analysis status, preservation logic.	It is a status flag, not a quality score.
confidence_score	Layer 1 schema and orchestrator aggregation	Confidence-style value for extracted/derived fields.	Stored CV analysis and AI insight display.	It is not a calibrated probability of employment success.

Signal	Source File or Component	Meaning	Used In	Limitation
completeness_score	CvProcessingService.php	Backend percentage derived from analysis confidence when available.	Profile/dashboard completeness display.	It depends on parser output availability.
Skill signal	Dashboard.jsx	Extracted skill count normalized and capped at ten skills.	Career Readiness Snapshot.	More skills do not automatically mean a better candidate.
Experience signal	Dashboard.jsx	Total parsed years mapped to a display percentage using a three-year reference.	Career Readiness Snapshot.	It is a UI signal, not a universal seniority formula.
Extraction metadata	Layer 1 metadata and Laravel analysis metadata	Source, spatial status, segmentation, gaps, action-verb and experience details.	Debugging, review, and future evaluation.	Metadata quality depends on successful text recovery.

Table 23. Confidence and readiness signal summary.

6.6 Skill Canonicalization With Practical Example

Skill extraction is noisy because the same skill can appear in different forms. The canonicalizer supports exact variant mapping, exact canonical matching, RapidFuzz matching when available, normalized-key fallback, semantic embedding fallback, and pass-through behavior. The current committed config is largely industry-agnostic, so the example below is labeled illustrative of the implemented mapping stages rather than proof that every alias is already configured in source data.

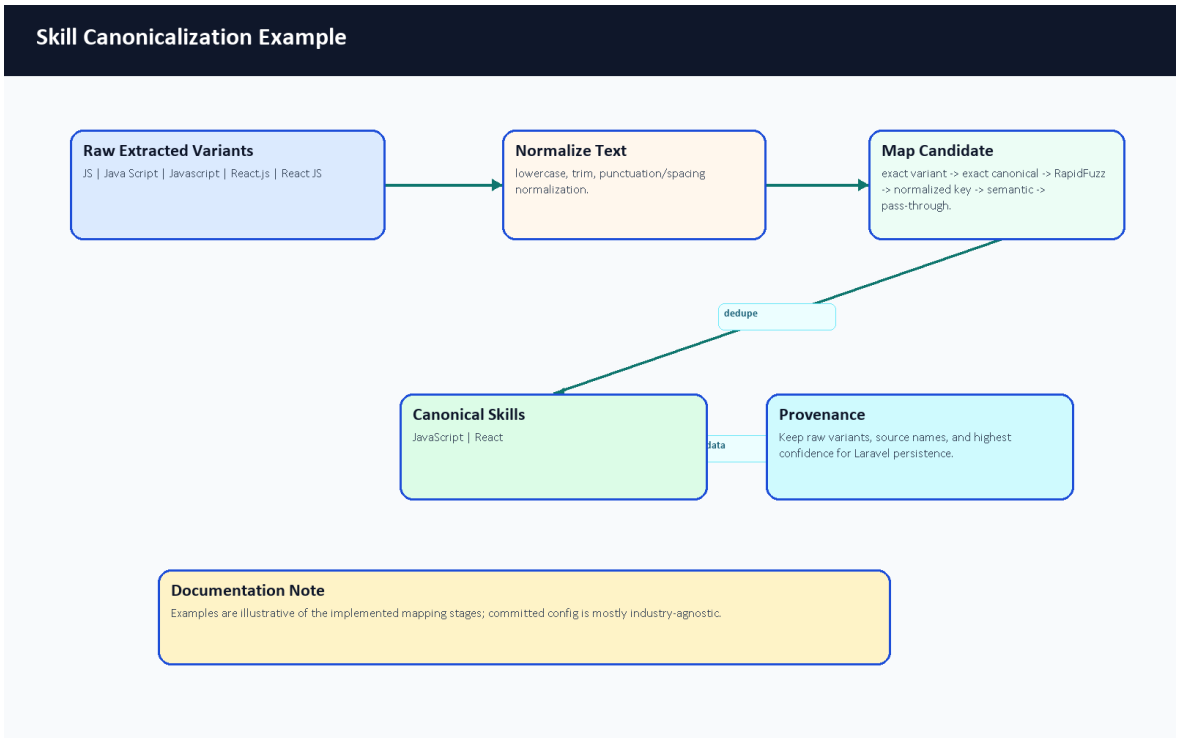


Figure 23. Skill canonicalization example.

Raw Extracted Skill	Normalized Skill	Why
JS	JavaScript	Illustrative abbreviation normalization.
Java Script	JavaScript	Illustrative spacing normalization.
Javascript	JavaScript	Illustrative casing/spelling normalization.

Raw Extracted Skill	Normalized Skill	Why
React.js	React	Illustrative framework alias normalization.
React JS	React	Illustrative punctuation and spacing normalization.

Table 24. Skill canonicalization example.

6.7 Fine-Tuned BERT NER Architecture

The NER architecture is a fine-tuning workflow, not a from-scratch language model. The training notebook uses bert-base-cased, tokenizes CV text with offsets, aligns character-span annotations to BIO token labels, trains a token-classification head, and exports career_compass_ner_final. At runtime, AdvancedNEREngine can load local ignored model weights if supplied; those weights are not committed to Git. A user-provided Colab export now provides recorded training-run metrics, but the repository alone still does not contain the final dataset, model weights, or a fully reproducible benchmark package.

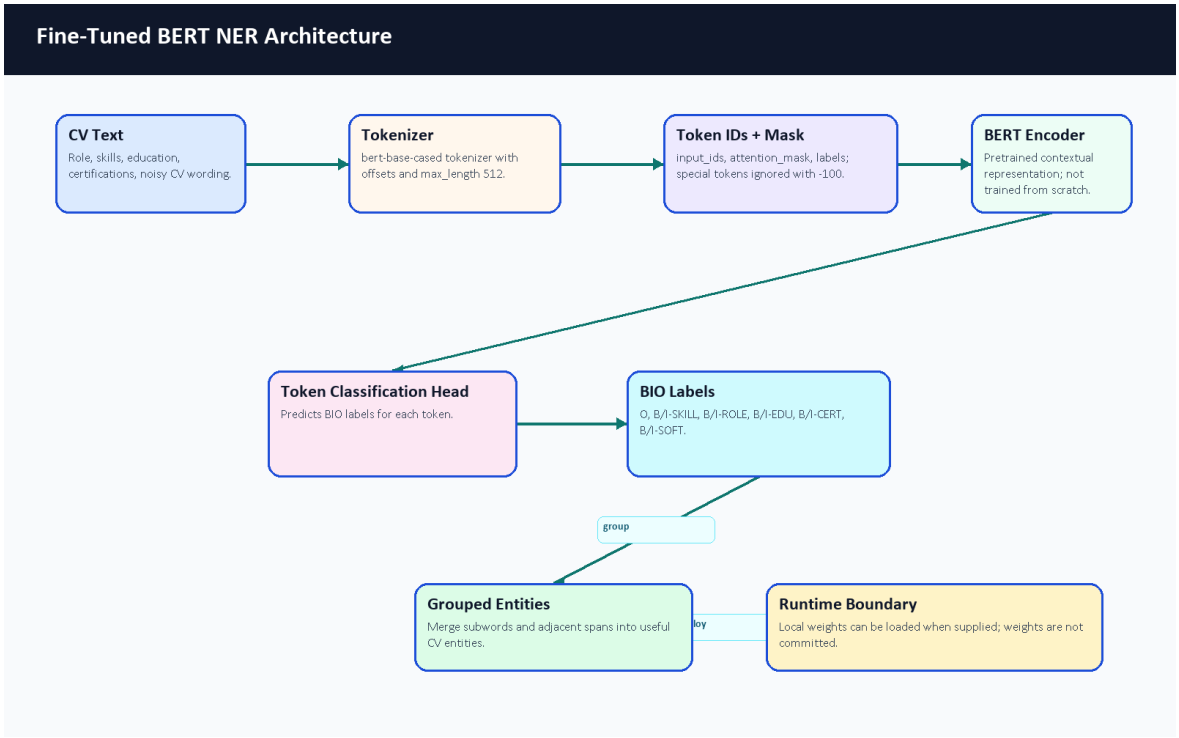


Figure 24. Fine-tuned BERT NER architecture.

The simplified BIO example in Table 19 remains valid for examiner explanation: Backend can start a ROLE entity, Developer can continue it, and Laravel or Docker can start SKILL entities. The actual model sees tokenized subwords and offsets rather than only human-readable words.

6.8 Detailed Training Pipeline

Synthetic training data is used because labeled CV NER data is not naturally available in the repository. The generator is designed to create varied technical CV snippets, including positive examples and negative decoys. Negative decoys matter because they teach the model not to tag every technical-looking phrase. The cleaner normalizes samples and validates entity spans before the notebook performs token alignment and fine-tuning.

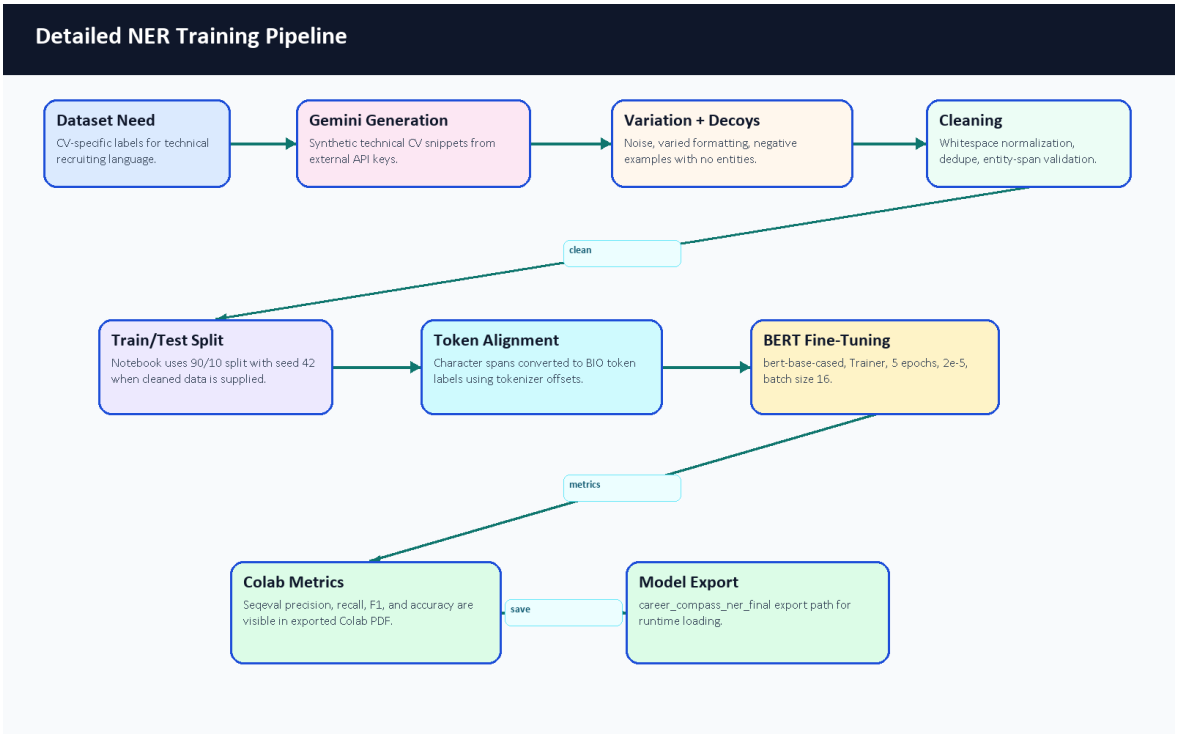


Figure 25. Detailed NER training pipeline.

The documentation pass also reviewed committed evaluation evidence and generated a dataset transparency note under docs/graduation-book/model-analysis/dataset_statistics.md. The user-provided Colab PDF gives recorded training-run output cells, including split counts and overall validation metrics. However, the cleaned dataset content and model weights are still not committed, and the PDF does not show per-label support counts. Therefore, the report includes the verified Colab metrics but still avoids a fake per-label distribution chart. Figure 29 records which evidence is available and what remains unavailable for reproducible academic review.

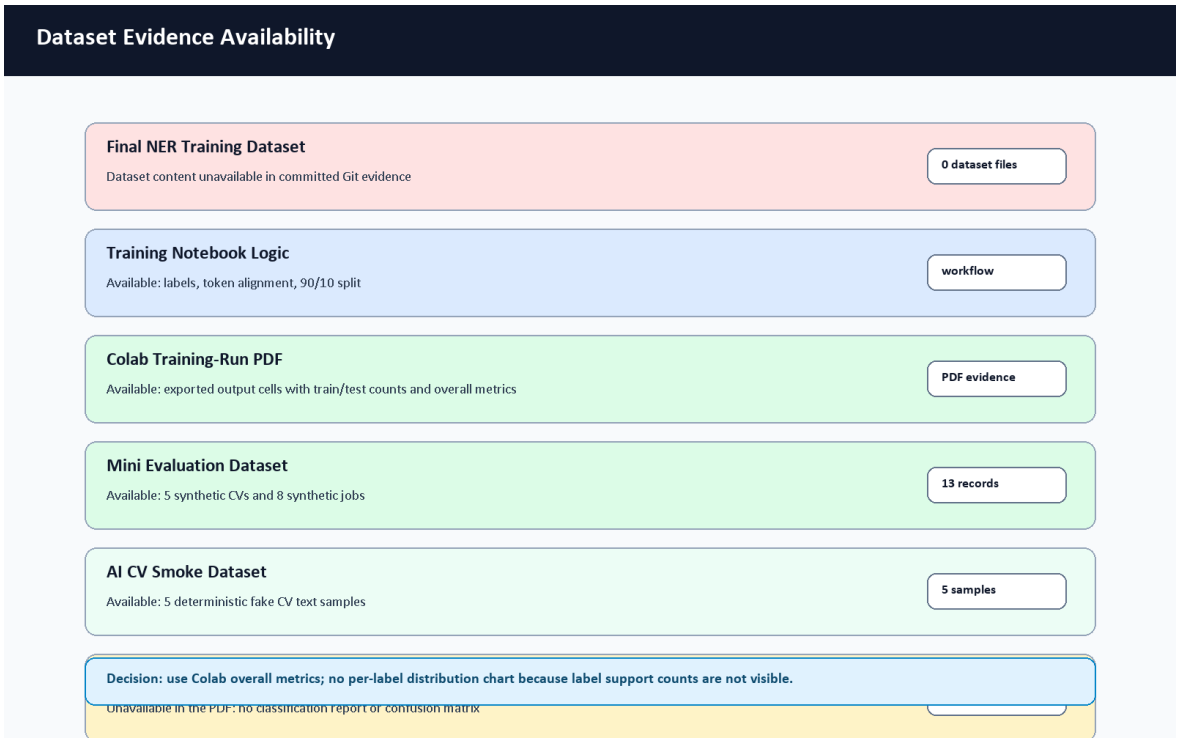


Figure 29. Dataset evidence availability summary.

Dataset Statistic	Status	Reason
Cleaned NER training samples	45,911 rows recorded in Colab PDF	The PDF shows the generated rows loaded from train_real_tech_cleaned.json; the dataset content itself is not committed.
Recorded train split	41,319 rows	Visible in the exported Colab PDF; split uses test size 0.1 and seed 42.
Recorded test split	4,592 rows	Visible in the exported Colab PDF and used for the notebook evaluation output.
Entity counts by label	Not visible in the PDF	The PDF shows labels but not per-label support counts.
Negative decoy count	Not available from committed evidence	Generator/cleaner support decoys, but final generated data is absent.
Mini evaluation CV samples	5	Generated documentation mini dataset under docs/graduation-book/evaluation/; separate from NER training.
Mini evaluation job samples	8	Generated documentation mini dataset; not a production benchmark.
AI CV Analyzer smoke samples	5	Deterministic text-only smoke set created for this evidence pass; evaluates parser-style labels and dependency availability, not transformer weights.
Final NER label distribution chart	Not generated	No per-label support counts are visible in the PDF, and no committed final labeled dataset exists to count SKILL/ROLE/EDU/CERT/SOFT/O labels honestly.

Table 25. Dataset availability and transparency.

6.8.1 Colab NER Fine-Tuning Results

The team exported the Google Colab notebook train_ner.ipynb as a PDF with visible output cells. This PDF was copied into docs/graduation-book/model-analysis/colab_train_ner_results.pdf after inspection. It is treated as supporting training evidence for the NER fine-tuning process. The PDF shows the notebook title train_ner.ipynb - Colab, timestamp 6/7/26, 3:55 AM, the heading CareerCompass AI Engine: Global Skill NER training (Autonomous), and a synthetic data augmentation strategy. It also shows the cleaned dataset path train_real_tech_cleaned.json, 11 BIO labels, train/test row counts, tokenization completion, model initialization from bert-base-cased, training arguments, and epoch-level metrics.

These numbers improve the academic evidence for the training workflow, but they should be interpreted carefully. They are Colab-run validation outputs for the generated/synthetic dataset and notebook split visible in the PDF. They are not production accuracy, not a large real-world CV benchmark, and not reproducible from the repository alone unless the same dataset, runtime, and exported model artifacts are supplied.

Parameter	Value from Colab PDF	Source
Notebook	train_ner.ipynb - Colab	PDF header
Export timestamp	6/7/26, 3:55 AM	PDF header
Dataset file	train_real_tech_cleaned.json	PDF data-loading cell
Total loaded rows	45,911	PDF dataset output
Train rows	41,319	PDF dataset output
Test rows	4,592	PDF dataset output
Split	test size 0.1, seed 42	PDF data-loading cell
Base checkpoint	bert-base-cased	PDF model initialization cell
Labels	O plus B/I for SKILL, ROLE, EDU, CERT, SOFT	PDF label configuration cell
Max length	512 tokens	PDF tokenization cell
Epochs	5	PDF training arguments
Learning rate	2e-5	PDF training arguments
Batch size	16 train / 16 eval	PDF training arguments
Weight decay	0.01	PDF training arguments
Best model metric	F1	PDF training arguments

Table 33. Colab NER training run configuration.

The full epoch-by-epoch numeric table is retained in docs/graduation-book/model-analysis/colab_ner_training_results_summary.md. In the main chapter, the same verified values are shown as charts so the trend is easier to read in the PDF.

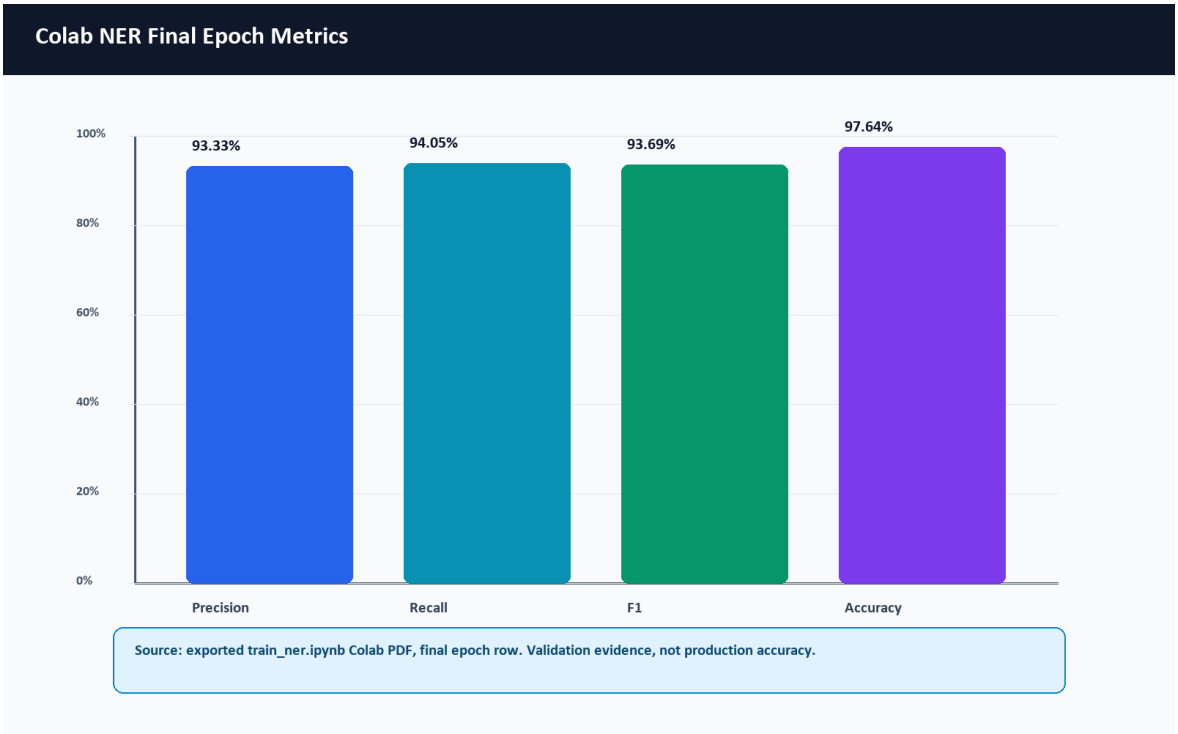


Figure 50. Colab NER final epoch metrics.

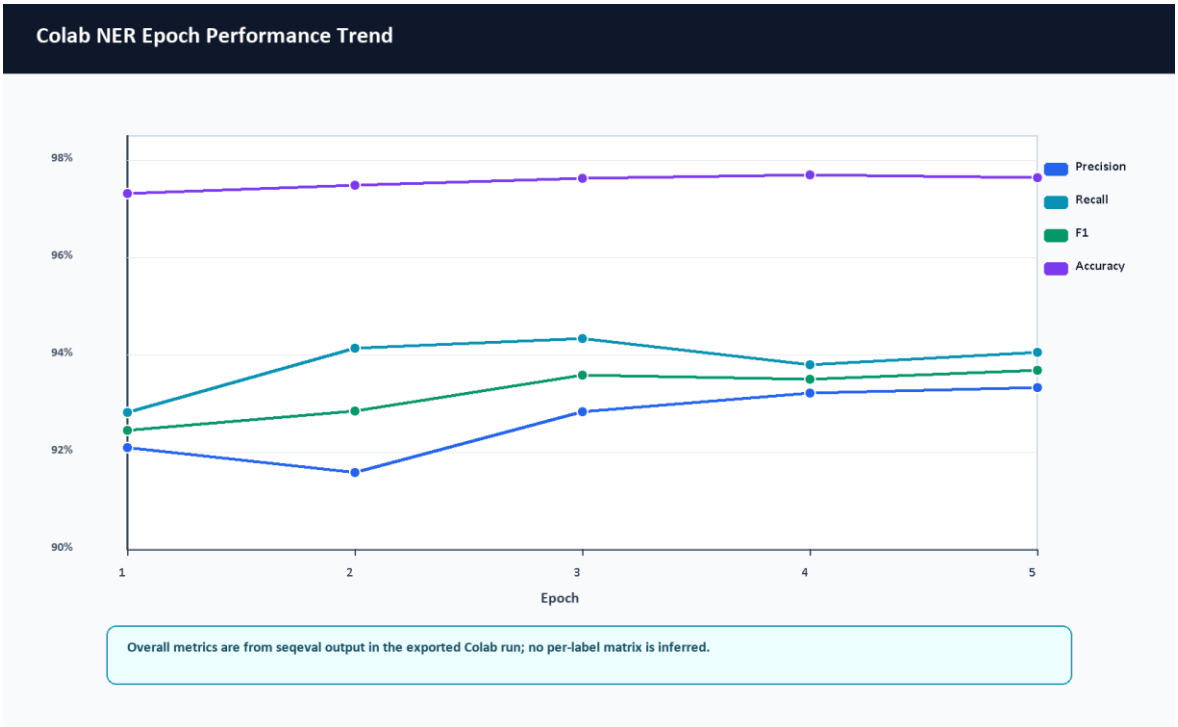


Figure 51. Colab NER epoch performance trend.

The performance chart shows that the overall F1 score increases from 0.924603 at epoch 1 to 0.936900 at epoch 5, while accuracy remains high throughout the run. These values are overall seqeval metrics from the notebook validation split; they are not per-label SKILL/ROLE/EDU/CERT/SOFT metrics.

Figure 52. Colab NER training and validation loss curve.

The notebook uses Hugging Face Trainer with AutoModelForTokenClassification and a token-classification dataset. It does not define a custom loss function in the visible code, so this report treats the reported training and validation loss as the Trainer's token-classification objective values rather than a separately designed loss formula. Training loss decreases steadily from 0.077623 to 0.037280. Validation loss remains low, with the lowest visible value at epoch 3 (0.063463) and a small increase by epoch 5 (0.068058). The final epoch still has the strongest visible F1 score, but the validation-loss movement is a reason to interpret the run cautiously rather than overclaiming model generalization.

Metric	Final Epoch Value	Source
Precision	0.933307	Colab PDF output, epoch 5
Recall	0.940521	Colab PDF output, epoch 5
F1-score	0.936900	Colab PDF output, epoch 5
Accuracy	0.976376	Colab PDF output, epoch 5
Training loss	0.037280	Colab PDF output, epoch 5
Validation loss	0.068058	Colab PDF output, epoch 5

Table 34. Colab NER final metric summary.

No per-label classification report or confusion matrix is visible in the exported Colab PDF, and the attached notebook output does not contain classification_report, confusion_matrix, sklearn.metrics, or matrix-like output cells. Therefore, this report does not invent per-label SKILL/ROLE/EDU/CERT/SOFT support, precision, recall, F1, or a confusion-matrix chart. Future training exports should save a token-level or entity-level confusion matrix plus a per-label classification report with support counts.

6.9 Matching Score Formula and Penalty Logic

The Layer 3 matcher exposes a clear score-composition formula in similarity.py and matching_config.json. It is exact for the code path implemented by IntelligentMatcher.calculate_match; however, it still depends on upstream component scores such as semantic similarity, skill similarity, and domain similarity.

```
BaseScore =
    semantic_score * w_semantic
  + skills_score   * w_skills
  + domain_score   * w_domain

FinalScore =
    clamp(BaseScore - total_penalty + bonus_boost, 0.0, 1.0)

MatchScorePercent = round(FinalScore * 100, 2)
```

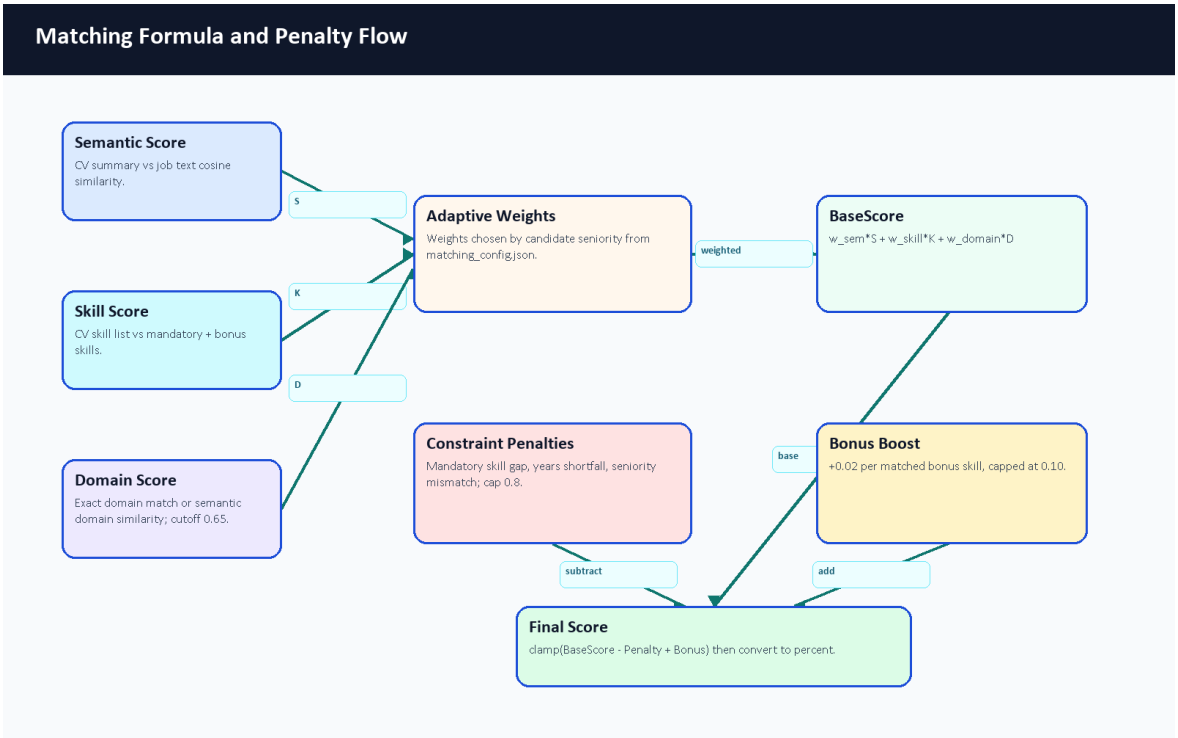


Figure 26. Matching formula and penalty flow.

Seniority	Semantic Weight	Skill Weight	Domain Weight	Notes
intern	0.30	0.60	0.10	Early roles emphasize concrete skill overlap.
junior	0.40	0.40	0.20	Balanced summary and skill evidence.
mid	0.35	0.35	0.30	Adds more domain importance.
senior	0.25	0.25	0.50	Domain alignment becomes more important.
lead	0.20	0.20	0.60	Leadership roles emphasize domain/role alignment.
default	0.35	0.35	0.30	Fallback profile.

Table 26. Seniority-aware matching weights.

Constraint penalties are also code-derived. Missing mandatory skills subtract 15 percent each, capped at 50 percent. Experience shortfall subtracts a proportional penalty capped at 30 percent. Seniority mismatch subtracts 20 percent. Total validation penalty is capped at 80 percent. Bonus skills add 2 percent each, capped at 10 percent. The /api/hybrid-match endpoint additionally blends the Layer 3 semantic/adaptive result with TF-IDF when TF-IDF is available: 60 percent semantic/adaptive and 40 percent TF-IDF.

6.10 Explainable AI Recommendation Output

The analyzer does not only return a single percentage. It also returns supporting evidence that can be shown to users and examiners: score breakdowns, missing mandatory skills, strengths, gaps, red flags, and a fit verdict. This is important academically because it makes the recommendation process inspectable rather than opaque.

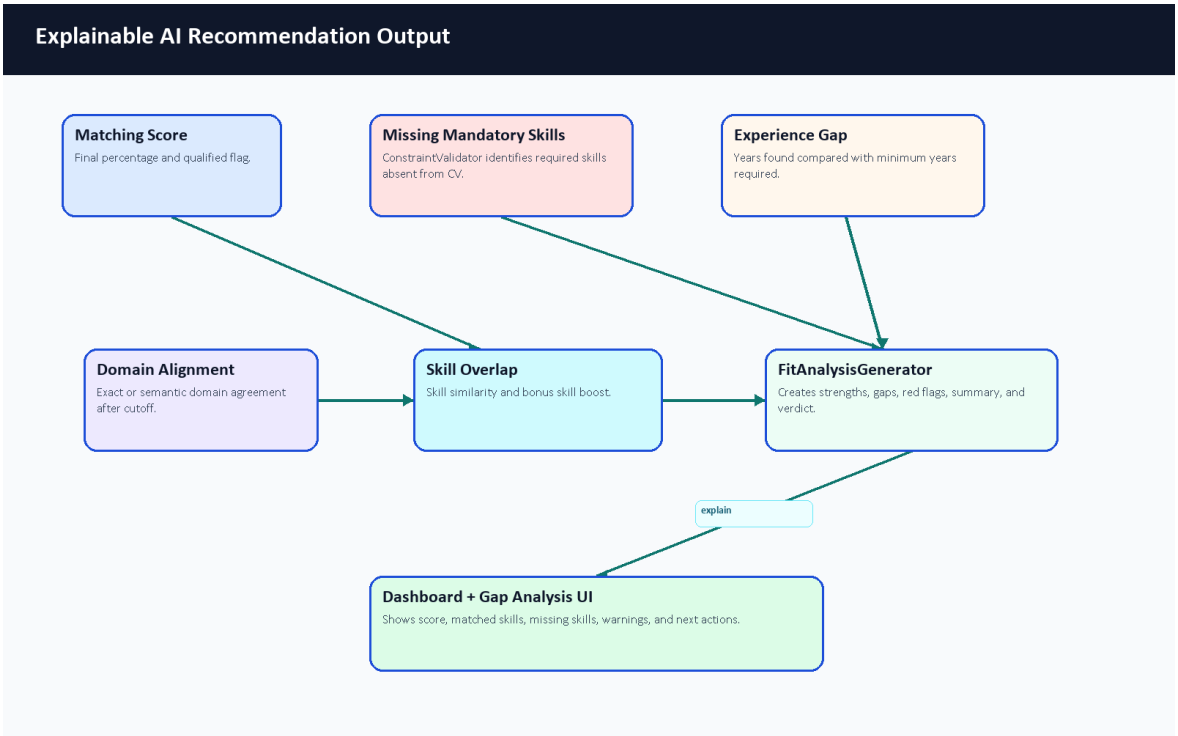


Figure 27. Explainable AI recommendation output.

Output Type	Example	Why It Helps
Score	78 percent	Gives a quick summary of estimated fit.
Matched skills	Laravel, Docker, MySQL	Shows evidence supporting the recommendation.
Missing skills	Kubernetes	Turns the gap into a learning target.
Red flags	Significant seniority mismatch	Warns that a numeric score should not be read alone.
Verdict	Strong Match or Potential Fit	Converts score ranges into readable guidance.
Gaps	Experience shortfall or missing mandatory skills	Explains why a candidate may need improvement before applying.

Table 27. Recommendation explanation output types.

6.11 AI Analyzer Sequence

The analyzer is synchronous during CV upload: Laravel calls FastAPI and receives a structured parse result before updating the returned user resource. The stored profile, skills, experiences, CV analysis, and private file metadata then support later dashboard, recommendation, and gap-analysis requests.

Figure 28. AI analyzer sequence diagram.

6.12 Computational Complexity Overview

The following complexity statements are approximate code-level descriptions, not formal proofs. They describe how cost grows with input size and component counts.

Component	Approximate Complexity	Explanation
Text cleanup and line processing	O(n)	Scans the recovered text and lines, where n is text length.
Spatial PDF row grouping	Approximately O(w log w)	Word positions are grouped/sorted, where w is extracted word count.
NER chunking	O(c x model_cost)	Long CV text is split into c chunks; transformer inference dominates per chunk.
Skill canonicalization	O(s x k) for fuzzy/semantic comparison	s extracted skills may be compared against k configured/canonical names.
Domain classification	O(d) embedding comparisons	CV context is compared against d taxonomy domain descriptions.
Matching one job	O(s + r) plus embedding calls	Compares candidate skills with job requirements r and computes semantic/domain signals.
Ranking many jobs	O(j x match_cost)	j jobs require repeated matching or backend fallback comparison.
TF-IDF fallback	O(t) vectorization plus sparse cosine	Depends on token count t in the two texts.

Table 28. Computational complexity overview.

6.13 Raw CV Input to Structured Output Example

The example below is an illustrative walkthrough designed for examiner readability. It is not a live model benchmark. It uses a small CV fragment and a small job description to show how the three layers cooperate. Because the full transformer/OCR runtime dependencies were not available in the documentation Python environment, the JSON block is labeled as an illustrative schema example based on the actual schema.py response structure rather than a live model run.

Sanitized raw CV fragment:

```
Demo Student
Laravel Backend Developer
Skills: Laravel, MySQL, RESTful APIs
Experience: Backend Developer at Demo Company, 2025-2026
Education: Computer Science
```

Raw Evidence	Extracted Output	Schema Area
Laravel Backend Developer	Predicted role/title evidence.	profile.current_title, analysis.predicted_role
Laravel, MySQL, RESTful APIs	Hard technical skills.	skills.items[]
Backend Developer at Demo Company, 2025-2026	One experience item with company, title, period, and technologies.	experience.items[]
Computer Science	Education evidence.	Profile/education metadata when recovered
Text CV fragment	Successful text parsing path; no OCR needed in this example.	parsing_status, stats, analysis.metadata

Table 29. Raw CV fragment extraction example.

Illustrative sanitized output based on an actual analyzer response structure:


```
{
  "parsing_status": "success",
  "profile": {
    "full_name": "Demo Student",
    "current_title": "Laravel Backend Developer",
    "alternative_titles": ["Backend Developer"],
    "headline": "Professional Summary",
    "contact": {
      "email": "student@example.com", "phone": "+20XXXXXXXXXX",
      "location": "Giza, Egypt",
      "linkedin_url": "https://example.com/linkedin",
      "github_url": "https://example.com/github", "portfolio_url": null
    },
    "summary": "Redacted summary text for a backend-focused student CV.",
    "confidence_score": 0.93
  },
  "stats": {"page_count": 2, "char_count": 4110, "word_count": 498, "language_hint": null},
  "skills": {
    "items": [
      {"confidence_score": 0.65, "name": "Laravel", "category": "hard", "evidence": "ner"},
      {"confidence_score": 0.65, "name": "MySQL", "category": "hard", "evidence": "ner"},
      {"confidence_score": 0.65, "name": "RESTful APIs", "category": "hard", "evidence": "rule"}
    ],
    "confidence_score": 0.65
  },
  "experience": {
    "items": [
      {
        "confidence_score": 0.85, "title": "Backend Developer",
        "company": "Demo Company", "location": "Remote",
        "start_date": "2025-12-01", "end_date": "2026-01-01", "is_current": false,
        "description": ["Developed Laravel APIs and database-backed features."],
        "technologies": ["Laravel", "MySQL"]
      }
    ],
    "confidence_score": 0.85
  },
  "analysis": {
    "summary": null,
    "predicted_role": "Laravel Backend Developer",
    "seniority": "Intern",
    "primary_domain": "Full Stack Development",
    "strengths": ["Diverse technical portfolio with multiple backend technologies."],
    "gaps": [], "red_flags": [],
    "confidence_score": 0.75,
    "metadata": {
      "segmentation": {
        "found_sections": ["profile_summary", "experience", "projects", "skills", "education"],
        "sections_missing": [], "anomalies": []
      },
      "experience": {"total_experience_years": 0.08, "action_verb_score": 0.3, "gap_details": []},
      "extraction": {"source": "spatial", "spatial_status": "ok", "word_count_spatial": 499},
      "layer2": {
        "seniority_details": {"level": "Intern", "semantic_match": "Intern"},
        "categorized_skills": {
          "hard_skills": ["Laravel", "MySQL", "RESTful APIs"],
          "soft_skills": [], "management_skills": []
        },
        "domain_scores": {"Backend Development": 0.3338, "Full Stack Development": 0.4616}
      }
    },
    "domain_scores": {"Backend Development": 0.3338, "Full Stack Development": 0.4616}
  }
}
```


Section	Meaning
profile	Normalized identity, contact, title, headline, summary, and confidence data.
stats	Document-level counts such as pages, characters, words, and language hint.
skills	Extracted and categorized skill items with confidence and evidence source.
experience	Parsed work-history items, dates, descriptions, technologies, and confidence.

Section	Meaning
analysis	Predicted role, seniority, domain, strengths, gaps, red flags, and confidence.
analysis.metadata.segmentation	CV sections detected or missing during document understanding.
analysis.metadata.layer2	Classification details such as seniority reasoning, categorized skills, and domain scores.

Table 35. AI CV Analyzer output schema sections.

Classification	Result	Reason
Domain	Full Stack Development	The sanitized sample preserves the attached schema's multi-domain scoring style, with backend and frontend scores visible.
Seniority	Intern	Short recorded experience and the Layer 2 seniority detail support an early-career estimate.
Skill category	Hard technical skills	Laravel, MySQL, and RESTful APIs are technical implementation skills.

Table 30. Layer 2 interpretation example.

Example job: Junior Backend Developer requiring Laravel, MySQL, Docker, and REST APIs.

Matching Evidence	Result
Matched skills	Laravel, MySQL, Docker, REST APIs
Missing skills	None in this simplified example
Fit interpretation	Good illustrative fit for a junior backend role
Explanation	Skill overlap is strong; seniority appears compatible; final production score would require live matcher execution.

Table 31. Layer 3 matching evidence example.

6.14 Why Not Use a Direct LLM-Only Approach?

Direct LLM analysis can be powerful, especially for summarizing complex CVs and producing natural-language feedback. CareerCompass still avoids a direct LLM-only runtime because the graduation/demo system must be reproducible, inspectable, containerized, and privacy-aware. The Gemini-based code is used for synthetic training-data generation, not for sending private uploaded CVs to an external LLM during the normal runtime flow. The runtime analyzer is decomposed into layers so each part can be tested, explained, and improved independently.

Direct LLM-Only Approach	CareerCompass Hybrid Analyzer
Requires an external inference service for each private CV unless self-hosted.	Runs the analyzer locally/containerized in the demo stack.
Responses can vary across prompts, model versions, and temperatures.	Uses deterministic rules, typed schema validation, and explicit status values.

Direct LLM-Only Approach	CareerCompass Hybrid Analyzer
Harder to benchmark each sub-step because extraction, reasoning, and explanation are blended.	Layers can be inspected separately: text recovery, NER, rules, classification, matching, and explanation.
Privacy risk is higher if real CVs are sent to a remote service.	Runtime CV analysis can stay inside the deployed services.
Can explain textually but may hallucinate unsupported fields.	Outputs structured strengths, gaps, red flags, metadata, and confidence-style signals from code paths.
May be faster to prototype but harder to reproduce academically.	Fits Docker-based graduation evaluation and component-level evidence.

Table 32. AI approach comparison.

6.15 AI Analyzer Limitations and Future Work

The limitations below make the AI contribution more academically honest, not weaker. They identify exactly what should be improved before the system is treated as a stronger research or production artifact.

- OCR quality affects scanned or image-heavy CV extraction.
- Synthetic data may not cover all real CV styles, languages, layouts, and informal wording.
- The Colab PDF provides recorded NER training-run metrics, but the final cleaned NER dataset and exported model weights are not committed, so the run is not reproducible from repository files alone.
- More real or human-reviewed labeled CV data is needed for trustworthy per-label precision, recall, and F1.
- A larger fixed benchmark should evaluate parsing, role prediction, domain classification, seniority, matching, and gap analysis together.
- Arabic and multilingual CV support can be improved beyond the current UI localization.
- The role/domain taxonomy and skill alias catalog should expand through reviewed labor-market evidence.
- A model card and dataset card should be added for any exported model artifact.
- Human-in-the-loop review can improve reliability for important student guidance decisions.
- Privacy-preserving training and evaluation workflows should be designed before using real CV data.

6.16 AI Chapter Summary

The standalone AI chapter was added because the analyzer is a core project contribution. The system is best described as a transparent, layered hybrid analyzer for a graduation/demo environment. It does not claim production-grade AI accuracy, a certified hiring probability, or repository-alone reproducible final NER benchmarking. The new Colab PDF strengthens training-run evidence, while the academic value remains the integration of document recovery, NER, rules, canonicalization, classification, matching, explanation, and honest fallback behavior.

Chapter 7: AI Job Miner and Scraping Deep Technical Analysis

7.1 Purpose of Job Mining in CareerCompass

CareerCompass needs job mining because career guidance becomes weak when job data is static. A student's CV skills, predicted role, and gap-analysis output are useful only when compared against job descriptions that contain current requirements. The job-mining subsystem supplies those job descriptions to the recommendation and gap-analysis workflows, while the admin interface gives operators visibility into sources, target roles, imported jobs, and failures.

The chapter uses the term "job mining" instead of claiming unrestricted web scraping. The repository contains a deterministic local demo source, API adapters, HTML parser adapters, and a Scrapy spider path. These are source adapters for a graduation/demo system. They do not prove complete labor-market coverage, production-grade crawling reliability, or permission to scrape every configured website.

7.2 Scraping Design Philosophy

The implementation separates responsibilities deliberately. Laravel remains the trusted application backend and system of record. It owns authentication, authorization, job records, skill synchronization, admin controls, and user-facing APIs. The Python AI Job Miner service owns network-heavy adapter work, public API parsing, HTML parsing, Scrapy execution, adapter quality checks, and callback payload construction. Queue workers sit between them so slow and failure-prone network work does not block normal browser requests.

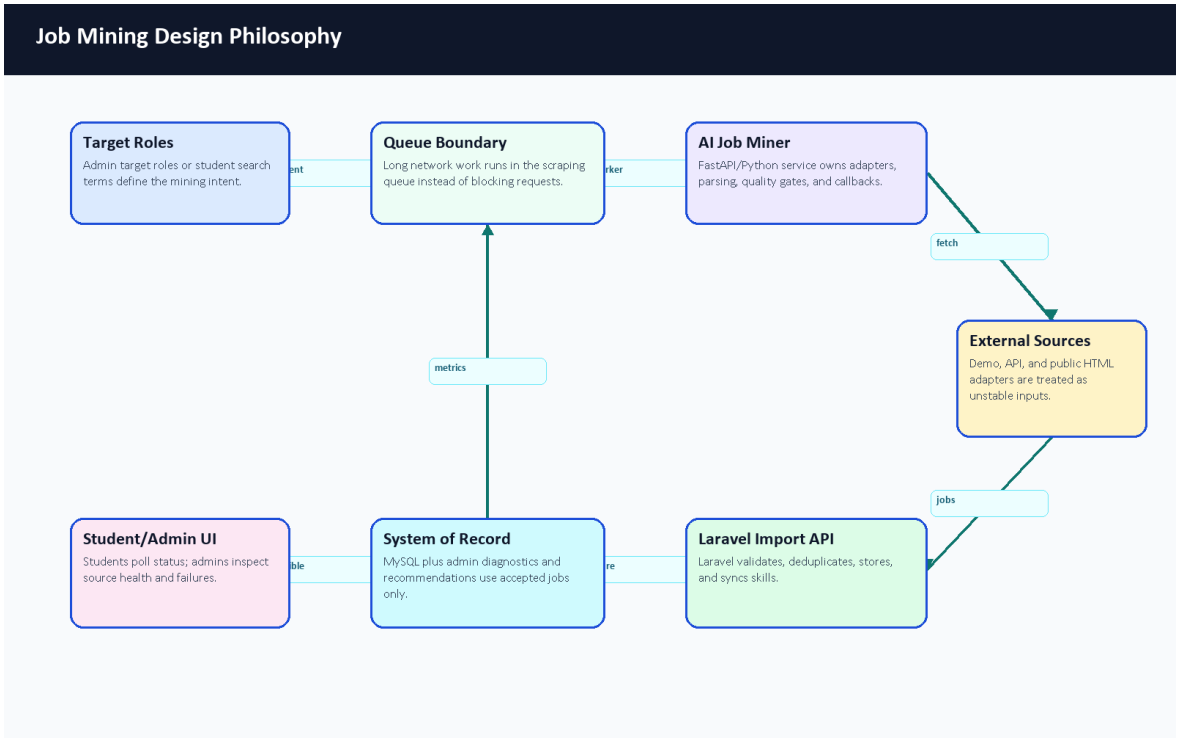


Figure 53. Job mining design philosophy.

Design Question	CareerCompass Decision	Reason
Why job mining?	Use imported job descriptions to support recommendations, gap analysis, and market context.	Static seed data becomes stale and cannot represent changing skill demand.

Design Question	CareerCompass Decision	Reason
Why Python/FastAPI?	Keep scraping/API ingestion dependencies in ai-job-miner.	Python has stronger parsing/scraping tooling and isolates unstable network work from Laravel.
Why Laravel as system of record?	Only Laravel validates, deduplicates, stores, and exposes accepted jobs.	Auth, admin controls, database consistency, and skill sync already belong to Laravel.
Why queues?	ProcessOnDemandJobScraping and market scraping jobs run on the scraping queue.	External sources can timeout, block, or return malformed data.
Why diagnostics?	Admin pages show source status, source tests, target roles, failed URLs, and batch progress.	Operators need evidence instead of assuming sources are healthy.

Table 36. Job mining design decisions.

7.3 AI Job Miner Runtime Architecture

The deployed runtime uses Docker Compose service separation. The AI Job Miner service is named cc-job-miner, maps host port 8003 to container port 8000, and exposes /health. Laravel reaches it through SCRAPER_SERVICE_URL, while the scraper calls Laravel callback endpoints through LARAVEL_API_BASE_URL. The production overlay also defines backend-worker-scraping, which runs the database queue with the scraping queue name and a longer timeout than ordinary request work.

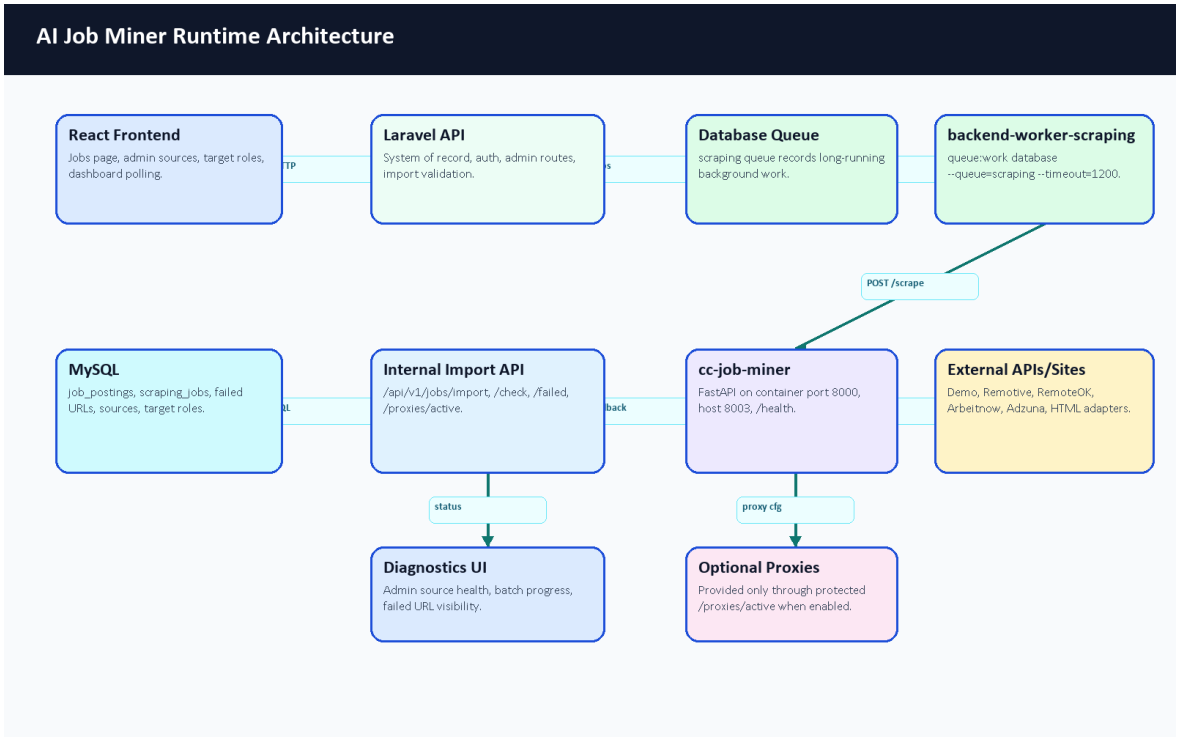


Figure 54. AI Job Miner runtime architecture.

Component	Implementation Evidence	Runtime Role
React frontend	frontend/src/api/endpoints.js, scrapingSources.js, admin pages	Starts user/admin actions and polls status.
Laravel API	JobController, ScrapedJobController, admin controllers	Creates jobs, protects routes, validates imports, and exposes results.

Component	Implementation Evidence	Runtime Role
Database queue	ProcessOnDemandJobScraping, ProcessMarketScrapingCategory	Runs slow scrape work outside request/response flow.
AI Job Miner	ai-job-miner/service_api.py	FastAPI adapter service with /health, /metrics, and protected /scrape.
Scrapy path	ai_job_miner/settings.py, linkedin_spider.py, pipelines	Public-page spider flow with robots obedience, delay, retries, dedupe, and Laravel export.
MySQL	migrations and models for jobs, sources, target roles, scraping jobs, failed URLs	Stores accepted jobs and operational state.
Optional proxies	InternalProxyController, SCRAPER_USE_PROXIES	Supplies active proxies only through a protected internal route when enabled.

Table 37. Scraping runtime component map.

7.4 Complete Job Mining Flow

The complete flow starts with a user search or an admin target role. Laravel checks whether usable stored jobs already exist. If stored data is enough for the user workflow, Laravel returns it. If not, Laravel creates a ScrapingJob record, dispatches a background worker, calls AI Job Miner, receives candidate jobs through protected import endpoints, deduplicates and stores them, syncs required skills, and exposes status through polling/dashboard endpoints.

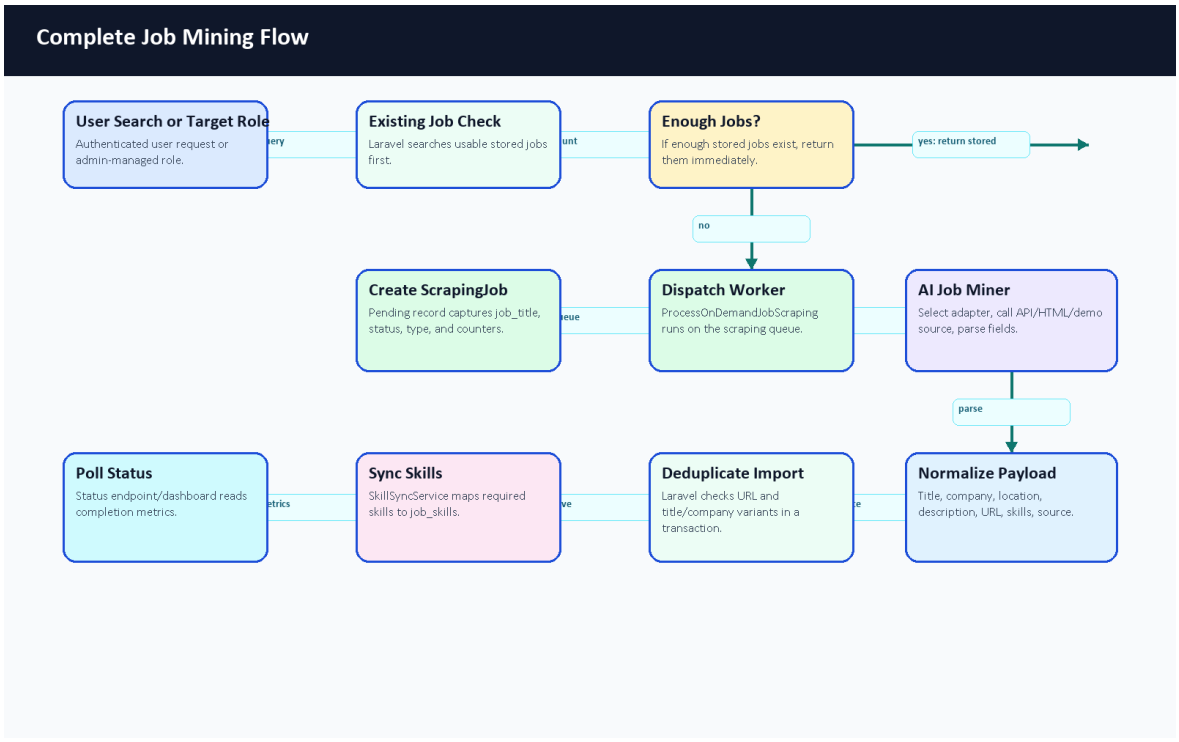


Figure 55. Complete job mining flow.

The important architectural point is that the Python service does not directly become the database owner. It is an ingestion service. Candidate jobs become CareerCompass jobs only after Laravel form requests validate the payload and the import transaction completes.

Figure 56. Scraping sequence diagram.

7.5 On-Demand Scraping and Status Polling

On-demand scraping is implemented in JobController. scrapeAndStore accepts a query and maximum result count, creates a pending ScrapingJob, dispatches ProcessOnDemandJobScraping, and returns the scraping job ID. scrapeJobTitleIfMissing first checks whether public usable jobs with a matching title already exist. If jobs exist, it returns data_exists: true; otherwise it queues a scrape and returns a polling URL. checkScrapingStatus returns the lifecycle state, counters, completed timestamp, matching stored jobs, or an error message.

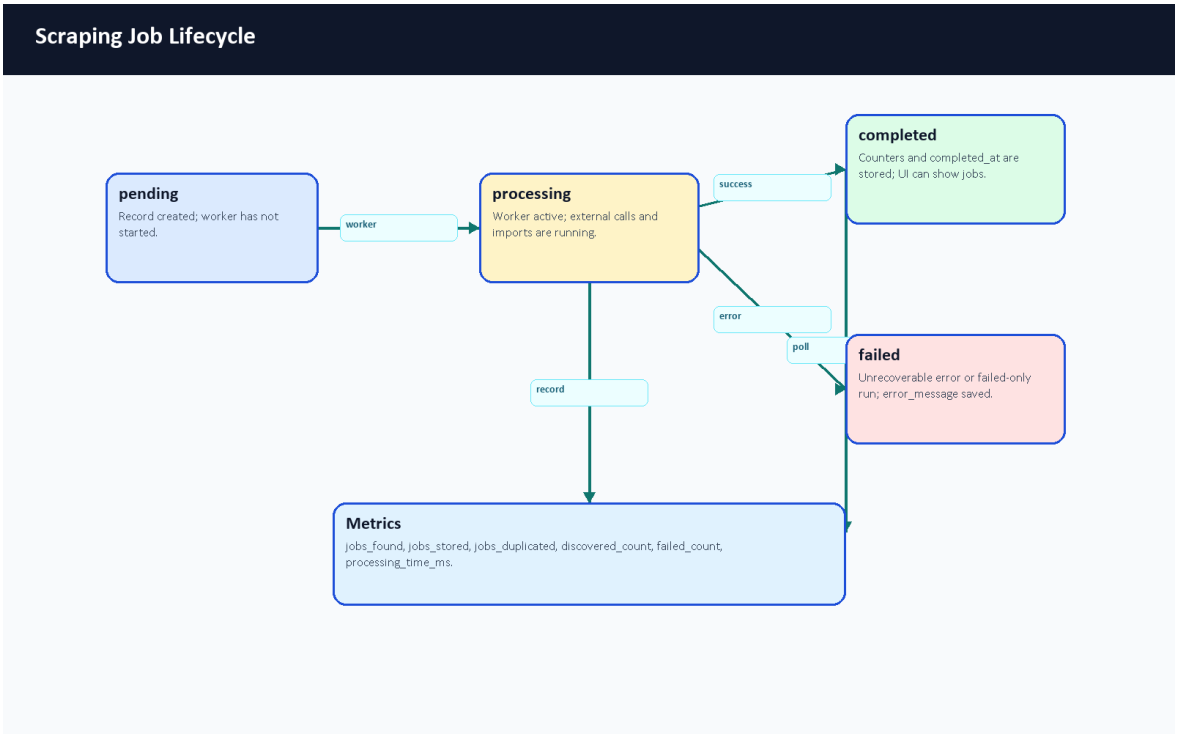


Figure 57. Scraping job lifecycle.

Status	Meaning	User/Admin Behavior
pending	The ScrapingJob record exists and is waiting for a worker.	Poll status and keep the UI non-blocking.
processing	The queue worker has started and external/import work is running.	Continue polling or show progress/admin diagnostics.
completed	The worker finished and stored counters such as jobs_found, jobs_stored, jobs_duplicated, discovered_count, failed_count, and processing_time_ms.	Display imported/stored jobs and final metrics.
failed	The run ended with an unrecoverable error or failed-only outcome and error_message is stored.	Show an error and use admin diagnostics/manual review.

Table 38. On-demand scraping lifecycle states.

7.6 Source Management and Target Roles

Admin source management is implemented through ScrapingSourceController, ScrapingSource, and AdminSources.jsx. Sources store endpoint, method, type, mode, status, headers, and params. The model computes adapter names and support metadata so the UI can distinguish demo/local sources, supported API adapters, missing credentials, external-risk HTML adapters, and unsupported configurations.

Target roles are implemented through TargetJobRoleController, TargetJobRole, and AdminTargets.jsx. A full scraping run combines active target roles with active/runnable sources. Unsupported sources and sources missing required credentials are skipped instead of being counted as successful.

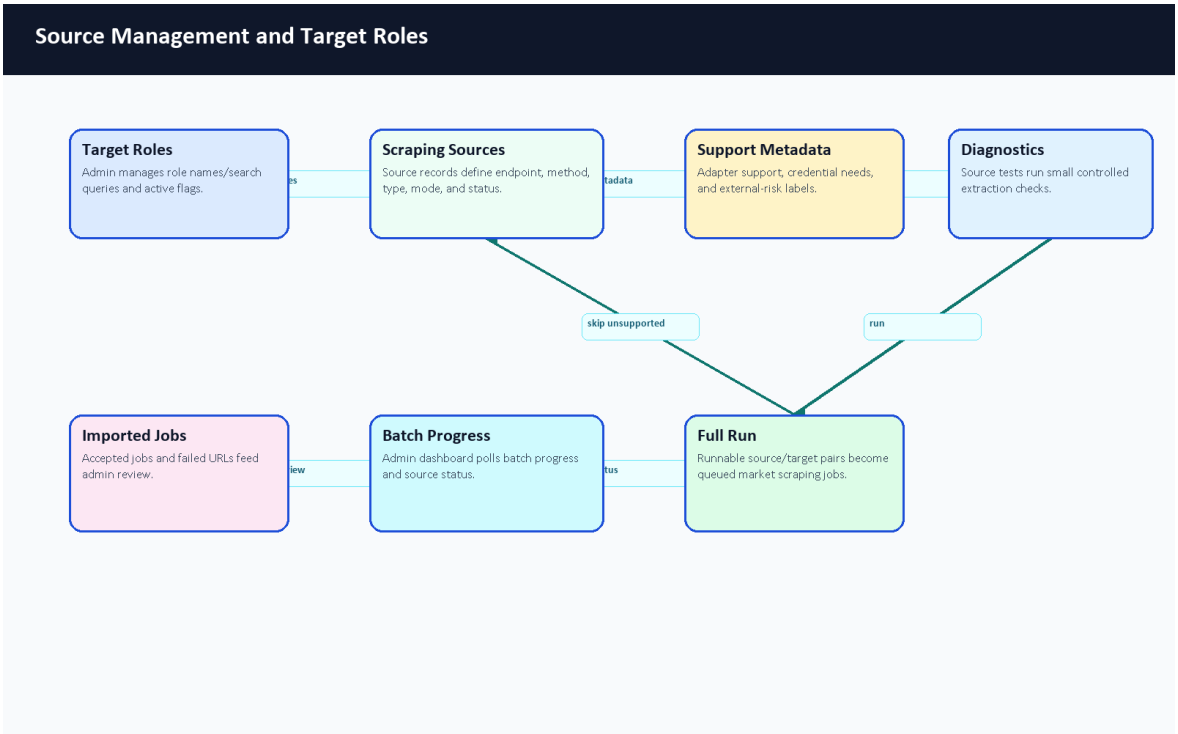


Figure 58. Source management and target-role flow.

Control	Code Evidence	Purpose
Source CRUD/status	ScrapingSourceController, StoreScrapingSourceRequest, UpdateScrapingSourceRequest	Manage source definitions and active/inactive state.
Source support metadata	ScrapingSource::supportMetadata()	Label demo, supported APIs, config-required sources, external-risk sources, and adapter-missing sources.
Source diagnostics	test, testSingle, runSourceDiagnostic	Run a small extraction and report support status, jobs stored/rejected, failures, and elapsed time.
Target roles	TargetJobRoleController, TargetJobRoleSeeder	Manage role names/search queries for market scraping.
Full scraping run	runFullScraping, ProcessMarketScrapingCategory	Queue source/target pairs and record per-run ScrapingJob status.
Admin evidence	Figures 44-47	Dashboard, jobs, source diagnostics, and target-role screens show the operator workflow.

Table 39. Source management and target-role controls.

The admin operational evidence is already shown in Figures 44-47: dashboard, jobs, source diagnostics, and target roles. This chapter refers to those screenshots instead of repeating them, so Chapter 7 can focus on architecture, flows, and implementation behavior.

7.7 Laravel Import Pipeline and Deduplication

The import pipeline is centered in ScrapedJobController::import. It runs inside a database transaction and applies a layered duplicate strategy. First it checks URL, which is the strongest available source identity. Then it checks title/company candidates, including original and title-case variants. Finally it checks squished lowercase title/company values. If a job already exists, Laravel updates it; otherwise it creates a

new job_postings record. SkillSyncService then normalizes and links required skills without detaching prior evidence.

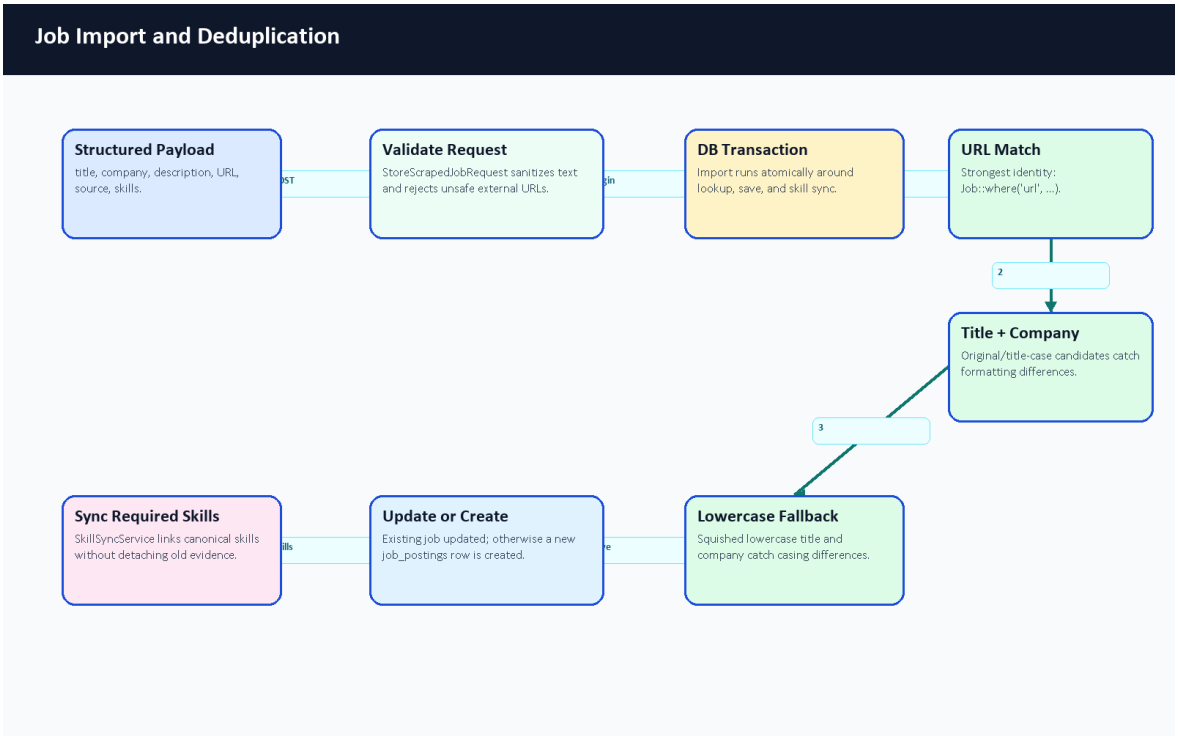


Figure 59. Job import and deduplication flow.

Deduplication Stage	Evidence	Reason
URL match	Job::where('url', ...)	Strongest available unique source identity.
Title/company variants	original title and title-case candidate with company	Catches common formatting differences.
Lowercase title/company	squished lowercase title and company comparison	Catches casing/spacing differences.
Update or create	Import runs inside DB::transaction	Keeps lookup, save, and skill sync atomic.
Skill sync	SkillSyncService::syncJobSkills(..., detaching: false)	Preserves and extends job-skill matching evidence.

Table 40. Import and deduplication stages.

The current duplicate strategy is appropriate for a demo system, but a stronger production importer should add source-specific IDs, canonical URLs, content hashes, expiration states, and reviewed merge rules.

7.8 Failed URL Handling and Dead Letter Queue

The failure path uses ScrapingFailedUrl records as a lightweight dead-letter style store. AI Job Miner reports failed source URLs to POST /api/v1/jobs/import/failed; Laravel validates the payload with ReportScrapingFailureRequest and stores URL, optional source/job IDs, error message, retried flag, and failed timestamp. Admin dashboard routes expose failed URLs for a scraping job.

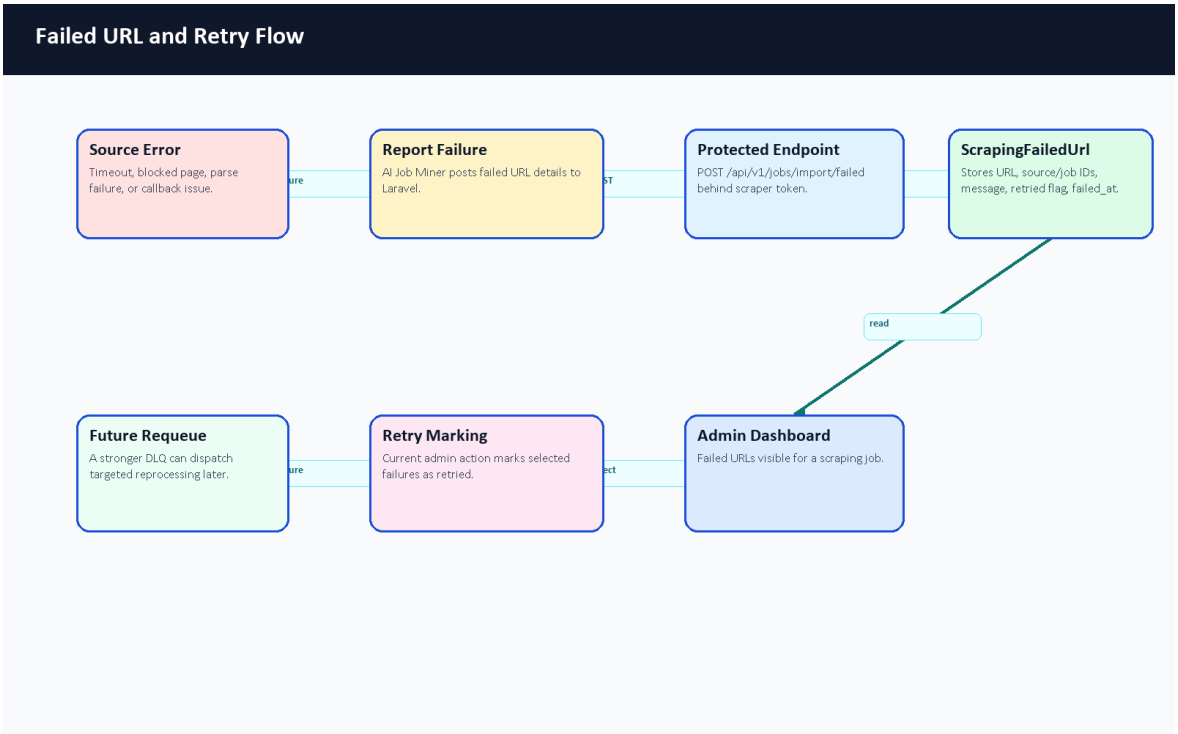


Figure 60. Failed URL and retry flow.

Failure Type	Handling	User/Admin Visibility
Timeout or network error	Operational category reported through failed URL callback when available.	Admin failed-URL list and source diagnostics.
Parse or quality failure	Adapter may classify empty, rejected, or data-quality failed outcomes.	Source diagnostic result and counters.
Duplicate candidate	Import check/import path tracks duplicate or non-created outcomes.	Batch/on-demand counters rather than a user-facing error.
Source disabled or unsupported	Source is skipped before full run or reported as adapter missing/config required.	Admin source status and planned/skipped run summary.
Missing internal token	scraper.token middleware rejects Laravel import callbacks.	Request rejected; should be reviewed through logs/config.

Table 41. Failed URL and operational failure handling.

The current retry-failures admin endpoint marks selected failed URLs as retried. It does not yet dispatch a targeted re-fetch job. The book therefore describes it as operational retry marking, not as a complete production DLQ processor.

7.9 Admin Diagnostics and Retry Operations

Admin diagnostics are more than a source list. ScrapingSourceController::runSourceDiagnostic creates a diagnostic ScrapingJob, calls the scraper with a small query, captures adapter classification, records elapsed time, and returns support metadata, job preview counts, quality rejections, failed URL counts, and an output excerpt. DashboardController exposes scraper health, batch progress, failed URLs, and retry

marking. This gives examiners a visible way to discuss what happened rather than only whether a scrape produced jobs.

The diagnostic design is especially important for external sources. Public APIs can require credentials, and websites can change HTML or block automated access. A mature demo should show those outcomes honestly: skipped, config required, blocked, empty, failed, partial, or successful.

7.10 Security, Tokens, Rate Limits, and Proxy Configuration

Scraping uses multiple security boundaries. User and admin actions go through authenticated Laravel routes. Laravel-to-miner calls use X-Scraper-Service-Token on AI Job Miner /scrape. Miner-to-Laravel callbacks use the protected scraper.token middleware and throttle:scraper; in the current Laravel middleware this is checked through the bearer token against SCRAPY_API_TOKEN. Import logs are redacted by ScrapedJobController::redactForLogs, and the Python service redacts token/API-key-like fields in adapter output.

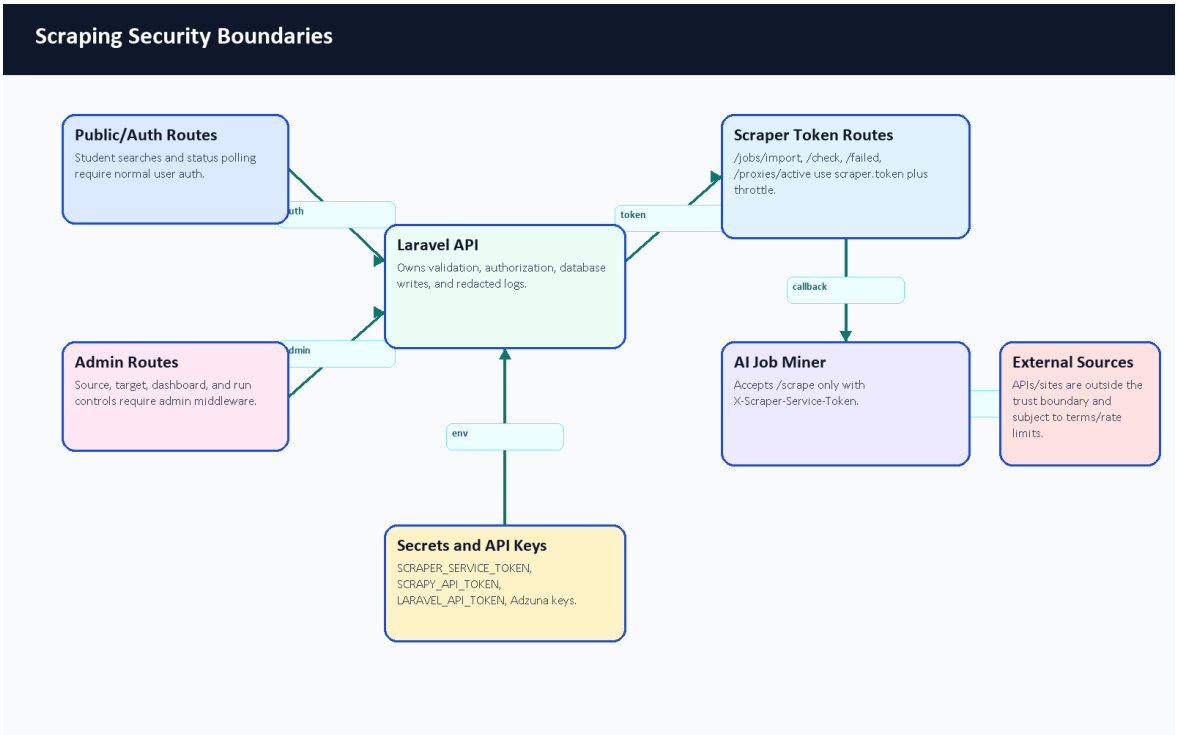


Figure 61. Scraping security boundaries.

Control	Code / Configuration Evidence	Purpose
Laravel import token	VerifyScraperToken, SCRAPY_API_TOKEN, LARAVEL_API_TOKEN	Protects /jobs/import, /jobs/import/check, /jobs/import/failed, and /proxies/active.
Laravel-to-miner token	SCRAPER_SERVICE_TOKEN, X-Scraper-Service-Token	Prevents public calls to AI Job Miner /scrape.
Scraper throttling	throttle:scraper route middleware	Limits protected callback traffic.
Secret redaction	redactForLogs, _sanitize_sensitive	Avoids logging tokens, API keys, app IDs, authorization headers, passwords, and secrets.
External API keys	ADZUNA_APP_ID, ADZUNA_APP_KEY	Enables Adzuna adapter when configured; missing keys are reported honestly.

Control	Code / Configuration Evidence	Purpose
Proxy configuration	SCRAPER_USE_PROXIES, /proxies/active	Optional operational feature; not a permission bypass or reliability guarantee.
Rate-limit configuration	SCRAPER_RATE_LIMIT_PER_MINUTE and Scrapy delay/retry settings	Documents conservative runtime policy, but source-specific terms still matter.
Robots and terms	Scrapy ROBOTSTXT_OBEY = True; RFC 9309 context [41]	Ethical scraping requires respecting source rules and terms.

Table 42. Scraping security and configuration controls.

7.11 Job Mining API Contracts

The scraping API surface has three groups: authenticated user endpoints, protected internal scraper endpoints, and admin endpoints. Detailed examples are included in Appendix A so maintainers can reuse them without exposing real tokens. The examples follow an OpenAPI-style documentation pattern [39].

Group	Method and Path	Auth / Middleware	Purpose
User/Auth	POST /api/v1/jobs/scrape	Authorization: Bearer <user-token>	Queue on-demand scraping for a query.
User/Auth	POST /api/v1/jobs/scrape-if-missing	Authorization: Bearer <user-token>	Return existing jobs or queue scraping when missing.
User/Auth	GET /api/v1/scraping-status/{jobId}	Authorization: Bearer <user-token>	Poll lifecycle state and counters.
Internal scraper	POST /api/v1/jobs/import/check	Authorization: Bearer <internal-token>	Check duplicate URL before import.
Internal scraper	POST /api/v1/jobs/import	Authorization: Bearer <internal-token>	Validate, deduplicate, save/update job, and sync skills.
Internal scraper	POST /api/v1/jobs/import/failed	Authorization: Bearer <internal-token>	Store failed source URL evidence.
Internal scraper	GET /api/v1/proxies/active	Authorization: Bearer <internal-token>	Return active proxy definitions when enabled.
Scraper service	POST /scrape on AI Job Miner	X-Scraper-Service-Token: <internal-token>	Execute adapter work for Laravel worker.
Admin	/api/v1/admin/scraping-sources*, /api/v1/admin/target-roles*, /api/v1/admin/scraping/run-full	User token plus admin middleware	Manage sources, target roles, diagnostics, and full runs.

Table 43. Job mining API contract summary.

On-demand request example:

```
{
  "query": "Backend Developer",
  "max_results": 10
}
```

Status response example:

```
{
  "success": true,
```

```
  "status": "completed",
  "jobs_found": 8,
  "jobs_stored": 5,
  "jobs_duplicated": 3,
  "failed_count": 0,
  "processing_time_ms": 12640
}
```

Internal import example:

```
{
  "title": "Junior Backend Developer",
  "company": "Example Co",
  "location": "Remote",
  "description": "Build APIs with Laravel and MySQL.",
  "requirements": "Laravel, MySQL, REST APIs",
  "url": "https://example.com/jobs/123",
  "source": "remotive",
  "scraping_source_id": 5,
  "skills": ["Laravel", "MySQL", "REST APIs"]
}
```

7.12 Scraping Evaluation and Validation Evidence

The strongest current scraping evidence is architectural and test evidence, not live market coverage evidence. The repository includes AI Job Miner tests for service auth, health, metrics, adapter parsing, classification, redaction, blocked/empty outcomes, and skill extraction helpers. Laravel validates imports through form requests and transactions. Docker Compose wires the ai-job-miner service, long-running scraping worker, tokens, and callback URLs.

Figure 62. Scraping validation evidence.

Evidence	Result to Record	What It Proves	Limitation

Evidence	Result to Record	What It Proves	Limitation
python -m compileall ai-job-miner	Passed using the bundled Python runtime.	Python syntax/importability for the service files.	Not runtime source success.
python -m pytest in ai-job-miner	Passed inside the running container: 75 tests, 1 warning.	Service logic and adapter parser behavior under tests.	Mocked tests do not prove real website availability.
/health and /metrics	AI Job Miner /health returned 200 with {"status":"ok","service":"CareerCompass Job Miner"}; /metrics returned Prometheus-style scraper counters.	FastAPI job-miner service is alive and exposes basic scraper metrics.	Not source coverage or data quality.
Docker Compose config	Passed for base plus production overlay.	Service wiring, tokens, workers, and ports are valid YAML/config.	Not external scraping success.
Deterministic demo scrape	Protected /scrape call using CareerCompass Demo Jobs returned SUCCESS, previewed 3 jobs, stored 3, and reported 0 failed URLs; smoke rows were cleaned afterward.	Demo adapter and Laravel import path can work together without external websites.	Direct service call bypassed the queue worker, so the temporary ScrapingJob status stayed pending and was not status-polling evidence.
Import API validation	Documented from form requests and controller code.	Laravel accepts structured payloads and rejects unsafe data.	Not a complete benchmark.
Admin diagnostics screenshots	Figures 44-47.	Admin UI supports source, job, dashboard, and target-role operations.	Point-in-time demo evidence.

Table 44. Scraping validation evidence.

The final smoke test used a direct protected /scrape request to validate the deterministic demo adapter and Laravel import path. A full authenticated /jobs/scrape-if-missing queue lifecycle with browser polling remains a recommended final demonstration check because it exercises the user-facing queue trigger and status-polling path rather than only the internal scraper contract.

Component	Primary Evidence	Summary
FastAPI service/API layer	ai-job-miner/service_api.py	Provides /health, /metrics, and protected /scrape orchestration.
Demo/local adapter	CareerCompass Demo Jobs source path and tests	Deterministic source used for safe smoke evidence without external websites.
Adzuna/API adapter	Adzuna source code and environment keys	Optional external API source when credentials and quotas are configured.
HTML/Scrapy-related path	ai_job_miner/settings.py, spiders, pipelines	Public-page crawling path with robots/delay/retry configuration boundaries.
Laravel JobController	JobController	Starts on-demand scraping and exposes status polling.
Laravel ScrapedJobController	ScrapedJobController	Validates imports, checks duplicates, records failures, and redacts logs.
Queue jobs	ProcessOnDemandJobScraping, market scraping jobs	Move long network tasks out of browser request time.
Skill sync	SkillSyncService	Connects imported job requirements to canonical skills.

Component	Primary Evidence	Summary
Admin source/target controllers	ScrapingSourceController, TargetJobRoleController	Manage active sources, diagnostics, target roles, and full-run triggers.
Frontend admin/user pages	frontend/src/pages/admin, user job pages	Surface jobs, sources, targets, status, and retry/diagnostic views.

AI Job Miner source and function inventory summary.

No source coverage percentage, success rate, or every-job-board claim is made. External-source behavior should be retested shortly before the final defense if the team wants live demonstration evidence.

7.13 Limitations, Ethics, and Future Work

The scraping subsystem is useful because it connects CV analysis to job requirements, but it must remain academically honest. Public websites can change HTML, block requests, or impose rules. APIs can require credentials and quotas. Proxy usage can introduce reliability and compliance risks. Stored jobs can become stale. Duplicate detection can be improved.

Area	Current Boundary	Future Work
Source coverage	Demo/local source is deterministic; external adapters are partial and unstable.	Add reviewed source policies and source-specific adapters.
External terms	Code cannot prove permission for every source.	Record robots/terms review per source before enabling external runs.
Rate limits	Scrapy delay/retry settings and rate-limit config exist.	Add per-source rate-limit dashboard and enforcement.
API keys	Adzuna uses environment credentials when configured [40].	Add secret rotation and key-status diagnostics.
Proxies	Optional protected proxy route exists.	Use only compliant proxies and document source permission.
Duplicate detection	URL and title/company transaction logic.	Add canonical IDs, URL normalization, and content hashes.
Data freshness	Imported jobs remain until managed.	Add expiration, archival, and human review queue.
Failed URLs	Failed URL records and retry marking.	Add targeted DLQ reprocessing with attempt counts.
Evaluation	Tests and structural validation.	Add reproducible live-source health checks without inflated claims.

Table 45. Scraping limitations, ethics, and future work.

Ethical operation should respect robots.txt and website/API terms, prefer official APIs where available, avoid private/login/CAPTCHA bypasses, keep request rates conservative, and avoid presenting imported data as exhaustive or assured. This framing keeps the system useful for a graduation demonstration while making its real boundaries clear.

Chapter 8: Testing and Evaluation

8.1 Introduction

Evaluation was performed using repository-aware commands and browser evidence. The goal was to verify that each major part of the graduation/demo system runs and to document limitations honestly.

8.2 Testing Strategy

The testing strategy combined automated tests, build checks, configuration checks, service health probes, and manual functional evaluation. Automated tests provide repeatable evidence. Screenshots provide visible workflow evidence. Manual tables document behavior that is difficult to fully automate in the available environment.

8.3 Backend Testing

Backend validation was executed inside the backend container. Composer dependencies were already installed. php artisan config:clear, php artisan route:list, migrations, and tests passed. The route list confirmed 131 routes. The Laravel test suite passed with 39 tests and 297 assertions.

8.4 Frontend Testing

The frontend was validated using the existing frontend/node_modules and the bundled Node runtime. ESLint passed with 9 warnings and 0 errors. The warnings were related to React fast-refresh export conventions and hook dependency notes. The Vite production build passed and transformed 2904 modules.

8.5 Python Services Testing

Python syntax compilation passed for both AI services. The AI Job Miner pytest suite was rerun inside the running ai-job-miner Docker container and passed with 75 tests and 1 warning. The local bundled Python runtime still did not include pytest, so the successful pytest evidence is specifically container-based. The AI CV Analyzer pytest suite was not rerun in this cleanup pass; AI CV Analyzer syntax compilation passed, and earlier smoke/Colab evidence remains documented separately.

8.6 Docker and Integration Testing

Docker Desktop was initially unavailable to the shell, then was started through docker desktop start. Docker Compose configuration validation passed for the base plus production overlay files. docker compose up -d was used without a rebuild to start the existing local stack. After startup settled, docker compose ps showed the main app containers running, with backend, frontend, Nginx, job miner, database, and queue workers healthy. Health probes returned 200 for /api/health, /api/ready, /status, AI Job Miner /health, and the AI CV Analyzer root endpoint. These checks prove service availability in the local demo stack, not external source reliability.

Figure 49. Validation evidence summary.

8.6.1 Module Validation Coverage Matrix

The following matrix summarizes validation coverage by system area. Items marked as previously passed are preserved from the latest documented validation notes rather than claimed as fresh reruns in this backend/frontend/database polish pass.

System Area	Evidence	Latest Result	Limitation
Laravel backend	Container route list, migrations, and test suite evidence.	Previously passed: 131 routes and 39 tests / 297 assertions.	Not rerun in this polish pass because application code was not changed.
Database, migrations, and seeders	Migration inspection, ERD review, and backend migration/test evidence.	Schema relationships and constraints were re-reviewed against migrations.	A destructive fresh migration reset was not performed in this pass.
React frontend	ESLint and Vite production build evidence.	Previously passed with 9 lint warnings, 0 errors, and 2904 Vite modules transformed.	Frontend source was inspected; build was not rerun unless noted in generation notes.
AI CV Analyzer	compileall, Colab output PDF, model-analysis notes, and API examples.	Syntax compilation passed in this validation; Colab metrics remain recorded evidence.	Full training and live model inference were not rerun from repository files alone.
AI Job Miner	Container pytest, compileall, service health, and deterministic demo scrape.	75 tests passed with 1 warning in container validation.	Tests do not prove live external website availability.
Docker runtime	Compose config and local health probes.	Compose config and local health endpoints passed in this validation pass.	Health checks prove availability, not business correctness or external-source success.
API health/readiness	/api/health, /api/ready, /status, service health endpoints.	Returned 200 in this validation pass.	These are shallow liveness/readiness probes.

System Area	Evidence	Latest Result	Limitation
PDF/DOCX generation	Generator run, structural checks, link/bookmark inspection, and PDF page count.	Revalidated after every documentation generation pass.	Visual manual review is still recommended for final submission.
GitHub Actions	Workflow files and PR checklist.	Manual PR review remains required after push.	CI status depends on GitHub scheduling after branch updates.

Table 46. Module validation coverage matrix.

8.7 CI/CD Validation

GitHub Actions workflow files were reviewed as part of repository inspection. A live GitHub Actions status screenshot was not captured before the draft PR because PR checks only become meaningful after the branch is pushed and GitHub schedules workflows. The manual review checklist asks the team to inspect CI status on the opened draft PR.

8.8 AI CV Analyzer Model Evidence

The AI CV Analyzer training workflow was inspected from repository files, helper documentation, and the exported Colab training-results PDF. Full model training was not rerun during this documentation update because the generator requires external Gemini API keys, the cleaned training dataset content is not committed, the runtime dependencies for transformer inference were not installed in the bundled documentation Python environment, and the notebook is designed for a Colab/T4-style GPU runtime.

The important distinction is reproducibility scope. The repository alone still does not provide the final labeled evaluation dataset, final model weights, or a one-command reproducible training run. However, the user-provided Colab PDF does provide recorded output cells from the actual notebook run, including train/test counts and overall epoch metrics. Those values are reported as Colab-run training evidence on the generated/synthetic notebook split, not as production accuracy. A local ignored artifact folder was present on this workstation and safe metadata was inspected, but model binaries remain outside Git. The mini evaluation below remains useful for regression-style demonstration, but it is separate from the Colab NER training metrics.

Evidence Item	Status	What It Proves	What It Does Not Prove
Runtime NER artifact path	Code checks ai-cv-analyzer/models/ner_weights/career_compass_ner_final; the folder is ignored by Git.	The service can load a local token-classification model when deployed.	It does not prove the artifact is committed or provide a final held-out F1 score.
Local ignored metadata	config.json and tokenizer metadata were inspected locally without copying weights.	The local artifact uses a BERT token-classification configuration and cased tokenizer.	It is not a portable repository artifact.
Training notebook	Present under ai-cv-analyzer/training/train_ner.ipynb	The training process, label map, token alignment, Trainer setup, metrics code, and export steps are documented.	It does not include the cleaned dataset or model weights.

Evidence Item	Status	What It Proves	What It Does Not Prove
Colab training-results PDF	Exported PDF under docs/graduation-book/model-analysis/colab_train_ner_results.pdf	Shows recorded NER fine-tuning/evaluation outputs: 41,319 train rows, 4,592 test rows, and final epoch precision 0.933307, recall 0.940521, F1 0.936900, accuracy 0.976376.	It does not prove readiness for real deployment and may not be reproducible without the same dataset, runtime, and model artifacts.
Dataset generator	Present under ai-cv-analyzer/training/generate_tech_dataset.py	Synthetic labeled data can be generated from Gemini with key rotation and negative decoys.	It was not run here because it requires API keys and would generate a large dataset.
Dataset cleaner	Present under ai-cv-analyzer/training/clean_dataset.py	Dataset normalization, deduplication, and span validation are part of the workflow.	The cleaned dataset file itself is not committed.
API tests	Present under ai-cv-analyzer/tests/test_service_api.py	FastAPI status handling and hybrid-match formula behavior are covered with fakes.	These tests do not measure real NER model accuracy.
Dependency probe	Local bundled Python lacked transformers, torch, sentence_transformers, OCR/PDF packages, and Gemini client libraries.	Explains why live model inference and training were not rerun during documentation generation.	It does not reflect what the Docker image may install at runtime.
Mini evaluation	Generated under docs/graduation-book/evaluation/	Synthetic skill/recommendation/gap logic can be checked repeatably.	It is not a statistical live-model benchmark.

Table 47. Model evaluation evidence.

8.9 NER Extraction Examples

The table below documents expected extraction behavior from the inspected NER labels and runtime post-processing. It is intentionally marked as example evidence rather than measured per-label accuracy because transformer dependencies were not available in the local documentation runtime and the Colab PDF reports overall metrics rather than a per-label classification report.

Example CV Text	Expected NER Entities	Downstream Use	Evidence Type
Experienced Backend Developer with Laravel, Docker, and MySQL.	ROLE: Backend Developer; SKILL: Laravel, Docker, MySQL	Predicted role, extracted skills, backend/domain matching.	Illustrative example from label schema and code path.
Graduated from Faculty of Computers and Information, Kafr El-Sheikh University.	EDU: Faculty of Computers and Information, Kafr El-Sheikh University	Education/profile evidence.	Illustrative example from EDU label behavior.
AWS Cloud Practitioner certified with Kubernetes deployment experience.	CERT: AWS Cloud Practitioner; SKILL: Kubernetes	Certification and cloud/DevOps skill evidence.	Illustrative example from CERT/SKILL labels.

Example CV Text	Expected NER Entities	Downstream Use	Evidence Type
Leadership, communication, and teamwork across agile projects.	SOFT labels exist in training setup; runtime grouping mainly returns SKILL/ROLE/EDU/CERT.	Soft-skill interpretation is handled mostly by taxonomy and rule layers.	Limitation observed from runtime grouping code.

Table 48. NER extraction examples.

8.10 Semantic Matching vs TF-IDF Fallback Examples

The semantic matching path could not be executed locally during this documentation update because sentence-transformer dependencies were unavailable in the bundled Python environment. The pure Python TF-IDF matcher was executed directly as a small deterministic fallback check. It gave a positive score for overlapping backend skills and zero for an unrelated mobile-role comparison.

Pair	Semantic Path Status	TF-IDF Fallback Result	Interpretation
CV: Laravel Docker MySQL REST APIs; Job: Backend developer with Laravel Docker MySQL	Not executed locally; dependencies unavailable.	0.4316	Keyword overlap confirms a backend-oriented match signal.
CV: Flutter Dart mobile UI; Job: Backend developer with Laravel Docker MySQL	Not executed locally; dependencies unavailable.	0.0000	No meaningful keyword overlap, so fallback does not inflate score.
Expected runtime behavior	Sentence embeddings plus TF-IDF in /api/hybrid-match when both paths are available.	60 percent semantic/adaptive plus 40 percent TF-IDF in that endpoint.	The fallback helps explainable matching but is not a substitute for full semantic evaluation.

Table 49. Semantic matching and TF-IDF example results.

8.11 Model Evaluation Limitations

- The Colab PDF provides recorded overall training-run metrics, but the repository alone still does not include the dataset/model artifacts needed to reproduce the run.
- The cleaned labeled dataset used for final training is not committed.
- The model-weight folder is ignored by Git; safe local metadata was inspected, but binary weights were not copied or benchmarked.
- Local documentation Python did not include transformer, sentence-transformer, OCR, PDF, or Gemini packages, so live model inference and training were not rerun here.
- The Colab PDF does not show a per-label classification report, per-label support counts, or a confusion matrix.
- The examples in Table 48 are expected-behavior examples, while the TF-IDF values in Table 49 are actual small local fallback checks.
- A stronger final defense package should add a fixed labeled CV test set, saved per-label NER metrics, and CI-friendly inference smoke tests.

8.12 CV Analyzer Mini Dataset Evaluation

A sample PDF CV was generated for the screenshot workflow and uploaded through the running system. The upload succeeded, and the dashboard showed parsed CV data, backend role inference, extracted skills, and profile completeness. To strengthen the evaluation beyond that smoke test, this revision adds a mini synthetic dataset under docs/graduation-book/evaluation/.

The mini CV evaluation is explicitly offline and deterministic. It uses fake CV text, expected skill labels, and a keyword/role inference evaluator. It does not claim live model accuracy. The live AI CV Analyzer endpoint can be added to this mini-evaluation later, but the current document records only metrics that were actually computed from the synthetic dataset.

8.13 Recommendation Mini Dataset Evaluation

The recommendation mini evaluation ranks synthetic jobs for each synthetic CV using skill overlap plus domain and seniority bonuses. This validates the recommendation concept and provides a repeatable regression check for report evidence. It is not a production recommender benchmark, and the report does not claim complete job-market coverage.

8.14 Gap Analysis Mini Dataset Evaluation

The gap-analysis mini evaluation compares expected matched and missing skills with computed matched and missing skills for selected CV/job pairs. This directly validates the explanation structure used by the gap-analysis workflow: matched skills should be shown separately from missing skills.

Mini Dataset Files

The mini evaluation uses fake synthetic CVs and fake synthetic job records stored under docs/graduation-book/evaluation/. It is intentionally small and preliminary. It is useful for graduation validation and regression checks, but it is not statistically representative and should not be used as a production benchmark.

Sample ID	Expected Role	Seniority	Domain	Expected Skills
cv_backend_laravel	Backend Laravel Developer	junior	backend_web	PHP, Laravel, MySQL, REST API, Docker, Git
cv_frontend_react	Frontend React Developer	intern	frontend_web	React, JavaScript, Vite, HTML, CSS, API integration
cv_data_ml	Data and ML Student	student	data_ml	Python, pandas, scikit-learn, NLP, data analysis
cv_full_stack	Full Stack Developer	junior	full_stack_web	Laravel, React, MySQL, Docker, REST API, Git
cv_qa_testing	QA Testing Engineer	intern	quality_assurance	testing, test cases, pytest, API testing, bug reporting

Table 50. Mini CV dataset.

Job ID	Title	Domain	Required Skills
job_laravel_backend	Junior Laravel Backend Developer	backend_web	PHP, Laravel, MySQL, REST API, Docker, Git
job_react_frontend	React Frontend Intern	frontend_web	React, JavaScript, Vite, HTML, CSS, API integration

Job ID	Title	Domain	Required Skills
job_full_stack_web	Full Stack Web Developer	full_stack_web	Laravel, React, MySQL, Docker, REST API, Git
job_data_analyst	Junior Data Analyst	data_ml	Python, pandas, scikit-learn, NLP, data analysis
job_qa_intern	QA Automation Intern	quality_assurance	testing, test cases, pytest, API testing, bug reporting
job_devops_docker	DevOps Docker Intern	devops	Docker, Git, CI, Linux, monitoring
job_php_api	PHP API Developer	backend_web	PHP, Laravel, REST API, MySQL, Git, Docker
job_nlp_assistant	NLP Assistant Intern	data_ml	Python, NLP, scikit-learn, data analysis, testing

Table 51. Mini job dataset.

Metric Definitions

Skill precision measures how many extracted skills are expected labels. Skill recall measures how many expected skills were extracted. Skill F1 is the harmonic mean of precision and recall [30]. Recommendation top-1 and top-3 relevance compare ranked jobs against manual relevance labels. Gap agreement compares computed matched/missing skills against expected matched/missing skills.

Area	Metric	Value	Notes
CV offline	Macro skill precision	1.000	Keyword extraction over synthetic CV text
CV offline	Macro skill recall	1.000	Compared with expected skill labels
CV offline	Macro skill F1	1.000	F1 computed from precision/recall [30]
CV offline	Role match rate	1.000	Rule-based role inference on synthetic data
CV offline	Seniority match rate	1.000	Rule-based seniority inference
CV offline	Domain match rate	1.000	Rule-based domain inference
Recommendation offline	Top-1 relevance	1.000	Top recommendation belongs to manual relevant set
Recommendation offline	Top-3 relevance	1.000	Any top-3 job belongs to manual relevant set
Recommendation offline	Mean precision@3	0.800	Relevant jobs among top three
Gap offline	Matched skill agreement F1	1.000	Computed matched skills vs. expected matched skills
Gap offline	Missing skill agreement F1	1.000	Computed missing skills vs. expected missing skills

Table 52. Mini evaluation metrics.

Recommendation Ranking Details

CV Sample	Expected Relevant Jobs	Top 3 Offline Recommendations	P@3
cv_backend_laravel	job_laravel_backend, job_php_api, job_full_stack_web, job_devops_docker	job_laravel_backend, job_php_api, job_full_stack_web	1.000

CV Sample	Expected Relevant Jobs	Top 3 Offline Recommendations	P@3
cv_frontend_react	job_react_frontend, job_full_stack_web	job_react_frontend, job_full_stack_web, job_qa_intern	0.667
cv_data_ml	job_data_analyst, job_nlp_assistant	job_data_analyst, job_nlp_assistant, job_laravel_backend	0.667
cv_full_stack	job_full_stack_web, job_laravel_backend, job_php_api, job_react_frontend, job_devops_docker	job_full_stack_web, job_laravel_backend, job_php_api	1.000
cv_qa_testing	job_qa_intern, job_nlp_assistant	job_qa_intern, job_nlp_assistant, job_react_frontend	0.667

Table 53. Recommendation ranking details.

Gap Analysis Pair Details

CV / Job Pair	Matched Skills	Missing Skills	Agreement
cv_backend_laravel -> job_laravel_backend	Docker, Git, Laravel, MySQL, PHP, REST API	None	matched F1=1.000; missing F1=1.000
cv_frontend_react -> job_full_stack_web	React	Docker, Git, Laravel, MySQL, REST API	matched F1=1.000; missing F1=1.000
cv_data_ml -> job_nlp_assistant	NLP, Python, data analysis, scikit-learn	testing	matched F1=1.000; missing F1=1.000
cv_qa_testing -> job_qa_intern	API testing, bug reporting, pytest, test cases, testing	None	matched F1=1.000; missing F1=1.000
cv_full_stack -> job_react_frontend	React	API integration, CSS, HTML, JavaScript, Vite	matched F1=1.000; missing F1=1.000

Table 54. Gap analysis pair details.

8.15 AI CV Analyzer Smoke Evaluation

The smoke evaluation under docs/graduation-book/evaluation/ uses five short, fake CV text samples: backend, data analyst, frontend, DevOps/cloud, and low-information/noisy input. It is deterministic and useful as a small reproducibility check. It does not run the full transformer NER model and must not be reported as final model accuracy.

The script also probes the local documentation runtime. In this run, full analyzer import was unavailable because ModuleNotFoundError: No module named 'pdfplumber'. The pure Python TF-IDF matcher was available, with a backend-overlap probe score of 0.5101.

Package	Status
easyocr	Unavailable
pdfplumber	Unavailable

Package	Status
pydantic	Available
sentence_transformers	Unavailable
torch	Unavailable
transformers	Unavailable

Dependency probe for the AI CV Analyzer smoke evaluation.

Area	Metric	Value	Notes
AI analyzer smoke	Macro skill precision	0.971	Five manually labeled text samples
AI analyzer smoke	Macro skill recall	1.000	Expected skill labels defined in smoke sample JSON
AI analyzer smoke	Macro skill F1	0.985	Deterministic text smoke metric, not NER benchmark
AI analyzer smoke	Role match rate	1.000	Simple role inference over sample text
AI analyzer smoke	Domain match rate	1.000	Simple domain evidence markers
AI analyzer smoke	Seniority match rate	0.800	One mismatch preserved as evidence of limitation
AI analyzer smoke	Parsing status match rate	1.000	Includes low-information abstention sample

AI CV Analyzer smoke evaluation metrics.

Sample	Skill F1	Role	Domain	Seniority	Status
smoke_backend_laravel	1.000	Pass	Pass	Check	Pass
smoke_data_analyst	1.000	Pass	Pass	Pass	Pass
smoke_frontend_react	0.923	Pass	Pass	Pass	Pass
smoke_devops_cloud	1.000	Pass	Pass	Pass	Pass
smoke_low_information	1.000	Pass	Pass	Pass	Pass

Per-sample AI CV Analyzer smoke evaluation results.

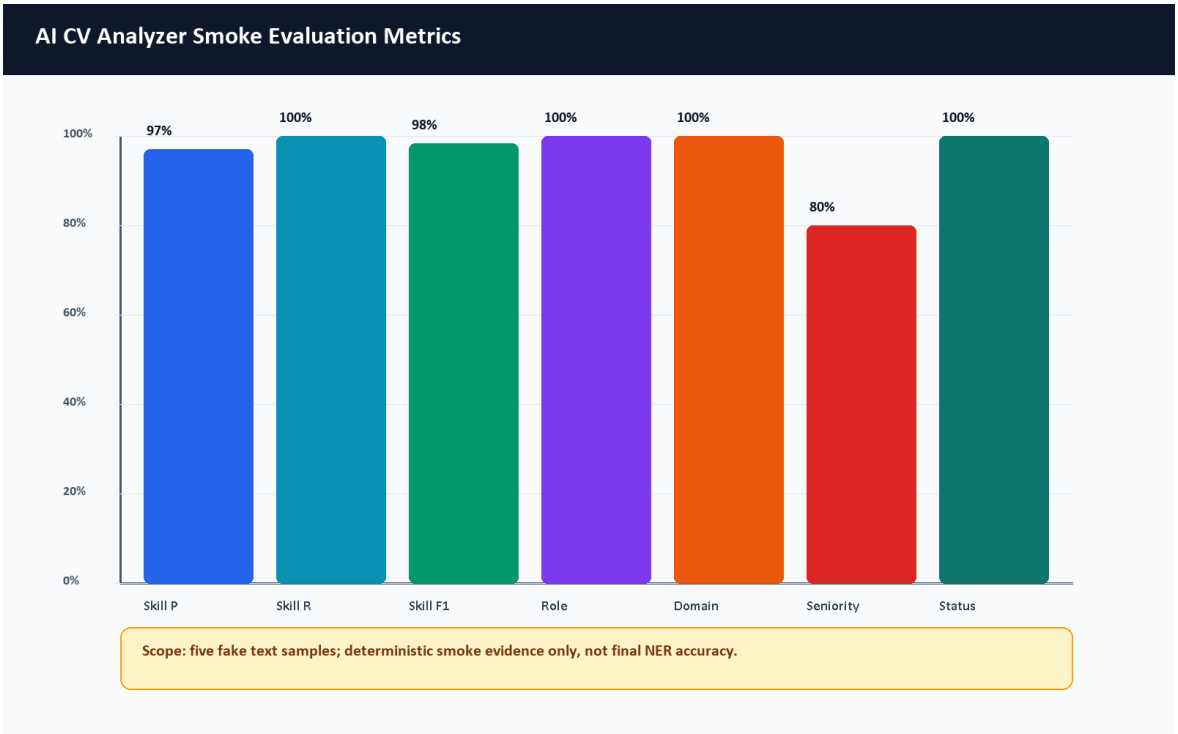


Figure 30. AI CV Analyzer smoke evaluation metrics.

8.16 Job Miner Evaluation

The AI Job Miner is evaluated in more detail in Chapter 7. This testing chapter records only validation outcomes: syntax compilation, the AI Job Miner pytest run when available, Docker Compose wiring, health checks when the stack is running, import validation, and admin diagnostics evidence. Because external job sources can change, throttle, or require keys, long-term data quality evaluation should be repeated near the final defense and should not be replaced by source-template counts.

8.17 Application Tracker Evaluation

The application tracker was evaluated by saving a selected job to the tracker and loading the Applications page. The screenshot shows saved opportunity state. Backend tests also include application tracker behavior.

8.18 Admin Dashboard Evaluation

Admin login and admin dashboard access were tested with the demo admin account. The admin dashboard, admin jobs, admin sources, and admin target roles pages were captured. The dashboard displayed 22 users, 209 imported jobs, 9 active sources, and 13 target roles at capture time.

8.19 Performance Observations

The local Docker stack is heavy because it runs frontend, backend, multiple Laravel workers, MySQL, MinIO, two Python AI services, Prometheus, and Grafana. Initial full build can exceed a short command timeout on a Windows laptop. Once images are built, targeted service startup and HTTP checks are practical for a graduation demo.

8.20 Evaluation Limitations

- The AI CV Analyzer pytest suite was not executed because pytest was absent in that container.

- The browser CV upload remains a smoke test, and the mini dataset is synthetic rather than statistically representative.
- The AI CV Analyzer training notebook and exported Colab PDF were inspected, but full model training was not rerun because the cleaned training dataset and external generation keys are not committed and the workflow is designed for Colab GPU execution.
- The Colab metrics are recorded training-run evidence on the notebook split, not a production benchmark on a large real-world CV dataset.
- The AI CV Analyzer smoke evaluation uses five deterministic text samples and does not measure the transformer NER model weights.
- The recommendation score shown in screenshots is an estimated local demo output.
- External scraping reliability depends on source availability and changing website/API behavior.
- Production security, privacy, and performance audits remain future work.

8.21 Summary of Results

Area	Command or Scenario	Result	Evidence
Docker config	docker compose -f docker-compose.yml -f docker-compose.prod.yml config --quiet	Passed	Final cleanup command output
Docker stack	docker desktop start; docker compose up -d; docker compose ps	Running; main app containers healthy after startup settled	Service availability evidence
Backend routes	php artisan route:list	Previously passed, 131 routes	Earlier validation evidence retained
Backend tests	php artisan test	Previously passed, 39 tests, 297 assertions	Earlier validation evidence retained; backend code unchanged in cleanup pass
Frontend lint	ESLint	Previously passed, 9 warnings, 0 errors	Earlier validation evidence retained; frontend code unchanged in cleanup pass
Frontend build	Vite build	Previously passed, 2904 modules transformed	Earlier validation evidence retained; frontend code unchanged in cleanup pass
AI Job Miner tests	docker compose exec -T ai-job-miner python -m pytest	Passed, 75 tests, 1 warning	Fresh container command output
AI Job Miner demo scrape	Protected /scrape call with demo/local source	SUCCESS; previewed 3, stored 3, failed URLs 0; smoke rows cleaned afterward	Validates demo adapter plus Laravel import path, not queue status polling
AI CV Analyzer syntax	python -m compileall ai-cv-analyzer	Passed with non-fatal .pytest_cache listing warning	Final cleanup command output
AI CV Analyzer pytest	Not rerun	Skipped in cleanup pass	Colab/smoke evidence retained separately
HTTP probes	/api/health, /api/ready, /status, :8003/health, :8000/	200 responses after Docker startup settled	Service availability only, not external scraping success

Table 55. Automated validation results.

Test ID	Module	Scenario	Status	Evidence

Test ID	Module	Scenario	Status	Evidence
M-01	Authentication	Register demo user	Passed	Register screenshot/API output
M-02	Authentication	Login student	Passed	Figure 34
M-03	CV upload	Upload valid PDF	Passed	Figures 35-37
M-04	CV upload	Invalid file handling	Not Run Manual	Backend validation tests
M-05	Recommendations	Open jobs page after CV	Passed	Figure 38
M-06	Gap analysis	Analyze selected job	Passed	Figure 40
M-07	Tracker	Save job	Passed	Figure 41
M-08	Admin	Login admin and open dashboard	Passed	Figure 44
M-09	Admin sources	Open diagnostics	Passed	Figure 46
M-10	Status	Open system status page	Passed	Figure 43

Table 56. Manual functional evaluation matrix.

Test ID	Expected vs Actual Observation
M-01	Expected user creation and token return; actual user was created with an accepted Gmail-style address.
M-02	Expected dashboard access after login; actual dashboard loaded.
M-03	Expected CV acceptance and parsing; actual upload parsed successfully.
M-04	Expected validation error for invalid file; actual manual browser repetition was not run because backend validation/tests already cover the rule.
M-05	Expected recommendations with estimated matches; actual jobs page loaded.
M-06	Expected match and skill breakdown; actual gap page loaded.
M-07	Expected saved job in tracker; actual application page loaded with saved item.
M-08	Expected admin-only dashboard; actual dashboard visible after admin login.
M-09	Expected source diagnostics; actual diagnostics page visible.
M-10	Expected health UI; actual system status page visible.

Table 57. Manual functional observations.

Chapter 9: Security and Privacy

9.1 Introduction

Security in CareerCompass is implemented for a graduation/demo context. The system includes meaningful controls, but it should not be presented as production-grade without future hardening, legal review, and operational security work.

9.2 Authentication and Authorization

User authentication uses token-based API access through Laravel. Laravel Sanctum supports API token and SPA authentication use cases [2]. CareerCompass stores an auth token in the browser and attaches it to API requests. Authorization is role-aware: student routes require authentication, while admin routes require the admin role. Authentication security should follow established password and session guidance in production [28].

9.3 Admin Access Control

Admin access is enforced server-side by admin middleware. Frontend route protection improves user experience, but the backend check is the important control. The demo admin account is generated by a seeder and should be changed or disabled in any non-demo environment.

9.4 CV File Privacy

CV files contain personal data. CareerCompass validates file type and size, stores files privately, and avoids public direct file exposure. OWASP's file upload guidance emphasizes validation, restricted storage, and safe handling [27].

9.5 Private Storage and Signed Downloads

Uploaded CV files are stored through private storage and accessed through signed or temporary URLs. MinIO/S3-compatible storage supports object-based file storage and access control patterns [9]. The graduation demo should still avoid uploading real sensitive CVs unless the environment is controlled.

The model-training workflow is intentionally documented separately from runtime CV processing. Synthetic training snippets may be generated through Google AI developer tooling [34], [35], but real student CV uploads should not be sent to external AI APIs without explicit consent, a privacy notice, retention rules, and supervisory approval. In the demonstrated runtime, Laravel sends the uploaded file to the local FastAPI analyzer service and stores the file privately; the Gemini-based generator is a training-support script, not the normal CV-upload path.

9.6 Internal Service Tokens

Scraper import routes use a service token middleware. This reduces accidental public ingestion, but service tokens must be rotated, stored securely, and monitored in production.

9.7 Payload and File Validation

Laravel form requests validate registration, login, CV upload, applications, and profile updates. File validation includes MIME type, extension, and size checks. Validation reduces risk but does not replace malware scanning, deep file inspection, or content disarm in production.

9.8 Logging and Request IDs

The API client attaches request IDs, and backend logging records important events such as CV processing status and AI gateway errors. For production, logs should avoid sensitive CV content and should be retained according to a privacy policy.

9.9 Demo Security Limitations

Area	Current Demo Control	Production Hardening Needed
Admin account	Demo seeder account	Secret rotation, SSO/MFA, audit logs
CV files	Private storage and signed URLs	Malware scanning, retention policy, consent model
Tokens	Bearer tokens	Token rotation, secure cookie strategy, revocation review
Scraper service	Internal token	Secret manager, network isolation, rate limits
Monitoring	Local Prometheus/Grafana	Auth, TLS, dashboard access control
Privacy	Local demo posture	Legal review, privacy notice, data minimization

Table 58. Security and privacy controls.

9.10 Future Production Hardening

Future work should include HTTPS-only deployment, secure cookie/session strategy, administrator MFA, centralized secrets management, object scanning, retention policies, audit logging, rate-limit review, CSRF/CORS review, dependency vulnerability scanning, and a privacy impact assessment.

Chapter 10: Conclusion and Future Work

10.1 Conclusion

CareerCompass demonstrates a practical AI-assisted career guidance workflow for a graduation project. It integrates CV parsing, profile normalization, skill extraction, imported job records, estimated recommendation, gap analysis, application tracking, admin diagnostics, and containerized deployment.

10.2 Project Achievements

- Built a working multi-service web application with frontend, backend, AI services, database, object storage, proxy, and monitoring.
- Implemented student authentication, dashboard, CV upload, profile view, jobs, gap analysis, and application tracking.
- Implemented admin dashboard, job management, source diagnostics, and target role management.
- Documented the AI CV Analyzer runtime architecture, optional local NER artifact path, synthetic data-generation workflow, and Colab training notebook.
- Documented the AI Job Miner runtime architecture, queue flow, source management, import/deduplication logic, failed URL handling, and ethical scraping boundaries.
- Added tests and validation commands across backend, frontend, Python services, Docker, and HTTP probes.
- Captured browser screenshots from the running local system.
- Generated a formal report with diagrams, references, and evaluation notes.

10.3 Educational Value

The project demonstrates practical learning in software architecture, service decomposition, secure file handling, API design, database schema design, frontend state management, AI service integration, testing, Docker operations, monitoring, and academic documentation.

10.4 Current Limitations

- The system is a graduation/demo platform and not a production product.
- Recommendation and gap analysis outputs are estimates.
- AI evaluation needs larger labeled datasets, committed training artifacts, per-label reports, and repeatable model scoring.
- The exported NER model artifact path is supported by the runtime. The repository includes the exported Colab PDF with recorded overall NER metrics, but it does not include the cleaned dataset and model weights required to rerun the same training experiment from repository files alone.
- The Colab PDF records overall NER training metrics, but the final labeled NER dataset was not available in committed evidence, so label distribution and final per-label NER precision/recall/F1 were not claimed.
- OCR and PDF text-recovery quality can affect scanned CVs, image-heavy layouts, multi-column documents, and unusual fonts.
- Synthetic training examples may not represent all real student CV styles, Arabic/multilingual CVs, or informal job-market wording.
- External scraping sources can be unstable, require keys, change page structure, enforce rate limits, or impose terms that must be reviewed before use.
- The AI CV Analyzer container needs pytest installed to run its test suite.
- Security and privacy controls need production hardening before real deployment.

10.5 Future Work

- Add a larger CV/job evaluation dataset with manual labels.
- Add a reproducible NER evaluation pipeline that runs on a fixed labeled test set and records per-label precision, recall, and F1.
- Store model cards, dataset cards, and training-run summaries for each exported model artifact.
- Add a privacy-preserving workflow for any future human-reviewed real CV dataset.
- Improve Arabic/multilingual parsing, OCR, and field extraction.
- Expand the role taxonomy, skill aliases, and labor-market evidence with periodic review.
- Add human-in-the-loop review for high-impact guidance and ambiguous CV parsing results.
- Improve skill normalization through reviewed aliases and false-positive checks.
- Add production-grade authentication and administrator controls.
- Add malware scanning and retention policies for uploaded CV files.
- Improve recommendation explanations and calibration.
- Strengthen job-mining evaluation with reproducible source-health checks, canonical source IDs, data freshness rules, and targeted failed-URL reprocessing.
- Add automated browser end-to-end tests.
- Extend CI to run all containerized test suites consistently.
- Improve observability dashboards and alerting.
- Add deployment documentation for a secure cloud environment.

10.6 Final Remarks

CareerCompass is an original graduation project that connects academic software engineering requirements with a practical career guidance problem. Its strongest value is the integration of many realistic system parts into one demonstrable workflow, while preserving honest language about limitations and future production work.

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Appendices

Appendix A: API Request and Response Examples

This appendix expands the endpoint summary with JSON-oriented examples. The examples are intentionally small and use placeholders such as Authorization: Bearer <token>. They document the implemented API shape for examiners and future maintainers; they do not expose real tokens, private CV contents, or production secrets. The format follows an OpenAPI-style request/response documentation pattern [39].

Group	Example Endpoints	Purpose
Health	/api/health, /api/ready, /api/metrics	Liveness, readiness, and Prometheus metrics.
Auth	/api/v1/register, /api/v1/login, /api/v1/logout, /api/v1/user	User identity and token lifecycle.
CV	/api/v1/upload-cv, /api/v1/user/skills, /api/v1/user/cv-analysis	CV upload, parsed analysis, and extracted skills.
Jobs	/api/v1/jobs, /api/v1/jobs/recommended, /api/v1/jobs/{id}, /api/v1/jobs/scrape, /api/v1/jobs/scrape-if-missing, /api/v1/scraping-status/{jobId}	Job listing, details, recommendations, on-demand scraping, and status polling.
Gap Analysis	/api/v1/gap-analysis/job/{jobId}, /api/v1/gap-analysis/role/{roleId}	Skill comparison and recommendations.
Applications	/api/v1/applications	Save and update tracked opportunities.
Admin	/api/v1/admin/dashboard/stats, /api/v1/admin/dashboard/batch-progress, /api/v1/admin/dashboard/failed-urls/{scrapingJobId}, /api/v1/admin/dashboard/retry-failures, /api/v1/admin/scraping-sources, /api/v1/admin/target-roles, /api/v1/admin/scraping/run-full	Admin dashboards, diagnostics, source management, target roles, full runs, and failed URL operations.
Internal Scraper	/api/v1/jobs/import, /api/v1/jobs/import/check, /api/v1/jobs/import/failed, /api/v1/proxies/active	Service-token protected import, duplicate check, failure report, and proxy routes.

Table 59. API endpoint summary.

A.1 Core Authentication and Current User Endpoints

Login method and URL: POST /api/v1/login

Purpose: Authenticate a student or admin user and issue a bearer token for later API calls.

Login request example:

```
{
  "email": "student@example.com",
  "password": "<password>"
}
```

Login response example:

```
{
  "success": true,
  "message": "Login successful",
  "data": {
    "token": "<user-token>",
    "user": {
      "id": 7,
      "name": "Demo Student",
      "email": "student@example.com",
      "role": "student",
      "profile": {
        "headline": "Backend Developer",
        "location": "Giza, Egypt"
      }
    }
  }
}
```

Current user method and URL: GET /api/v1/user

Request header:

```
Authorization: Bearer <user-token>
```

Current user response example:

```
{
  "data": {
    "id": 7,
    "name": "Demo Student",
    "email": "student@example.com",
    "role": "student",
    "job_title": "Backend Developer",
    "headline": "Backend Developer",
    "summary": "Sanitized profile summary.",
    "location": "Giza, Egypt",
    "total_experience_years": 0.5,
    "seniority": "Intern",
    "primary_domain": "Backend Development",
    "phone": "+20XXXXXXXXX",
    "linkedin_url": "https://example.com/linkedin",
    "github_url": "https://example.com/github",
    "profile": {
      "headline": "Backend Developer",
      "location": "Giza, Egypt",
      "contact_info": {}
    },
    "experiences": [],
    "skills": []
  }
}
```

A.2 Student CV Upload Endpoint

Method and URL: POST /api/v1/upload-cv

Purpose: Upload a PDF/image CV to Laravel, send it to the local FastAPI AI CV Analyzer, persist successful structured outputs, and return warnings/status for the dashboard.

Request example:

```
Authorization: Bearer <token>
Content-Type: multipart/form-data
```

```
cv=@sample_cv.pdf
```

Successful response example:

```
{
  "success": true,
  "message": "CV parsed successfully.",
  "parsing_status": "success",
  "analysis_id": 12,
  "skills_count": 4,
  "predicted_role": "Backend Developer",
  "profile_updated": true,
  "retry_available": false,
  "download_url": "https://example.local/api/cv-files/12?signature=...",
  "data": {
    "analysis_id": 12,
    "parsing_status": "success",
    "skills_count": 4,
    "predicted_role": "Backend Developer",
    "warnings": []
  }
}
```

Error response example:

```
{
  "success": false,
  "message": "The AI engine is currently unavailable. Please try again in a moment.",
  "parsing_status": "error",
  "retry_available": true,
  "warnings": [
    {
      "code": "ai_unavailable",
      "message": "The AI engine could not be reached. No profile data was changed."
    }
  ]
}
```

A.3 AI Analyzer Parse-CV Endpoint

Method and URL: POST /api/parse-cv

Purpose: FastAPI service endpoint that accepts an uploaded CV file and returns the typed parse schema used by Laravel.

Request example:

```
Content-Type: multipart/form-data

file=@sample_cv.pdf
```

Response example:

```
{
  "parsing_status": "success",
  "profile": {
    "full_name": "Demo Student",
    "current_title": "Backend Developer",
    "email": "student@example.com"
  },
  "stats": {
    "page_count": 1,
    "word_count": 340,
    "language_hint": "en"
```

```
  },
  "skills": {
    "items": [
      {"name": "Laravel", "category": "hard", "confidence_score": 0.65},
      {"name": "Docker", "category": "hard", "confidence_score": 0.65}
    ],
    "confidence_score": 0.65
  },
  "experience": {
    "items": [],
    "confidence_score": 0.0
  },
  "analysis": {
    "predicted_role": "Backend Developer",
    "seniority": "junior",
    "primary_domain": "Backend Development",
    "strengths": [],
    "gaps": [],
    "red_flags": [],
    "confidence_score": 0.65
  },
  "request_id": "example-request-id"
}
```


Error response example:


```
{
  "detail": "Empty file uploaded."
}
```


A.4 AI Hybrid Match Endpoint

Method and URL: POST /api/hybrid-match

Purpose: Compare CV text/skills with a job description using Layer 3 adaptive matching plus TF-IDF when available.

Request example:


```
{
  "cv_skills": ["Laravel", "Docker", "MySQL"],
  "cv_text": "Backend developer with Laravel Docker MySQL REST APIs.",
  "job_description": "Junior backend developer required with Laravel, Docker, MySQL, and REST API
experience.",
  "job_skills": ["Laravel", "Docker", "MySQL", "REST APIs"]
}
```


Response example:


```
{
  "hybrid_match_score": 78.4,
  "semantic_match_pct": 82.0,
  "tfidf_score_pct": 73.0,
  "missing_skills": [],
  "formula": "Final = (Adaptive Layer 3 x 60%) + (TF-IDF x 40%)",
  "matching_mode": "hybrid",
  "request_id": "example-request-id"
}
```


Validation error example:


```
{
  "detail": "cv_text must not be empty."
}
```

A.5 Recommended Jobs Endpoint

Method and URL: GET /api/v1/jobs/recommended

Purpose: Return personalized job recommendations for an authenticated student when CV/profile context exists, otherwise return recent usable jobs.

Request example:

```
Authorization: Bearer <token>
Accept: application/json
```

Response example:

```
{
  "success": true,
  "job_title": "Backend Developer",
  "data": [
    {
      "id": 101,
      "title": "Junior Backend Developer",
      "company": "DemoTech",
      "location": "Cairo",
      "source": "demo",
      "match_percentage": 84.5,
      "skills_count": 4
    }
  ],
  "meta": {
    "total": 1,
    "based_on": "Your CV title: Backend Developer"
  }
}
```

A.6 On-Demand Job Scraping Endpoint

Method and URL: POST /api/v1/jobs/scrape

Purpose: Start an authenticated on-demand scraping job for a query. Laravel validates the request, creates a ScrapingJob, dispatches the scraping queue worker, and returns a status identifier.

Request example:

```
Authorization: Bearer <user-token>
Content-Type: application/json
```

```
{
  "query": "Backend Developer",
  "max_results": 10
}
```

Successful response example:

```
{
  "success": true,
  "message": "Jobs scraping dispatched to background process",
  "data": {
    "query": "Backend Developer",
    "scraping_job_id": 42
  }
}
```

Validation error example:

```
{
  "message": "The query field is required.",
  "errors": {
    "query": ["The query field is required."]
  }
}
```

A.7 Scrape If Missing and Status Polling

Method and URL: POST /api/v1/jobs/scrape-if-missing

Purpose: Check whether matching stored jobs exist before queueing an external scrape.

Request example:

```
{
  "job_title": "Laravel Developer",
  "max_results": 10
}
```

Existing-data response example:

```
{
  "success": true,
  "data_exists": true,
  "message": "Job data already available",
  "jobs_count": 5
}
```

Queued response example:

```
{
  "success": true,
  "data_exists": false,
  "message": "Analyzing market data for this role. Please wait...",
  "scraping_job_id": 43,
  "status": "pending",
  "poll_url": "http://localhost/api/v1/scraping-status/43"
}
```

Method and URL: GET /api/v1/scraping-status/{jobId}

Status response example:

```
{
  "success": true,
  "scraping_job_id": 43,
  "job_title": "Laravel Developer",
  "status": "completed",
  "type": "on_demand",
  "started_at": "2026-06-08T00:00:00.000000Z",
  "results": {
    "jobs_found": 8,
    "jobs_stored": 5,
    "jobs_duplicated": 3,
    "discovered_count": 8,
    "failed_count": 0,
    "processing_time_ms": 12640,
    "completed_at": "2026-06-08T00:00:12.000000Z"
  },
  "jobs": [
    {
```

```
      "id": 101,
      "title": "Junior Backend Developer",
      "company": "Example Co"
    }
  ]
}
```

A.8 Internal Scraper Duplicate Check and Import

These Laravel endpoints are protected by scraper.token and throttle:scraper. In the current middleware, the scraper supplies Authorization: Bearer <internal-token>.

Method and URL: POST /api/v1/jobs/import/check

Request example:

```
Authorization: Bearer <internal-token>
Content-Type: application/json
```

```
{
  "url": "https://example.com/jobs/123"
}
```

Response example:

```
{
  "exists": false
}
```

Method and URL: POST /api/v1/jobs/import

Request example:

```
{
  "title": "Junior Backend Developer",
  "company": "Example Co",
  "location": "Remote",
  "description": "Build APIs with Laravel and MySQL.",
  "requirements": "Laravel, MySQL, REST APIs",
  "url": "https://example.com/jobs/123",
  "source": "remotive",
  "scraping_source_id": 5,
  "skills": ["Laravel", "MySQL", "REST APIs"],
  "work_type": "remote",
  "job_type": "full_time"
}
```

Response example:

```
{
  "success": true,
  "job_id": 101,
  "created": true
}
```

A.9 Failed URL Reporting, Proxies, and Admin Scraping

Method and URL: POST /api/v1/jobs/import/failed

Purpose: Store a failed source URL for diagnostics and retry visibility.

Request example:

```
{
  "url": "https://example.com/jobs/broken",
  "scraping_source_id": 5,
  "scraping_job_id": 43,
  "error_message": "Timeout while fetching public job detail page."
}
```

Response example:

```
{
  "success": true
}
```

Method and URL: GET /api/v1/proxies/active

Purpose: Return active proxy definitions to the scraper only when proxy configuration is enabled and authorized.

Admin source diagnostic request:

```
Authorization: Bearer <admin-token>
POST /api/v1/admin/scraping-sources/5/test
```

Admin full scraping request:

```
Authorization: Bearer <admin-token>
POST /api/v1/admin/scraping/run-full
```

Example full-run response shape:

```
{
  "success": true,
  "batch_id": "example-batch-id",
  "planned_jobs": 12,
  "skipped_sources": []
}
```

A.10 Gap Analysis Endpoint

Method and URL: GET /api/v1/gap-analysis/job/{jobId}

Purpose: Compare the authenticated student's extracted skills/profile with one job and return matched skills, missing skills, recommendations, and CV-analysis context.

Request example:

```
Authorization: Bearer <token>
Accept: application/json
```

Response example:

```
{
```

```
  "success": true,
  "data": {
    "job": {
      "id": 101,
      "title": "Junior Backend Developer",
      "company": "DemoTech"
    },
    "analysis": {
      "match_percentage": 80.0,
      "match_level": "Good Match",
      "matched_skills": ["Laravel", "Docker", "MySQL"],
      "missing_skills": ["Kubernetes"]
    },
    "recommendations": [
      "Practice Kubernetes deployment basics before applying."
    ],
    "cv_analysis": {
      "parsing_status": "success",
      "completeness_score": 75,
      "strengths": [],
      "gaps": [],
      "red_flags": []
    }
  }
}
```


Error response example:


```
{
  "success": false,
  "message": "Upload a CV first so the system can extract skills and profile data."
}
```


A.11 Application Tracking Endpoint

Method and URL: POST /api/v1/applications

Purpose: Let an authenticated student save or update an opportunity status without changing the shared job record.

Request example:


```
{
  "job_id": 101,
  "status": "saved",
  "notes": "Review Laravel and Docker requirements before applying."
}
```


Response example:


```
{
  "success": true,
  "data": {
    "id": 55,
    "job_id": 101,
    "status": "saved",
    "notes": "Review Laravel and Docker requirements before applying.",
    "job": {
      "title": "Junior Backend Developer",
      "company": "DemoTech"
    }
  }
}
```

A.12 Admin Dashboard Summary Endpoint

Method and URL: GET /api/v1/admin/dashboard/stats

Purpose: Give administrators a compact operational summary of users, jobs, CV activity, scraping runs, and source state.

Request example:

```
Authorization: Bearer <admin-token>
Accept: application/json
```

Response example:

```
{
  "success": true,
  "data": {
    "total_students": 39,
    "total_jobs": 128,
    "total_sources": 4,
    "total_targets": 12,
    "jobs_by_month": [
      {
        "month": "Jun 2026",
        "month_key": "2026-06",
        "count": 18
      }
    ],
    "scraper_overview": {
      "jobs_last_24h": 3,
      "avg_health_score": 91.5,
      "active_sources": 4,
      "total_sources": 4,
      "recent_failures": 0
    }
  }
}
```

A.13 Health and Readiness Endpoints

Methods and URLs: GET /api/health, GET /api/ready

Purpose: Confirm whether Laravel is alive and whether dependent services are ready.

Health response example:

```
{
  "success": true,
  "status": "ok",
  "service": "CareerCompass API",
  "request_id": "example-request-id"
}
```

Readiness response example:

```
{
  "success": true,
  "status": "ready",
  "checks": {
    "database": {"ok": true},
    "cache": {"ok": true},
    "ai": {"ok": true, "status": 200},
    "scraper": {"ok": true, "status": 200}
  },
  "request_id": "example-request-id"
}
```

Appendix B: Quick Start Guide for Examiners

This guide is intended for a local graduation/demo run. It assumes Docker Desktop is installed and running. It should not be read as a production deployment guide.

B.1 Start the Stack

```
git clone https://github.com/YousefAlTohamy/CareerCompass.git
cd CareerCompass
cp .env.example .env
docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build
docker compose exec backend-api php artisan migrate --seed
```

B.2 Useful URLs

- Frontend: http://localhost
- Backend health: http://localhost/api/health
- Backend readiness: http://localhost/api/ready
- AI Job Miner health, when the stack is running: http://localhost:8003/health
- System status page: http://localhost/status
- Admin dashboard: http://localhost/admin/dashboard
- Grafana, if exposed by the local compose stack: http://localhost:3000

B.3 Demo Accounts and Flow

The demo admin account is seeded from environment variables. The current example values are:

Variable	Demo Value
DEMO_ADMIN_EMAIL	careercompassadmin@gmail.com
DEMO_ADMIN_PASSWORD	CareerCompassAdmin2026

Demo admin environment values.

Student demo flow:

1. Open http://localhost.
2. Register a new student account or log in with an existing demo student.
3. Upload a small PDF CV from the dashboard.
4. Review dashboard readiness signals, extracted role, extracted skills, and profile page.
5. Open Jobs and review recommended jobs.
6. Open Gap Analysis for a selected job and review matched/missing skills.
7. Save a job to Applications.
8. Log in as the demo admin and review dashboard, jobs, sources, and target roles.

B.4 Validation Commands

```
docker compose config --quiet
docker compose -f docker-compose.yml -f docker-compose.prod.yml config --quiet
docker compose exec backend-api php artisan test
cd frontend
npm run lint
npm run build
```

Documentation/evaluation checks:

```
python docs/graduation-book/evaluation/run_mini_evaluation.py
python docs/graduation-book/evaluation/run_ai_cv_analyzer_smoke_eval.py
python docs/graduation-book/scripts/generate_graduation_book.py
python -m compileall ai-job-miner
python -m compileall ai-cv-analyzer
cd ai-job-miner
python -m pytest
cd ..
```

B.5 Troubleshooting Notes

- If the frontend appears stale after rebuild, hard-refresh the browser or rebuild the frontend/nginx containers.
- If Docker startup is slow, wait for MySQL, MinIO, Laravel workers, Python AI services, Prometheus, and Grafana to settle before judging readiness.
- AI model weights are not committed to Git; the runtime supports ai-cv-analyzer/models/ner_weights/career_compass_ner_final when supplied locally.
- NER training requires a cleaned labeled dataset and external API keys for synthetic-data generation; those are not included in the repository.
- Demo admin credentials should be changed for any non-demo environment.

Appendix C: Database Tables Summary

Table	Purpose
users	Authentication identity, role, and banned status.
user_profiles	Student profile details and contact metadata.
skills	Canonical skill catalog.
user_skills	User-to-skill relationship and proficiency metadata.
user_experiences	Extracted or entered experience records.
cv_analyses	CV parsing status, predicted role, confidence, file metadata, strengths, gaps, and warnings.
job_postings	Imported/demo job data.
job_skills	Job-to-skill relationship and requirement metadata.
applications	Saved opportunities and status tracking.
scraping_sources	Admin-configured job source definitions.
target_job_roles	Target role list for scraping and market exploration.
scraping_jobs	Scraping batch execution state.
scraping_failed_urls	Failed source URL diagnostics and retry marker records.
scraping_proxies	Optional active proxy definitions protected by internal scraper token.

Table 60. Database tables summary.

Appendix D: Docker Services Summary

Service	Role
nginx	Public gateway and reverse proxy.

Service	Role
frontend	React/Vite web UI container.
backend-api	Laravel API runtime.
backend-worker variants	Queue processing for default, high, AI, emails, and scraping work.
backend-scheduler	Scheduled Laravel commands.
db	MySQL database.
minio	S3-compatible CV object storage.
ai-cv-analyzer	FastAPI CV parsing service.
ai-job-miner	FastAPI job mining service.
prometheus	Metrics collection.
grafana	Metrics visualization.

Table 61. Docker services summary.

Appendix E: Screenshots

The screenshot set was reviewed during this pass. Some dashboard states are similar, but they document different examiner-visible states: before CV upload, upload UI, and after analysis. No screenshot merge was performed because combining them would reduce traceability and could disturb existing figure references in the generated List of Figures.

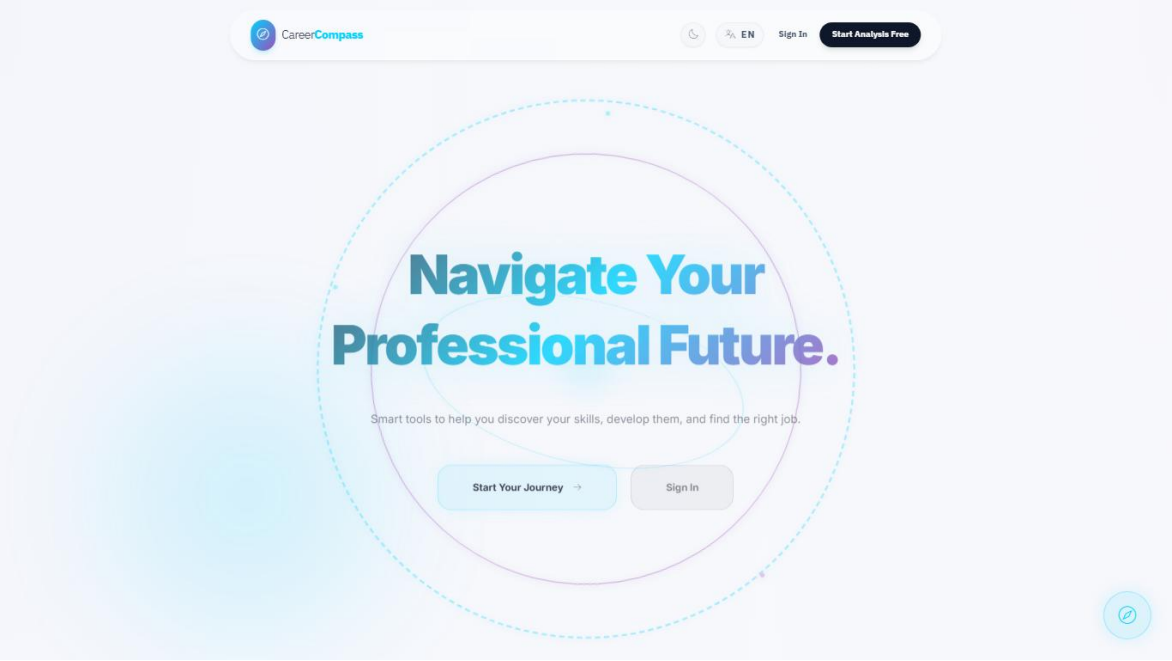

Figure 31. Home page.

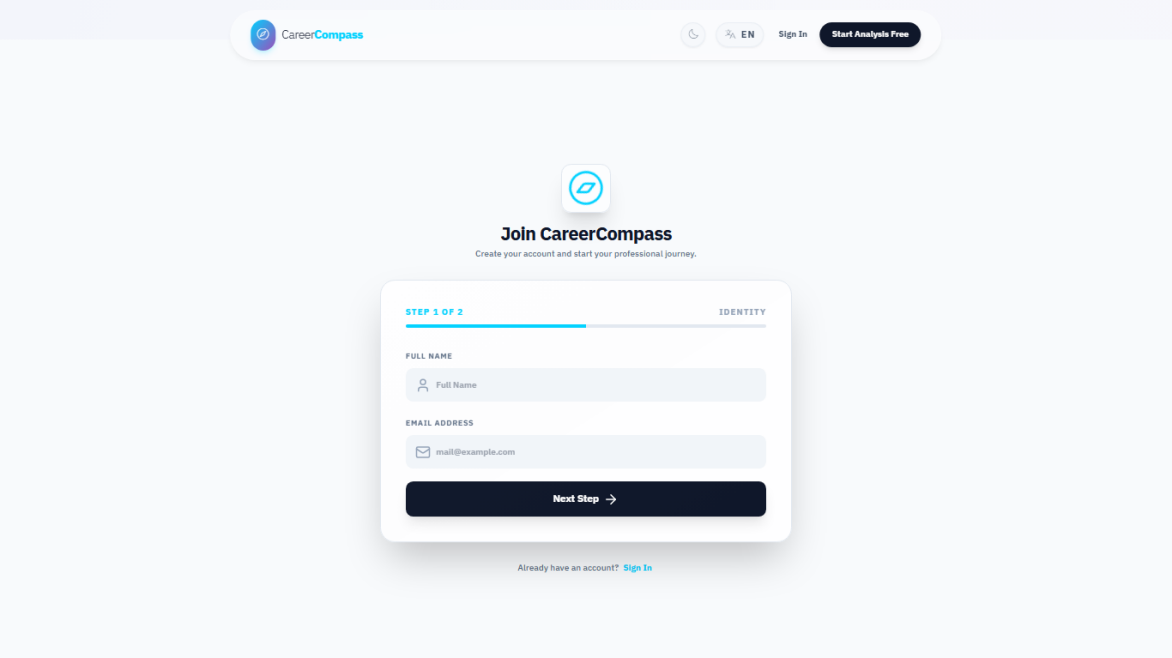

Figure 32. Register page.

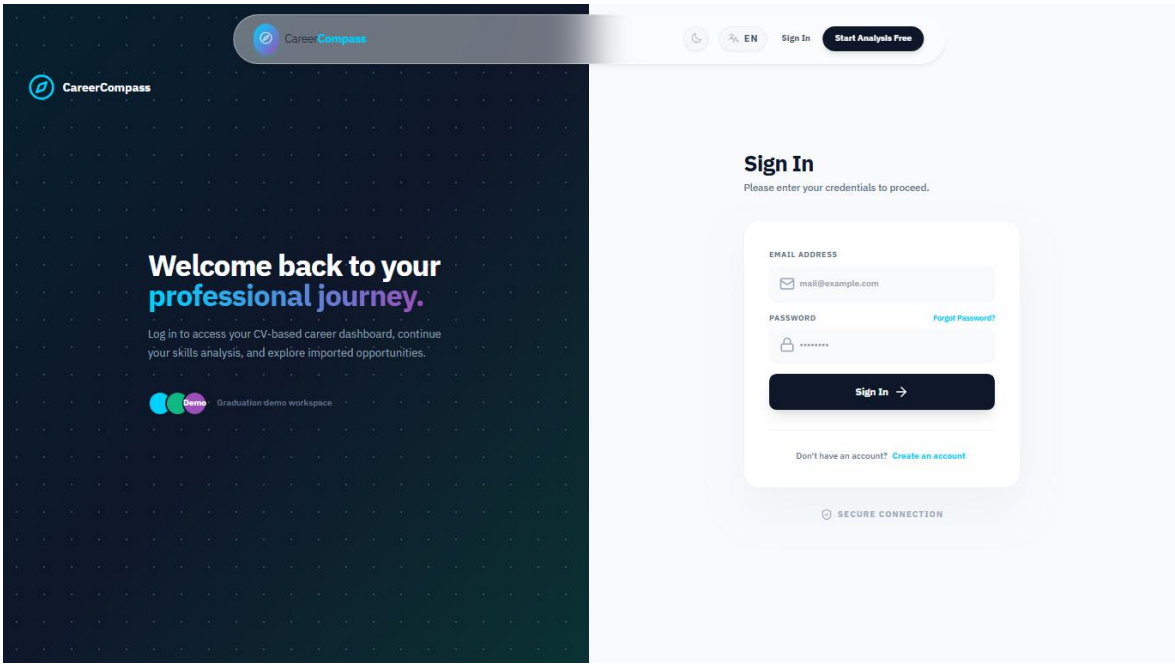

Figure 33. Login page.

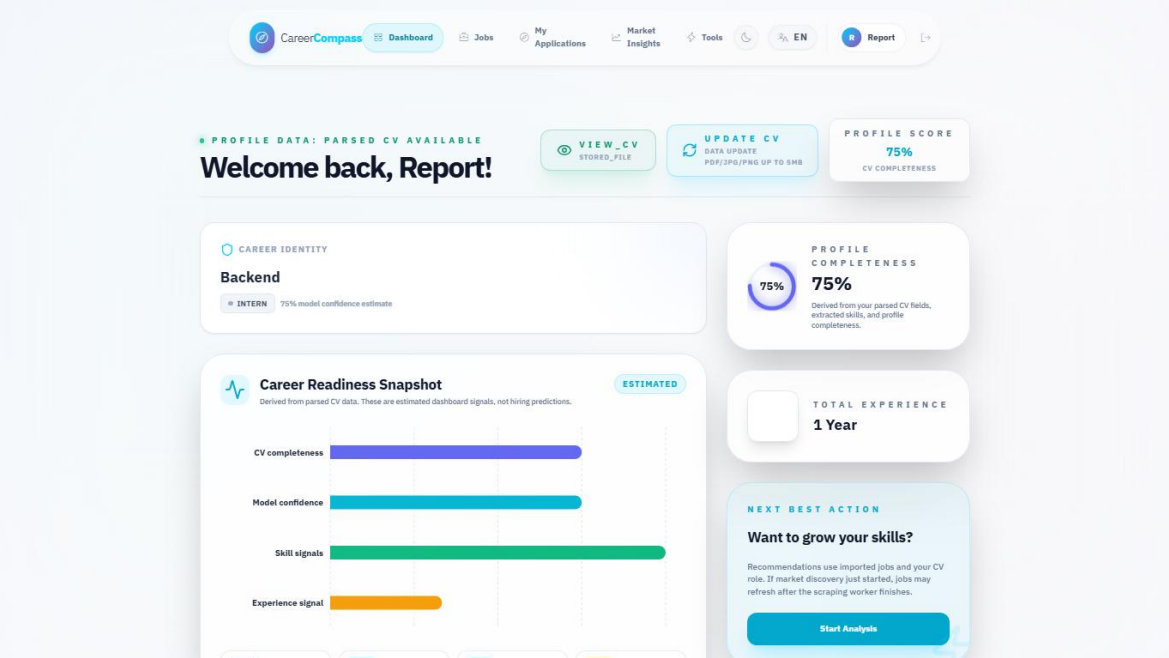

Figure 34. Student dashboard before CV upload.

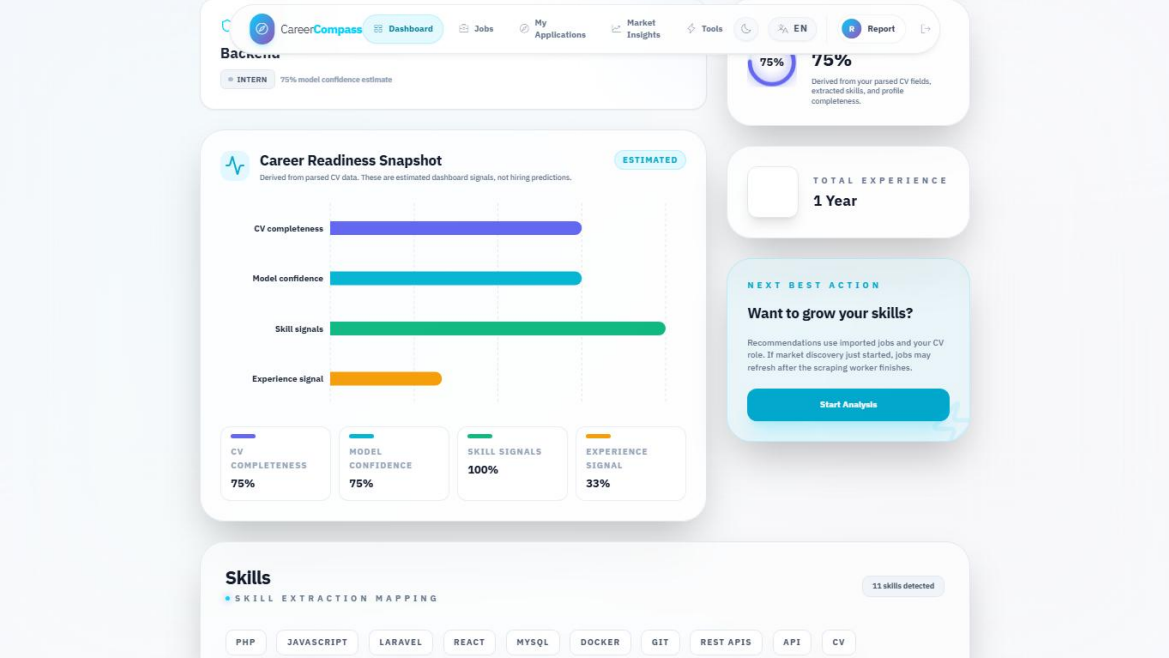

Figure 35. CV upload user interface.

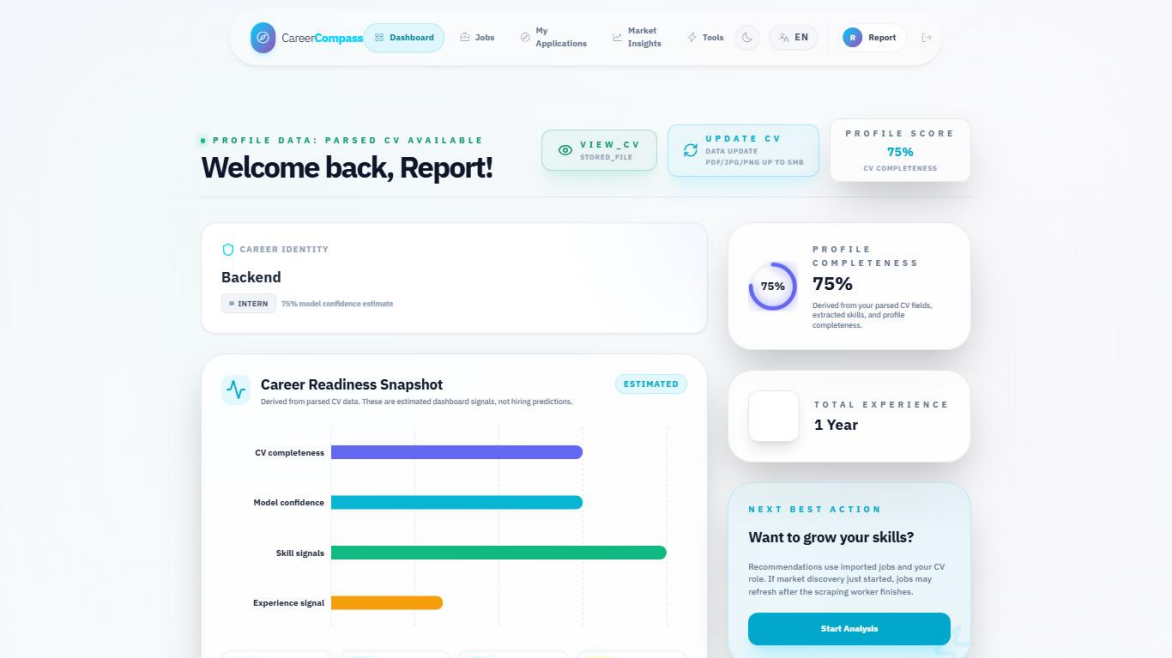

Figure 36. Dashboard after successful CV parsing.

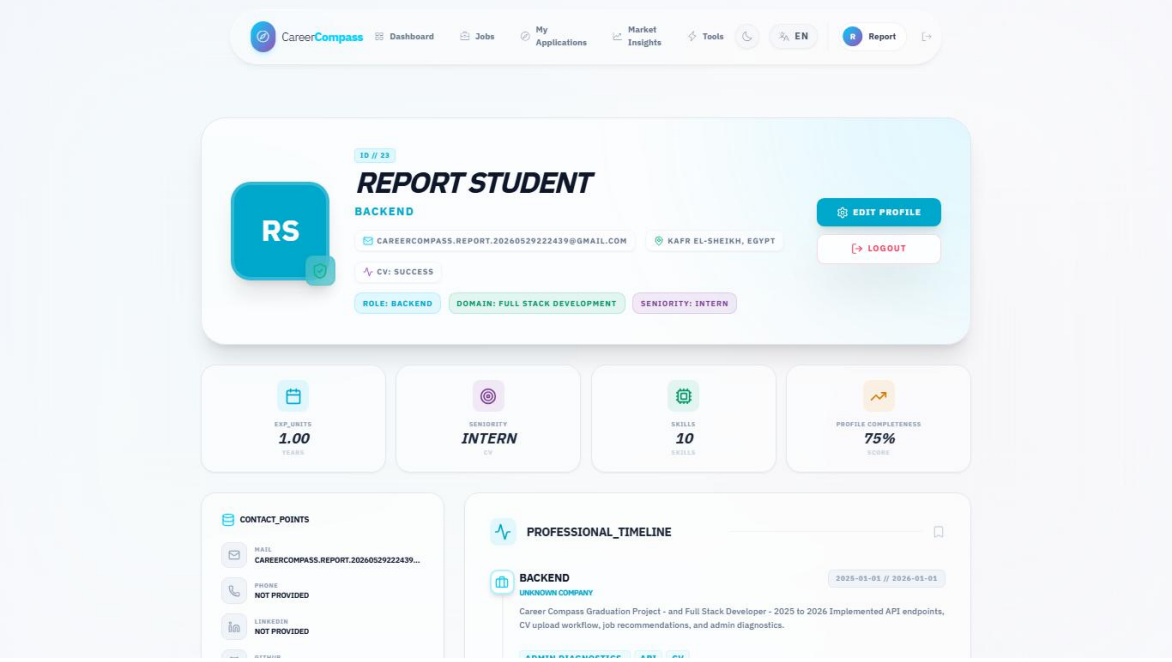

Figure 37. Extracted profile and skills page.

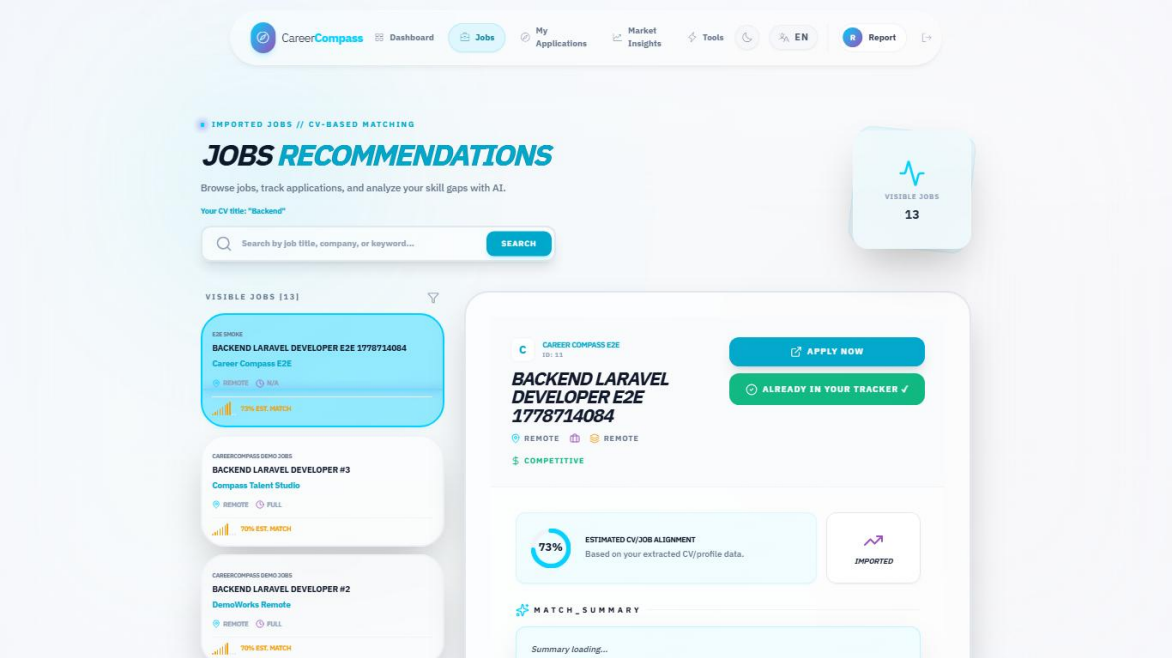

Figure 38. Jobs recommendations page.

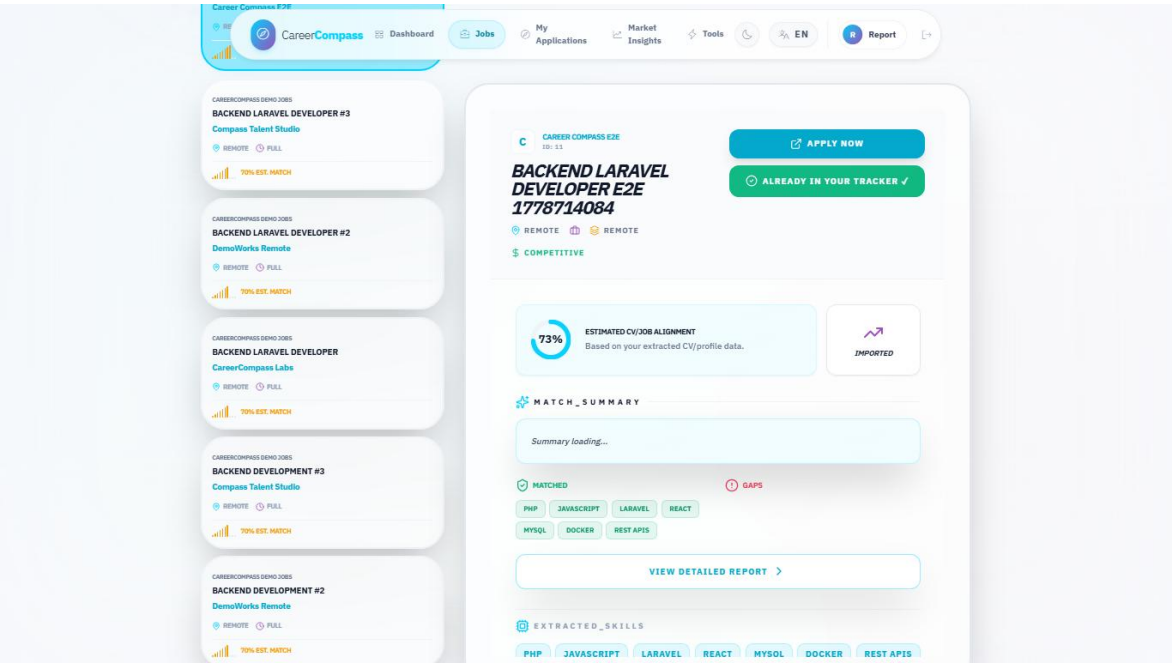

Figure 39. Job detail and inline gap panel.

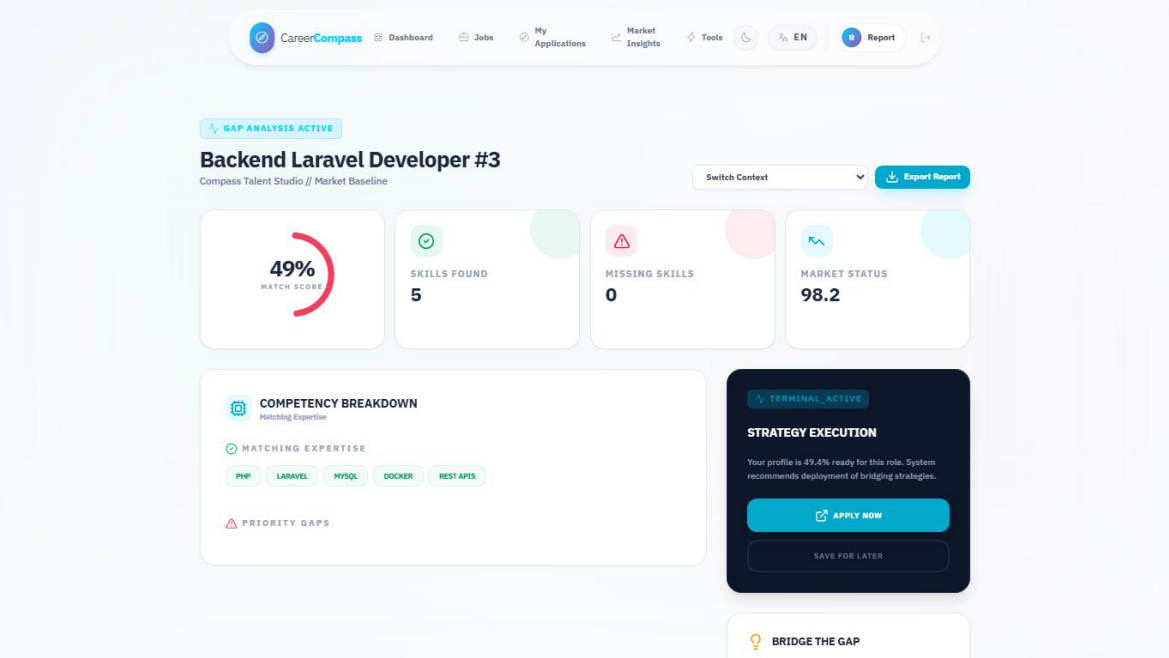

Figure 40. Gap analysis page.

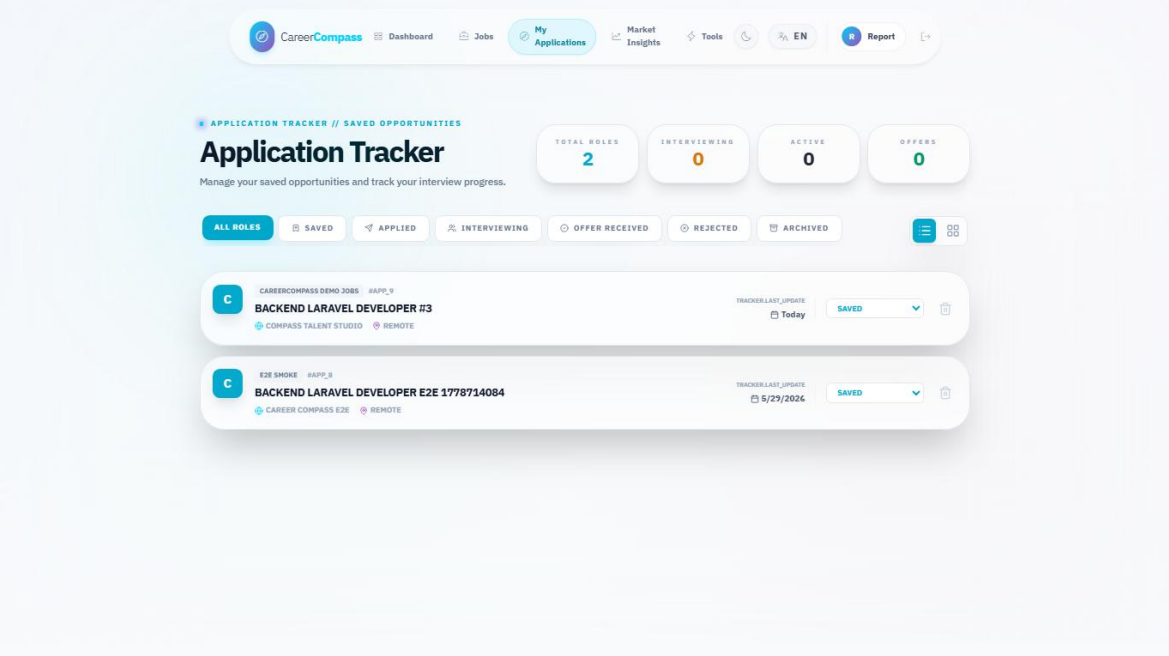

Figure 41. Applications tracker page.

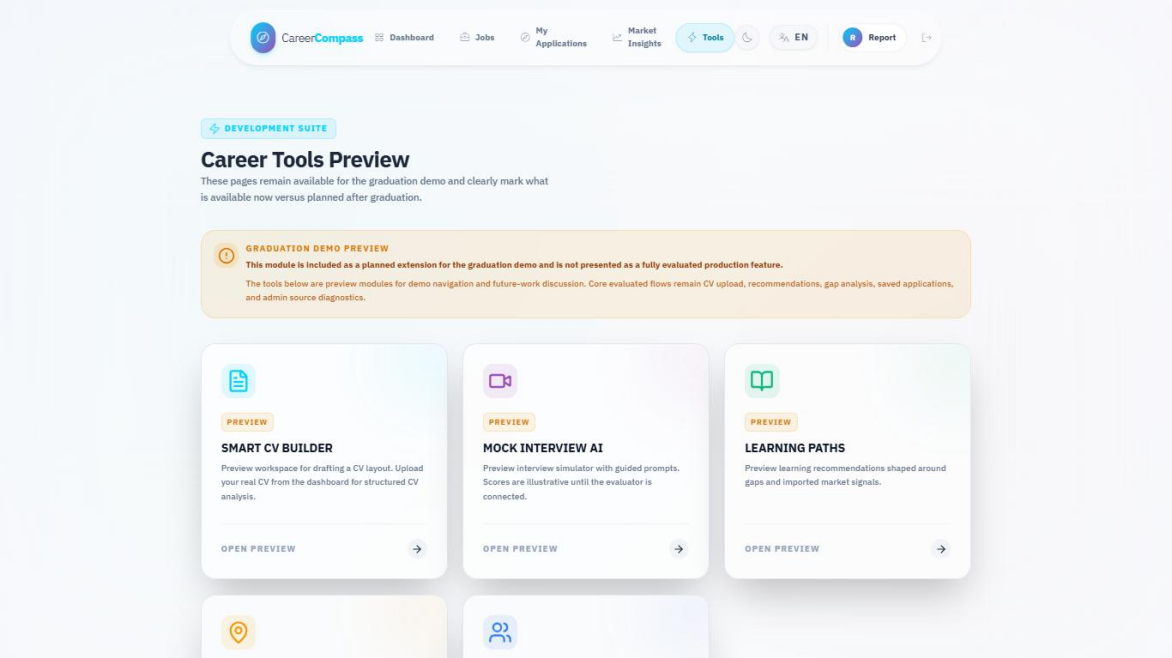

Figure 42. Tools Hub preview page.

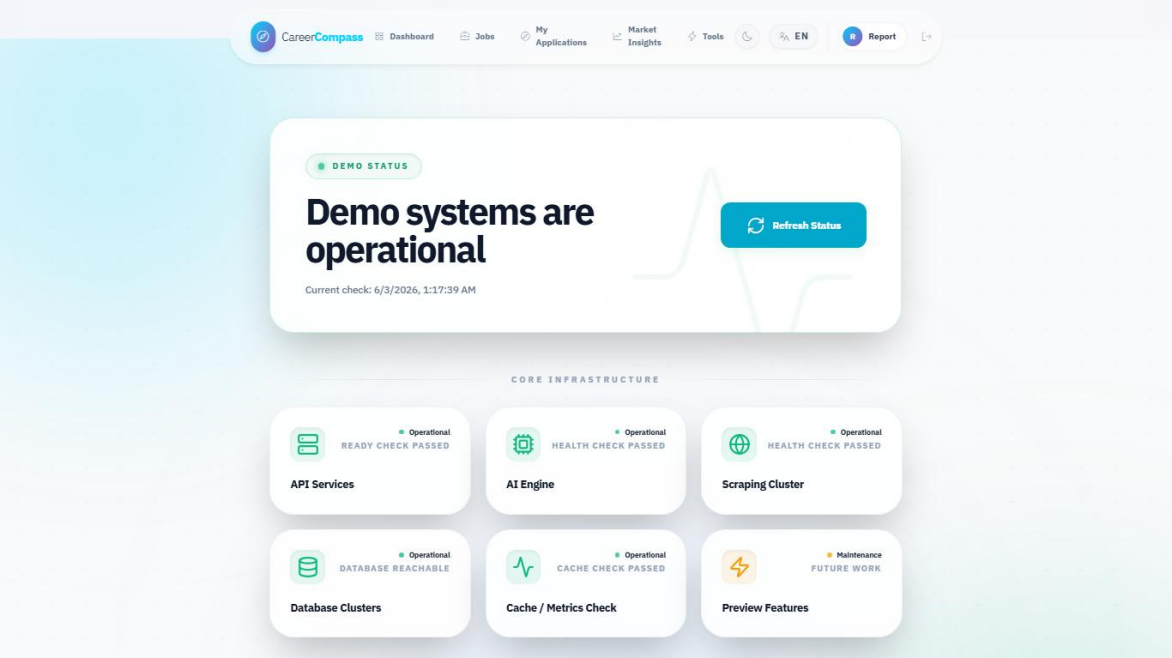

Figure 43. System status page.

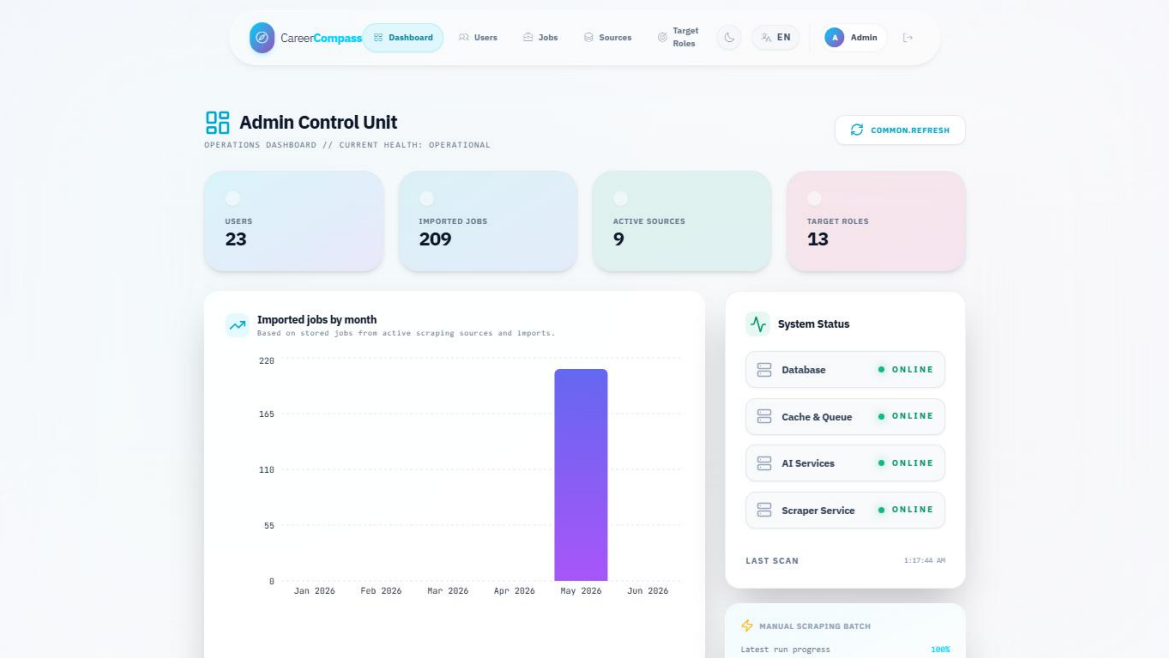

Figure 44. Admin dashboard.

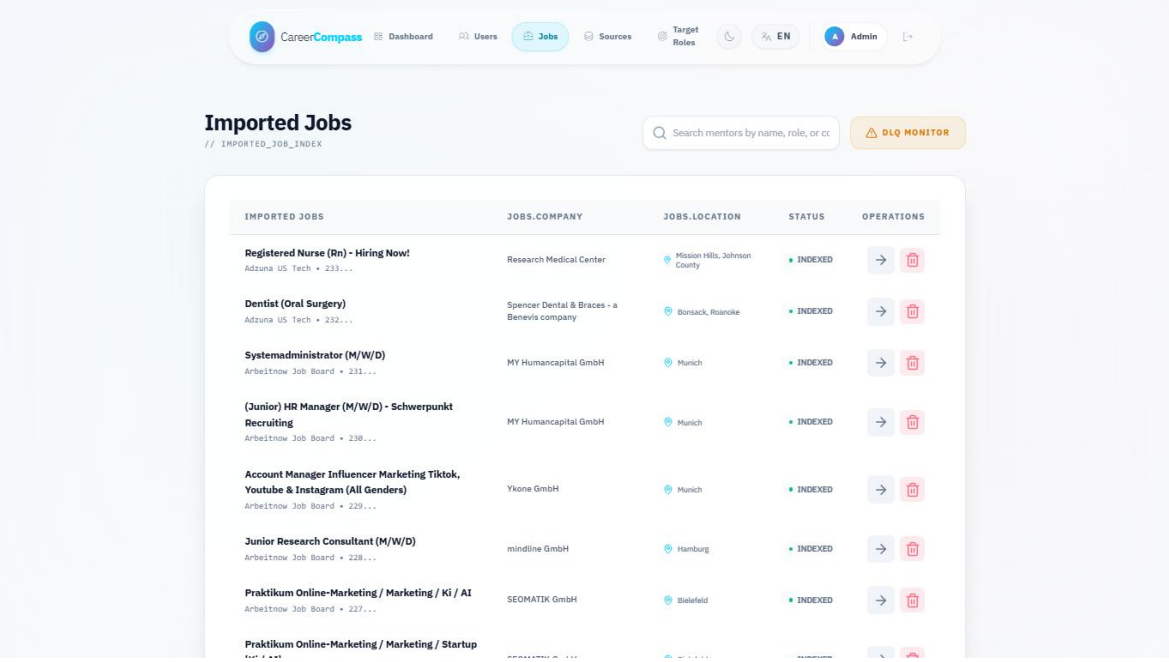

Figure 45. Admin jobs page.

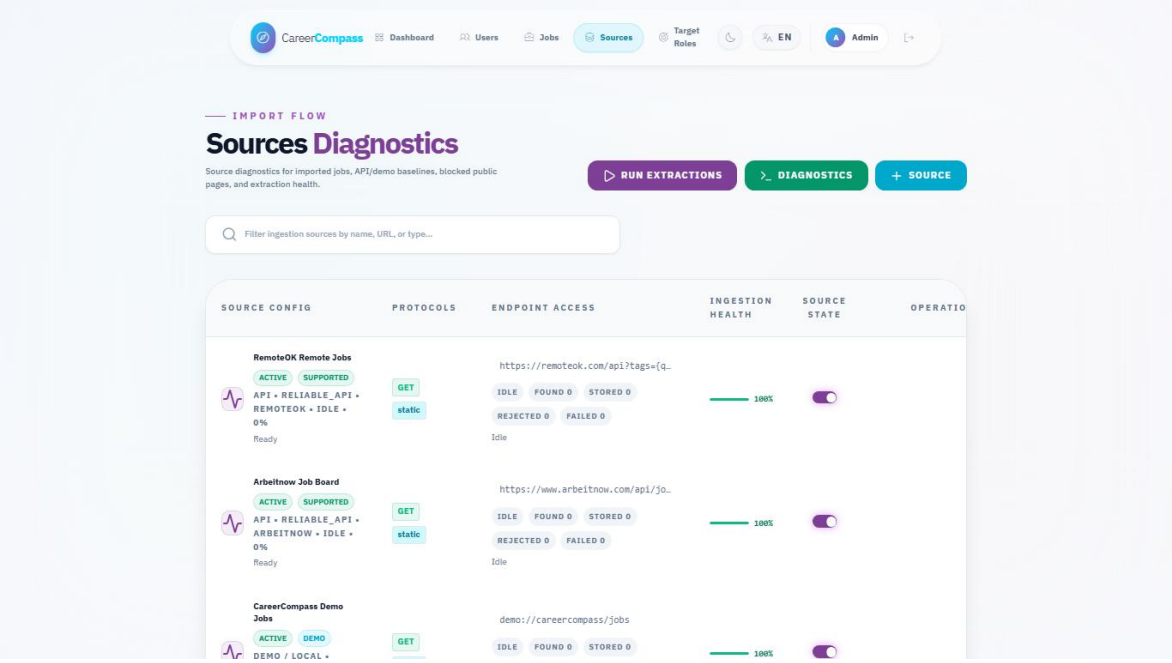

Figure 46. Admin sources diagnostics page.

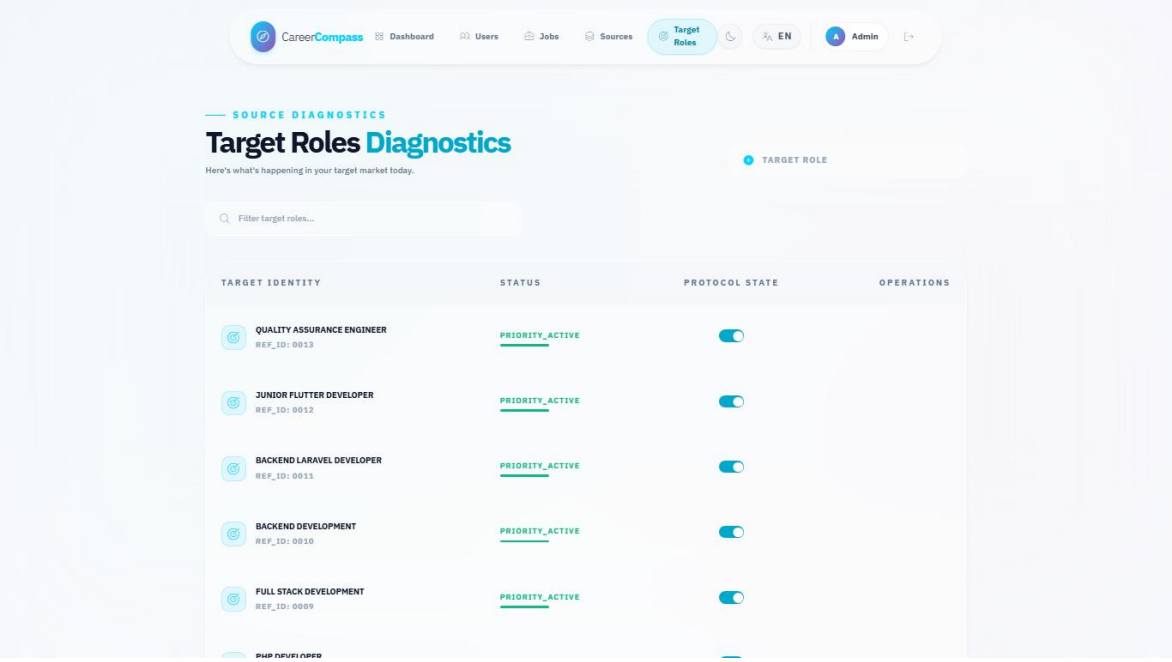

Figure 47. Admin target roles page.

Docker Compose Services Evidence

Service	Status	Meaning
nginx	running / healthy	Public HTTP gateway, host port 80.
frontend	running / healthy	React/Vite frontend, host port 5173.
backend-api	running / healthy	Laravel API, internal FPM port 9000.
backend-worker-scraping	running / healthy	Long-running scraping queue worker.
ai-cv-analyzer	running / healthy	FastAPI CV analyzer, host port 8000.
ai-job-miner	running / healthy	FastAPI job miner, host port 8003.
db	running / healthy	MySQL database, host port 3306.
minio	running	Private object storage, host ports 9000/9001.
prometheus	running	Metrics collection, host port 9090.

Readable evidence summary; raw command output remains in local validation logs.

Figure 48. Docker services evidence.

Validation Evidence Summary

Evidence	Result	Meaning
Compose config	PASSED	Base plus production overlay YAML/config validated.
Laravel health	200	/api/health and /api/ready verify API availability and dependencies.
Frontend status	200	/status returned the React/Vite page HTML.
AI Job Miner	75 passed, 1 warning	Container pytest validates service/API helper behavior.
AI CV Analyzer	PASSED	Python syntax compilation passed; root endpoint is operational.
Backend tests	RETAINED	Earlier container validation passed 39 tests / 297 assertions.
Frontend build	RETAINED	Earlier ESLint/build evidence passed with documented warnings.
Document build	PASSED	Markdown, DOCX, PDF, links, images, and JSON examples validated.

Readable evidence summary; raw command output remains in local validation logs.

Figure 49. Validation evidence summary.

Appendix F: Test Cases

The manual test matrix in Chapter 8 should be repeated before final submission. Additional recommended tests include invalid CV uploads, banned user login, expired signed download URLs, failed AI service behavior, scraper token rejection, admin route rejection for normal users, and browser checks on a clean database.

The supporting evaluation files are summarized below instead of listed as raw paths:

Evaluation Artifact	Purpose	Reader-Facing Evidence
Mini CV/job dataset	Deterministic synthetic records for recommendation and gap-analysis checks.	Chapter 8 mini evaluation tables.
Expected labels	Defines expected roles, domains, skills, and recommendation relevance for the mini dataset.	Table 52 and related detail tables.
Mini evaluation runner	Computes offline skill extraction, recommendation, and gap-analysis agreement.	Generated JSON/Markdown summaries.
AI CV smoke samples	Five sanitized fake CV text samples for analyzer smoke behavior.	Section 8.8 smoke evaluation.
AI CV smoke runner	Executes deterministic parser/check logic without exposing real CVs.	Figure 30 and smoke metric tables.

Evaluation support artifact summary.

Appendix G: GitHub Actions / CI Summary

The repository contains GitHub Actions workflow definitions. After the draft PR is opened, reviewers should inspect PR checks, rerun failed jobs if needed, and confirm that branch protection expectations are satisfied before merging. A CI screenshot was not embedded in this generated report because live PR checks were not available until after branch push.

Appendix H: Demo Script

9. Start Docker Desktop.
10. Run docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build.
11. Open http://localhost.
12. Register or login as a student.
13. Upload a sample PDF CV from the dashboard.
14. Review parsed profile and skills.
15. Open Jobs and select a recommendation.
16. Open Gap Analysis and explain matched/missing skills.
17. Save the job to Applications.
18. Login as the demo admin.
19. Open Admin Dashboard, Jobs, Sources, and Target Roles.
20. Open System Status and explain health checks.
21. Discuss limitations and future work honestly.

Appendix I: AI CV Analyzer Deep Inventory

This appendix summarizes the code audit that supports Sections 5.5, Chapter 6, and Sections 8.8-8.11. The detailed companion notes are stored in docs/graduation-book/model-analysis/ and should be kept with the generated book artifacts.

I.1 Runtime Call Path

22. Laravel receives the CV upload and stores the file privately.
23. Laravel sends the uploaded file to the FastAPI analyzer /api/parse-cv endpoint.
24. main.py chooses image or PDF handling and wraps processing in timeout/error fallbacks.

25. CVOrchestrator extracts ordered text, triggers OCR fallback when needed, segments sections, runs NER, extracts contacts and dates, canonicalizes skills, and validates the strict output schema.
26. Layer 2 enriches the result with domain, seniority, and skill-category classification.
27. Laravel persists normalized profile, skills, experiences, and analysis metadata.
28. Recommendations and gap analysis reuse the stored profile together with Layer 3 matching and backend services.

I.2 Function Inventory Summary

Area	Classes and Functions Audited	Main Purpose
main.py	_get_orchestrator, health_check, metrics, hybrid_match, analyze_cv, _process_with_timeout, process_file, _timeout_result, _error_result	API endpoints, service lifecycle, timeout/fallback behavior, and hybrid match composition.
Layer 1 NER/contact/sections	AdvancedNEREngine, extract_contacts, SemanticSegmenter, DataCanonicalizer, ExperienceEngine, spatial/OCR helpers	Converts raw CV text into profile, skills, experience, and confidence-style structured data.
Layer 2 classification	ClassificationOrchestrator, DomainEngine, SeniorityEngine, SkillEngine, load_taxonomy	Adds domain, seniority, and skill-category interpretation.
Layer 3 matching	SemanticEmbedder, IntelligentMatcher, ConstraintValidator, FitAnalysisGenerator, JobDescriptionEngine, RankingOrchestrator, tfidf.match_score	Produces explainable job-fit scores, penalties, verdicts, and ranking behavior.
Training and diagnostics	generate_tech_dataset.py, clean_dataset.py, train_ner.ipynb, verify_phase*.py, test_service_api.py, trace_cv.py	Supports synthetic dataset generation, cleaning, notebook training, phase verification, service tests, and trace output.

Table 62. AI CV Analyzer function inventory summary.

I.3 Training Summary

The notebook trains a BERT-family token-classification model from bert-base-cased, maps character spans to BIO token labels, uses a 90/10 train/evaluation split with seed 42, trains for five epochs with learning rate 2e-5 and batch size 16, computes seqeval precision/recall/F1/accuracy, and exports career_compass_ner_final. The exported Colab PDF records overall final-epoch metrics: precision 0.933307, recall 0.940521, F1 0.936900, and accuracy 0.976376. The final cleaned dataset content and model weights are not committed.

I.4 Generated Dataset Summary

The synthetic dataset script asks Google Gemini tooling to generate varied CV snippets across technical domains, rotates API keys/models from environment variables, includes negative decoys, and writes labeled examples for cleaning. The cleaner normalizes text, deduplicates exact samples, validates spans, and filters malformed or overly broad entities. This supports the training workflow, but it is separate from normal private CV upload processing.

I.5 Local Artifact and Secrets Boundary

.env and ai-cv-analyzer/models/ are ignored by Git. This protects secrets and avoids committing large model weights. Documentation may mention the runtime path and safe local metadata, but should not imply that the committed repository contains the model binary or private API keys.

Appendix J: AI Job Miner Deep Inventory

This appendix converts the job-mining audit notes into a compact reader-facing summary. The detailed companion files remain in docs/graduation-book/job-mining-analysis/ for repository traceability, but the important evidence is summarized below so the printed book is useful on its own.

Audit Area	What It Documents	Related Chapter
Source inventory	Demo/local source behavior, optional API adapters, HTML/Scrapy-related paths, unsupported or credential-gated sources, and external-risk boundaries.	Chapter 7 source management and limitations.
Function inventory	FastAPI service entry points, adapter orchestration, classification helpers, Laravel controllers, queue jobs, services, and frontend admin pages.	Chapter 7 architecture, import, and diagnostics sections.
Runtime flow	On-demand scraping, full/admin runs, protected service calls, Laravel import callbacks, database updates, and status polling.	Figures 53-57 and Table 38.
API contracts	Authenticated user scraping endpoints, internal scraper import/check/failure endpoints, proxy route, and admin source/target endpoints.	Appendix A and Section 7.11.
Evaluation summary	Compile checks, container pytest, health probes, deterministic demo-source smoke evidence, and validation boundaries.	Chapter 8 and Table 44.
Limitations and ethics	External site instability, API keys, rate limits, proxy risks, robots/terms considerations, data freshness, and duplicate-detection boundaries.	Section 7.13 and Chapter 9.

AI Job Miner audit support summary.

The appendix intentionally avoids copying large code blocks or raw path lists. Its purpose is to preserve the audit trail while keeping the graduation book readable in print.